SEMICONDUCTOR INTERVIEWS

ANSWER THE KING'S QUESTION

DATA-DRIVEN DEBATES INSPIRED BY KOREA'S ROYAL POLICY EXAMINATIONS

AI · MANUFACTURING · DESIGN · SUPPLY CHAINS

NAKCHO CHOI

SCHOLAR BRIDGE

Semiconductor Interviews: Answer the King's Question

Data-Driven Debates on AI, Manufacturing, Design, and Supply Chains, Inspired by Korea's Royal Policy Examinations

Nakcho Choi

Table of contents

Preface: From the King's Question to the Question of Silicon

Joseon's civil service examinations included an unfamiliar kind of prompt: the *chaekmun* (策問).[1] The king did not merely test how faithfully candidates had memorized the classics. He asked why a legal reform might produce a new abuse, how royal authority should be protected after power had been delegated, and how people should rebuild their lives after a war. The contradictions afflicting the country were placed squarely before the candidate. A strong answer drew on the past, but ultimately made a choice that its own time could bear.

What drew me to the *chaekmun* was not the novelty of an old examination. It was the recognition that **the questions that reveal talent test judgment among competing values, not the volume of a person's knowledge**. Speed and safety. Growth and distribution. Central control and frontline autonomy. Immediate results and costs passed to the next generation. The conflicts faced by officials in Joseon have changed their names, but they still enter today's interview rooms and conference rooms.

When that realization met the semiconductor industry, the first sentence of this book appeared.

[1] A *chaekmun* (策問) was a policy question through which the king asked candidates to address an urgent matter of state with evidence and concrete proposals. In this book, I call it "the king's question."

Semiconductors are usually explained in the language of engineering: process nodes, yield, bandwidth, power, and capital expenditure. Yet the decisions that produce semiconductors do not end inside engineering. The moment a nation treats its supply chain as a security concern, trade becomes a matter of ethics and diplomacy. When demand for AI accelerators surges, power grids, water, and the lives of nearby communities enter the cost sheet. When AI is introduced into process data, efficiency becomes a question of accountability and labor. When war disrupts the flow of industrial gases such as neon and helium, hidden dependencies redraw the world map. A chip is small, but war, exchange rates, climate, labor, education, inequality, and human desire are all etched into it.

This book widens the field of questions so that technical professionals do not think only inside technology. In 2026, semiconductors are strategic infrastructure, an instrument of security, and an industry that moves capital markets. Big Tech companies that lead design and manufacturers that bear responsibility for production cooperate closely while competing over price, allocation, data, and technological control. The effects do not stop at geopolitics or corporate income statements. They reach households through electricity bills, jobs, and the useful life of electronic products. A good engineer must therefore ask not only what can be built, but also where the technology will be used and how its benefits and risks will be shared.

Each chapter recasts this complicated reality in the form of a Joseon *chaekmun*. AI is used as a research tool to find candidate sources and compare opposing arguments, never as an authority that makes the decision for the reader. You will check sources, base years, and units; place the strongest opposing answers against one another; and state your own choice clearly. In one chapter, the strongest answer may be to stop immediately. In another, it may be to proceed as planned. Elsewhere, the best decision may be to divide the scope. The goal is not to memorize more current affairs. It is to develop **the ability to reason from data, withstand objections, and turn a responsible choice into words**.

These questions did not begin only in articles and statistics. They also arose when an organizational chart moved a person's place, or when authority and lines of communication disappeared before a project had formally ended. At such moments, I thought of Admiral Yi Sun-sin enduring disgrace before returning to battle, and of unnamed travelers who crossed the uncertain borders of the Silk Road to keep making a living. Early versions of that emotional record remain on Kakao's long-form publishing platform **Brunch Story**, in the magazines "Stories of a Display Engineer", "A Display Engineer's Coffee Experiments", and "Writing Practice".[2] This book takes from those records only an attitude: **measure, question, and finally judge in your own words**.

[2]Kakao describes Brunch Story as its writing and content-publishing platform. Kakao, "Kakao Story Services", January 9, 2024.

How to Use This Book in 15 Minutes

The primary readers are **applicants preparing for semiconductor and display-industry roles, including discussion-based interviews that connect technology with the humanities and current affairs**. Do not try to memorize an entire chapter. Read the "Today's Royal Policy Question" box and the data graphic first. Choose the strongest evidence from each of the three answers. Then close the sample 90-second response and state, in your own words, both your conclusion and the condition that would make you change it. Actual hiring formats and schedules vary by employer and country, so always follow the current job posting and instructions for your application.

A growing emphasis in Korean semiconductor recruitment on group discussions that connect industry, the humanities, and current affairs directly inspired this work. If candidates are asked whether autonomous tracking technology developed in war should be adapted for civilian disaster-response drones, they cannot reduce war to a single line about "rising demand" or "supply disruption." They must ask what is lost when human suffering is described as an industrial opportunity, and who bears responsibility for misuse if they ignore the dual-use nature of technology. More important than polished interview rhetoric is **the ability to read data, expose conflicting values, and name the conditions under which your own conclusion would be wrong**.

That is why I chose the *chaekmun* as the form through which to question today's semiconductor civilization.

> By what principles should the profits of the AI and data-center boom be divided among reinvestment, labor, shareholders, and the public?
>
> When autonomous tracking technology developed in war is adapted for civilian disaster-response drones, which functions should be retained and which should be blocked?
>
> If process AI raises productivity while reducing the place of skilled labor, whose progress is that innovation?
>
> Will public AI and advanced semiconductors lower the barrier to knowledge, or strengthen the power of those who own computational capital?

The originality of this book does not lie in using Joseon history as decoration. It lies in transplanting the *form of thought* embodied by the *chaekmun* into contemporary industry analysis. First, each chapter states an unavoidable dilemma in one sentence. It then tests the reality of that question through numbers and cases. It presents opposing answers in their strongest forms. Finally, it makes a choice and states the conditions under which that choice would fail. Exemplary historical answers are not preserved here as untouchable solutions; they serve as intellectual opponents that call forth new objections. Nineteen chapters apply eleven archetypal historical questions to different industrial cases. Even when the same source appears more than once, the contemporary question and conclusion do not repeat. The "Joseon Royal Policy Question Archive" mentioned in the text is an **editorially curated public archive** that makes open historical literature easier for modern readers to approach; it does not replace primary-source databases. Where possible, candidates, examination rankings, and answer summaries were checked against the *Veritable Records*, civil-service rosters, and materials from public institutions.

AI was an important tool in this process. It helped identify candidate sources, organize data tables, generate counterarguments I had missed, and compare multiple drafts. This book does not, however, present AI as the final judge of facts or values. AI can be a fast *seori* (書吏), or administrative clerk. The human author remains responsible for deciding which questions to keep, which numbers to trust, and how to speak about war and labor. I removed unsupported figures, labeled corporate claims as corporate claims, and rechecked time-sensitive numbers before publication. Once the author has reviewed and selected an AI-assisted sentence, responsibility for that sentence belongs to the author.

> **Data cutoff:** Policy and market information uses public material available through August 20, 2026, when the first-edition manuscript was frozen. Each statistic retains the source's base year and publication date. Values created for debate and scenario assumptions are distinguished from observed statistics in each chapter's "Data Note." Whenever you run one of the AI research prompts, recheck the original source and any corrections as of the date of your own research.

You will not find one model answer that works for every chapter. That is intentional. Some examples call for an immediate stop in the face of danger; others call for holding a decision until verification is complete; still others use staged judgment under deep uncertainty. Each example is a starting point that can be challenged, not an answer key. I hope the book becomes a foothold for an applicant reading a chapter on the way to an interview, a mirror that helps working engineers and executives see familiar choices afresh, a way for students to read technology alongside society, and a small map for citizens trying to understand invisible infrastructure.

The king once posed the question in an examination hall. There is no king today. Instead, shocks to the supply chain, the speed of AI, climate limits, the silence of workers, and public money ask questions of us. The reader of this book is both candidate and examiner. It is not enough to answer within the choices provided; we must also ask who has been erased from the question.

More important than predicting the future of semiconductors is asking what kind of future we intend to build. We now begin our own policy questions, etched in silicon.

> **Note**
>
> This book is not an official recruitment guide for any company. Company names and trademarks appear only in the context of criticism, education, and factual discussion based on public information.

Copyright

Semiconductor Interviews: Answer the King's Question

Author | Nakcho Choi

Publisher | Scholar Bridge

Publisher website | https://waterfirst.pro/

First Korean edition | August 30, 2026

Art direction and editorial design | Nakcho Choi

Cover and interior illustrations | Directed and edited by Nakcho Choi; generated with OpenAI GPT Image 2

Source names in the PDF and EPUB link to the original material. When a link is unavailable, the institution, document title, and year supplied in the text can be used to locate the same source.

Generative AI Disclosure

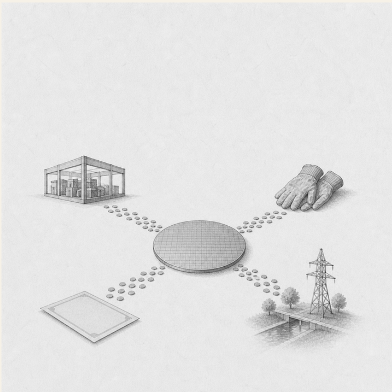
Generative AI technologies supported the planning, editing, and image-generation process for this book. The author is responsible for the final selection, arrangement, revision, and publication of all text and visual material.

Part I.

Forces Shaping the Industry

1. Who Should Share in the Gains from the AI Boom?

Today's Chaekmun



By what criteria should the gains from the AI and data-center boom be divided among reinvestment, labor, shareholders, and the public?

King Jeongjo's 1798 chaekmun on Honam[1] did not assume that abundant resources alone proved good governance. It demanded

[1]The historical chaekmun (策問) connected to this chapter asked how to address not only Honam's resources and talent, but also the burdens borne by its people and its accumulated abuses. The first sentence of the archival text reads, "王若曰□湖南一路卽我朝興王之地也。" In English: "The king speaks: the province of Honam is the land where our dynasty arose." Today's question is not a translation of that era's tax system into the setting of a modern corporation. It reinterprets the structure of the question, which asks us to consider both the total abundance created and the distribution of its burdens.

scrutiny of who enjoyed the benefits of production and who shouldered taxes, corvée labor, and entrenched local abuses.

Nor can the AI memory boom be reduced to "the company made a profit, so let us divide it fairly." The company bore the risks of R&D and failed capital investment; workers delivered yield, met schedules, and secured customer qualification; shareholders accepted capital losses during the downturn. Government and citizens also helped build the productive base through tax credits, transmission networks, water pipelines, education, and the cost of living imposed on host communities. The first task is not to rank their contributions. It is to decide **what counts as profit, how to separate items with different accounting characteristics, and whether the same rule should apply in booms and downturns**.

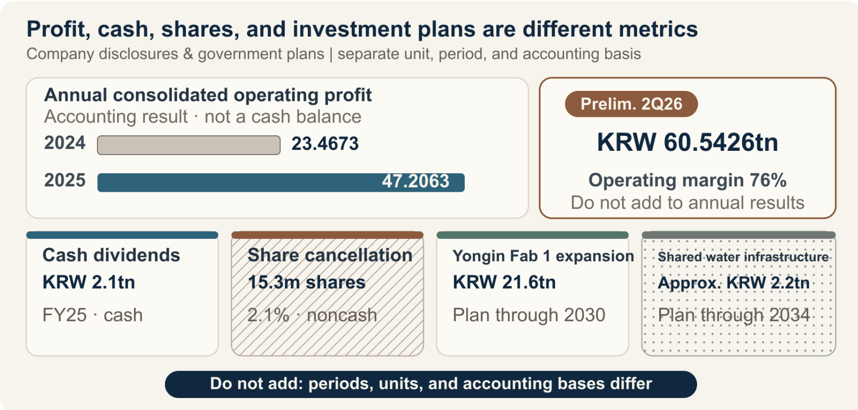
1.1. Why This Question, and Why Now?

According to the company's announcement, SK hynix reported 2025 consolidated revenue of KRW 97.1467 trillion, operating profit of KRW 47.2063 trillion, and net income of KRW 42.9479 trillion. Compared with 2024 operating profit of KRW 23.4673 trillion, operating profit increased by KRW 23.7390 trillion, or 101%. The company said HBM revenue more than doubled year over year and contributed substantially to its record results. The announcement, however, did not separately disclose absolute HBM revenue or operating profit by product. Calling the entire increase in company-wide operating profit "AI windfall profit" would overstate both causation and the relevant denominator.

For the second quarter of 2026, the company's preliminary consolidated results were KRW 79.3187 trillion in revenue, KRW 60.5426 trillion in operating profit, and a 76% operating margin. It announced quarter-end cash and cash equivalents of KRW 88 trillion, total borrowings of KRW 18.6 trillion, and net cash of KRW 69.4 trillion. These figures show the scale of the latest boom, but a single quarter's preliminary figures are not the same as a confirmed annual capacity for distribution. Price spikes, working capital, taxes, the timing of equipment payments, and nonrecurring gains and losses must all be examined.

"Profit" has at least four meanings. Operating profit is the accounting result of core operations; net income reflects financing, taxes, and nonrecurring items; and free cash flow deducts investment from actual cash generation. The performance-bonus base agreed between labor and management and the economic concept of excess profit are different again. When a new hire develops an allocation proposal, the task is not to choose one measure and conceal the others. It is to **separate statutory and contractual payments first, then calculate the cash that remains available for discretionary allocation**.

1.2. Data Lens



1.2.1. Evidence Table

No.	Disclosed figure, period, and denominator	Status	Meaning for the debate
1	SK hynix reported 2025 consolidated revenue of **KRW 97.1467 trillion**, operating profit of **KRW 47.2063 trillion**, and net income of **KRW 42.9479 trillion**. Operating profit in 2024 was KRW 23.4673 trillion.	Company's annual K-IFRS results announcement	Operating profit increased by KRW 23.7390 trillion, but that increase is neither cash available for distribution nor profit attributable solely to HBM.

No.	Disclosed figure, period, and denominator	Status	Meaning for the debate
2	The company announced preliminary consolidated revenue of **KRW 79.3187 trillion**, operating profit of **KRW 60.5426 trillion**, and an operating margin of **76%** for the second quarter of 2026, with quarter-end net cash of **KRW 69.4 trillion**.	Company's preliminary quarterly figures dated July 29, 2026	The figures show the latest boom and the company's financial capacity, but one quarter should not be fixed as the basis of a long-term allocation formula.

No.	Disclosed figure, period, and denominator	Status	Meaning for the debate
3	The company stated that **HBM revenue more than doubled** in 2025 from the previous year.	Company's explanation of causes	Because absolute HBM revenue and product-level cost and operating-profit denominators are unavailable, company-wide profit cannot be converted into a measure of HBM's contribution.

No.	Disclosed figure, period, and denominator	Status	Meaning for the debate
4	Total dividends for fiscal 2025 were **KRW 2.1 trillion**, and **15.3 million shares** were canceled. On August 19, 2026, the board resolved to acquire and cancel in full **KRW 40 trillion** of treasury shares; based on the previous day's closing price, the company described this as **24.07 million**	The 2025 share count and the 2026 acquisition amount are confirmed figures.	The confirmed figure and converted value switch places between the two years, so the "scale of cancellation" cannot be compared in a single column. The actual number of shares canceled will depend on share prices during the acquisition period.

No.	Disclosed figure, period, and denominator	Status	Meaning for the debate
5	In February 2026, the company announced an additional **KRW 21.6 trillion** investment in the first Yongin fab through December 2030, raising total investment to approximately **KRW 31 trillion** including the existing KRW 9.4 trillion. Equipment installation costs were excluded.	Multiyear capital plan approved by the board	KRW 21.6 trillion is neither annual capital expenditure for 2025 nor cash paid all at once. The time horizon, cancellation options, and equipment costs must be considered separately.

No.	Disclosed figure, period, and denominator	Status	Meaning for the debate
6	The company’s 2025 DBL measurement reports indirect economic contributions of **KRW 8.7114 trillion** from “employ-ment,” **KRW 9.5329 trillion** from “tax payments,” and **KRW 2.1117 trillion** from “dividends.” It states that the scope includes five subsidiaries and four social	Social-value metrics defined by the company	The employment figure cannot automatically be treated as total wages and performance bonuses, nor can the tax figure be treated as identical to income-tax expense in the consolidated income statement. The measurement scope and methodology must be checked.

No.	Disclosed figure, period, and denominator	Status	Meaning for the debate
7	The government's 2025 semiconductor support package proposed increasing fiscal investment in infrastructure from **KRW 3.1 trillion to KRW 5.1 trillion**, using the national budget to cover **70%** of the corporate share of undergrounding transmission lines in Yongin and	Nationwide support plan, limits, and tax provisions	This does not mean that any particular company actually received these amounts. A debate about public contribution requires company-specific tax credits and subsidies actually realized, along with their conditions and clawback provisions.

No.	Disclosed figure, period, and denominator	Status	Meaning for the debate
8	The Ministry of Environment and K-water announced plans to invest approximately **KRW 2.2 trillion** in the Yongin integrated water-supply project through 2034, supplying **1.072 million m³ per day**. Phase 1 is scheduled to supply **310,000 m³ per day** beginning in January 2031.	Long-term public infrastructure plan serving multiple companies and industrial complexes	These are not the current water use or subsidy receipts of one company. Water intake, supply, recycling volumes, and cost sharing must be examined separately.

1.2.2. How to Read These Numbers

First, **separate operating profit from cash**. Operating profit includes depreciation, while equipment payments, inventory, receivables, and tax payments move differently in the cash-flow statement. "A percentage of operating profit" is simple, but it overstates the capacity to pay if sustaining capital expenditure and working capital are not deducted.

Second, **do not add cash, share counts, and market value in a single column**. In 2025, the confirmed figure was the number of shares canceled; in the August 2026 resolution, it was the KRW 40 trillion acquisition amount, while 3.3% was a conversion based on the closing price. Even when figures carry the same label, what is confirmed and what is derived must be rechecked each year.

Third, **align annual results, preliminary quarterly figures, and multiyear plans to the same period**. Multiplying second-quarter 2026 operating profit by four and declaring it annual profit, or treating KRW 21.6 trillion through 2030 as investment in 2026 alone, distorts the amount available for allocation.

Fourth, **separate product contribution from company-wide performance**. The fact that HBM revenue more than doubled demonstrates the strength of AI demand. Yet legacy DRAM, NAND, eSSD, exchange rates, pricing, and costs also affect operating profit. If HBM-specific operating profit is not disclosed, the "share created by AI" must be expressed as a range.

Fifth, **consider both gross and net amounts for taxes, subsidies, and infrastructure**. Corporate income tax is a legal obligation; an investment tax credit is relief for qualifying investment; and support for transmission and water infrastructure may be shared by multiple users. Tax payments and support plans cannot simply be netted to manufacture a "government share."

Sixth, **do not substitute the company's single "employment" metric for labor's contribution**. Headcount, base pay, overtime, performance bonuses, contractor labor, occupational accidents, turnover, skill, and yield improvement must be examined separately. If figures from labor-management negotiations are not public, do not present estimated bonuses per employee as fact.

1.3. Competing Answers

1.3.1. Answer A — Distribute the Windfall Broadly

Record profits are not produced by technology and capital alone. Frontline employees met delivery commitments through shift work, yield improvements, and customer support; government and host communities shared the costs of tax credits, transmission networks, water, and transport; and shareholders accepted losses of capital during the downturn. Verified excess cash well above its historical midpoint should be distributed broadly through one-time performance bonuses, improved supplier pricing and safety, local infrastructure, and special dividends, reinforcing the legitimacy of the boom and trust within the organization.

Yet immediately distributing the increase in operating profit could overlook taxes not yet paid, maintenance investment, customer contracts, and the next process transition. Converting a one-time boom into permanent costs such as base pay or fixed dividends could create acute restructuring pressure in the next downturn.

1.3.2. Answer B — Prioritize Reinvestment and Protect Shareholder Rights

HBM and advanced packaging require companies to accept R&D failure, place equipment orders in advance, and endure long qualification cycles before returns are earned. The company and its shareholders continued to invest through the last loss-making period, and without next-generation capacity, today's jobs and tax base will not endure. After statutory taxes and agreed compensation have been paid, remaining cash should be allocated first to reinvestment and shareholder returns, based on long-term customer contracts, yields, and the likelihood of earning back the investment.

But the phrase "future investment" cannot indefinitely defer recognition of employees' and communities' contributions while cash accumulates. Unless the demand case, stop conditions, expected returns, and amount of public support actually realized are disclosed, reinvestment remains an unverifiable internal assertion.

1.3.3. Answer C — Use a Conditional Allocation Formula

I would begin by defining the denominator for discretionary allocation as follows:

Distributable pool = after-tax operating cash flow − sustaining and safety investment − committed growth investment − working-capital and net-debt safety buffer

Statutory taxes, base salaries, workplace-safety obligations, and existing contracts are not "shares" of this pool. They are obligations that must be honored first. Only the remaining distributable pool is divided among incremental reinvestment, employee profit-sharing, cash returns to shareholders, power and water efficiency and local benefits, and a downturn reserve. The market value of share cancellations should sit outside the cash-allocation table and be shown separately as an effect on share count.

The ratios should not be fixed automatically. The reinvestment share rises when noncancelable orders and downside cash flow are confirmed; the employee share rises when agreed performance indicators and contributions to safety and skill are verified. The industry, however, often **commits in advance** to returning a certain proportion of free cash flow to shareholders. That turns the shareholder share from a residual into an amount deducted from the denominator first, conflicting with the sequence above. Treat the commitment as a floor only, and adjust any amount above it within the distributable pool.

Burden sharing must also be symmetrical in a downturn. If the distributable pool is negative for two consecutive quarters, discretionary share repurchases, variable executive compensation, incremental performance bonuses, and cancelable growth investment should be reduced in a predetermined order. Base pay, safety, and essential training should not be the first adjustment items, while reserves accumulated during the boom should cushion a collapse in performance bonuses. Conversely, if investment and compensation are reduced during a downturn, they must be restored after the recovery using the same indicators.

1.4. AI Research Design

1.4.1. Prompt for the AI

> Design a profit-allocation proposal for a Korean semiconductor company benefiting from the AI and data-center boom. Research the following in order: □ the Financial Supervisory Service's DART annual reports, audit reports, and cash-flow statements; □ Korea Exchange disclosures and company board, shareholder-meeting, and earnings announcements; □ official wage and performance-bonus agreements issued by the company and labor union, or materials from the Ministry of Employment and Labor; □ Ministry of Economy and Finance and National Tax Service materials on corporate income tax and investment tax credits, and subsidy

and clawback conditions; and □ Ministry of Trade, Industry and Energy, Ministry of Environment, and local-government data on the cost and allocation of power and water projects. For every figure, record the relevant year, unit, consolidated or separate scope, cash or noncash status, actual or planned status, numerator, and denominator. Use separate columns for "verified fact, company claim, labor claim, government plan, debate assumption, and AI inference." Do not use operating profit as distributable cash. Present a formula that begins with after-tax operating cash flow and deducts sustaining and safety investment, committed growth investment, and working-capital and debt safety buffers. Create a table that sets out reinvestment, employee, shareholder, and public shares and the conditions for switching between boom and downturn rules. For every external source, provide the institution, document title, publication date, and direct URL.

1.4.2. Data Acquisition

Priority	Source	Items to verify
1	Audited financial statements, cash-flow statement, and notes	Consolidation scope, operating profit, operating cash flow, income taxes paid, sustaining and growth CAPEX, working capital, net debt

Priority	Source	Items to verify
2	Board, shareholder-meeting, and exchange disclosures	Dividend record date and total, treasury-share acquisition cost and shares canceled, investment period and cancellation conditions, actual cash outlay
3	Product and customer data	HBM, server DRAM, and eSSD revenue ranges; long-term contracts; cancellation rights; yields; customer concentration; downside pricing
4	Official labor-management materials	Base-pay and performance-bonus formulas, denominator, cap and deferral, eligible headcount, supplier coverage, downturn-adjustment rules
5	Primary tax and subsidy documents	Income-tax expense and cash taxes paid, tax credits realized, subsidy payment and clawback, employment and investment conditions

Priority	Source	Items to verify
6	Power, water, and local data	Government and corporate contributions, net withdrawal and recycling, peak power and consumption, costs borne by residents, local funds and disclosure scope

1.4.3. Assessing the Information

1. **Accounting character:** Distinguish profit, cash inflow, cash outflow, changes in share count, and market value.
2. **Period:** Convert preliminary quarterly figures, annual results, and five-year investment plans to a common period, clearly labeling estimates.
3. **Scope:** State whether each figure covers the consolidated group, separate entity, domestic entities, subsidiaries, or contractors.
4. **Causation:** Between HBM revenue growth and company-wide operating profit, control for pricing, exchange rates, legacy products, and costs.
5. **Additionality:** Determine whether public support increased investment or merely subsidized investment already planned.
6. **Symmetry:** Test whether boom-period payouts and downturn-period cuts operate through the same indicators and sequence.

7. **Source of claims:** Separate evaluative statements by the company, labor, and government from factual values, and record whether contrary evidence is public.

1.4.4. Core Questions for Debate

- Which measure should define "profit available for distribution": operating profit, net income, free cash flow, or the labor-management performance-bonus base?
- When HBM product-level profit is not disclosed, what range should be given for the AI boom's contribution?
- How should cash dividends and share cancellations, and multiyear CAPEX and current-year spending, be separated within the same allocation table?
- If tax credits and national funding for infrastructure are verified, to what should the company's public and local obligations be linked?
- When a downturn begins, in what order should shareholder returns, variable performance bonuses, and discretionary investment be adjusted?

1.5. Case Debate

The company and figures below are all **hypothetical conditions for debate**. They are not data from SK hynix or any other real company. You are a newly hired analyst assigned to the joint finance, labor-relations, and sustainability task force at "Hanbit Memory." The company forecasts KRW 10 trillion in after-tax operating cash flow for 2026. After deducting KRW 3 trillion of sustaining and safety CAPEX, KRW 2 trillion of committed growth CAPEX, and a KRW 1 trillion working-capital and net-debt safety buffer, the distributable pool is KRW 4 trillion.

Stakeholder demands are as follows:

- The investment division requests KRW 2.2 trillion for an additional, cancelable HBM line. Customer minimum-purchase commitments remain under negotiation.
- The labor union requests KRW 1.4 trillion in additional performance-based profit sharing, citing past yield improvements and the burden of shift work.
- Shareholders request KRW 1.8 trillion in cash for a special dividend and treasury-share acquisition.
- A local and public-sector council requests KRW 0.6 trillion for transmission-grid cost sharing, water recycling, supplier safety, and a community fund.

The demands total KRW 6 trillion, exceeding the distributable pool by KRW 2 trillion. Assume the company has actually realized KRW 0.8 trillion in investment tax credits and KRW 0.2 trillion in infrastructure support. If memory prices fall by 35% next year, after-tax operating cash flow would decline to KRW 5.5 trillion, and canceling the additional line could trigger a KRW 0.3 trillion penalty. All of these figures are also assumptions.

The new analyst must choose one of the three positions—**broad distribution, priority for reinvestment and shareholders, or a conditional formula**—and present a KRW 4 trillion allocation table. No party—the company, employees, shareholders, government, or community—may assign a value of zero to another party's contribution. The proposal must include the distributable-pool formula; the cash or noncash status of each share; investment-switching conditions tied to customer contracts and yield; disclosure and clawback conditions for public support actually realized; the adjustment sequence in the event of a 35% price decline; and the criteria for restoration in the next boom.

1.6. Sample 90-Second Response

"I choose **Answer C: allocate the distributable pool, not operating profit**. In the actual company announcement, 2025 operating profit was KRW 47.2063 trillion and dividends were KRW 2.1 trillion, but neither figure equals cash available for allocation. In the hypothetical case, I would distribute only KRW 4 trillion, calculated by deducting KRW 3 trillion in sustaining and safety investment, KRW 2 trillion in committed growth investment, and a KRW 1 trillion working-capital and debt safety buffer from KRW 10 trillion in after-tax operating cash flow. My proposal is KRW 1.6 trillion for reinvestment, KRW 1 trillion for employees, KRW 0.8 trillion for shareholders, KRW 0.4 trillion for power, water, and the community, and KRW 0.2 trillion for a downturn reserve. The counterargument that we must invest more during a boom to protect future jobs and earnings is valid. I would therefore reallocate the reserve and part of the shareholder share to reinvestment once customer minimum purchases are confirmed and 1.5 times the required investment cash remains even under a 35% price decline. Conversely, if the distributable pool is negative for two consecutive quarters, I would halt share repurchases and variable executive compensation first, while protecting base pay, safety, and essential training. Fair allocation does not mean equal percentages. It means agreeing in advance on a common denominator and symmetrical rules for booms and downturns."

1.7. Follow-Up Questions

Cross-Examination and Rebuttal

- If HBM product-level operating profit is not disclosed, how would you estimate the "AI windfall profit" within the KRW 23.7390 trillion increase?
- What objective indicators would cause you to change each proposed share—40%, 25%, 20%, 10%, and 5%—by 5 percentage points?
- Employees point to years of accumulated skill, while shareholders point to capital losses during the downturn. How would you compare the two contributions across one industry cycle?
- The confirmed figure in 2025 was the number of shares canceled, while in 2026 it was the KRW 40 trillion acquisition amount. What would you control for to compare the two years on the same basis?
- If you increase the community fund when tax credits and infrastructure support rise, would you double count corporate taxes, subsidies, and local contributions?

- If memory prices fall by 35% and the distributable pool turns negative, which would you cut first—shareholder returns, performance bonuses, or incremental investment—and which would you restore first after recovery?

1.8. Sources

- SK hynix, *SK hynix Announces FY25 Financial Results* (2026-01-28)
- SK hynix, *SK hynix Announces 2Q26 Financial Results* (2026-07-29)
- SK hynix, *New Facility Investment for Yongin Semiconductor Cluster* (2026-02-25)
- SK hynix, *DBL Performance—2025 SV and EV Measurements* (accessed 2026)
- Korea Policy Briefing and Ministry of Economy and Finance, *Plan to Strengthen Fiscal Investment for Leadership in Global Semiconductor Competitiveness* (2025-04-17)
- Presidential Water Commission and Ministry of Environment, *Yongin Semiconductor Cluster Water-Supply Project Moves into Full Implementation* (2025-05-16)
- National Tax Service, *Major Corporate Tax Credits and Exemptions*—National Strategic Technologies and Semiconductor Investment Tax Credits (accessed 2026)
- Joseon Dynasty Chaekmun Archive, King Jeongjo's 1798 *Seven Abuses in Honam and Their Remedies*

Data note: The 2025 and 2026 results, dividends, treasury-share counts and market values stated by the company, multiyear capital plan, and DBL measurements are actual disclosed values or official company plans and assessments drawn from the respective primary company sources. The claim that HBM contributed to performance and the DBL categories are company claims and measurements; they are not treated as equivalent to product-level profit or income-tax expense. The government's KRW 5.1 trillion, 70%, and 5-percentage-point figures and the water project's KRW 2.2 trillion and 1.072 million m^3 per day are nationwide support and infrastructure plans, not the amount received or used by a specific company. Every figure for Hanbit Memory—KRW 10 trillion, KRW 3 trillion, KRW 2 trillion, KRW 1 trillion, KRW 4 trillion, KRW 2.2 trillion, KRW 1.4 trillion, KRW 1.8 trillion, KRW 0.6 trillion, KRW 0.8 trillion, KRW 0.2 trillion, KRW 5.5 trillion, and KRW 0.3 trillion—as well as the 35% price decline, allocation amounts, and switching thresholds are hypothetical assumptions for debate.

2. Does War Advance Humanity?

Today's Chaekmun

Should a newly hired design engineer adapt autonomous tracking technology developed in wartime for a civilian disaster-response drone?

King Seonjo's 1568 chaekmun[1] demanded that a ruler choose neither war out of anger nor peace out of fear, but judge principle, capability, and circumstances together. Borrowing that structure, this chapter separates the fact that war may expand technological capability from the judgment that the resulting technology has improved human life.

> Weapons have grown more sophisticated by the day, but has the life of the people improved to the same degree? By what standard should a technology born of war be called progress?

This is not a quotation translated from the words of a particular king. It is a new question written for this chapter in the form of a Joseon-era chaekmun.

2.1. Why This Question, and Why Now?

The claim that war "invented" a technology must be distinguished from the claim that it "accelerated" the development, mass production, and diffusion of an existing technology. The airplane was invented before World War I, but reconnaissance, communications,

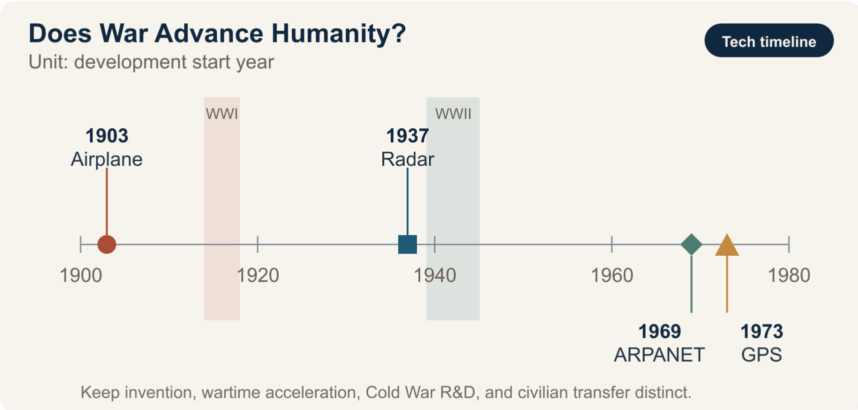
[1]The chaekmun (策問), or "royal policy question," that inspired this chapter asked: "When confronting a foreign enemy, should one choose war or peace, and how should principle, strength, and circumstances be weighed together?" The Chaekmun Archive reconstructs the meaning of the published translation in this concise Classical Chinese form: "禦外之道□戰與和而已。義、力、勢□何所先後□" It is not quoted as a literal transcription of the original. Nor is this chapter's question a modern-language translation of that text. It reinterprets the structure of the original question—which demands that war be judged not by principle alone, but also by strength, circumstances, and eventual consequences—and applies it to a contemporary technology problem.

high-altitude flight, and engine technology advanced rapidly during the war, while the aviation industry's manufacturing base expanded. World War II accelerated research and large-scale deployment in radar, electronic computing, and jet propulsion. ARPANET, an ancestor of the internet, began in 1969, while integrated development of GPS began in 1973. Rather than direct products of the world wars, both are examples of Cold War defense research later diffusing into civilian infrastructure.

The war in Ukraine has advanced the rapid-cycle integration of low-cost drones, satellite communications, electronic warfare, image recognition, and battlefield software. Underlying all of these systems are image sensors, communications chips, power semiconductors, memory, and processors. The same semiconductors can find people in a wildfire zone and prevent traffic accidents, yet they can also become the eyes and brains of weapons that identify and strike targets.

The issue in this chapter therefore extends beyond whether war advances technology. It asks **whether progress in technological capability can be equated with human progress**, and who bears responsibility for converting technology accumulated in war into benefits for peace.

2.2. Data Lens



2.2.1. Evidence Table

No.	Verified fact	Meaning for the debate
1	The Smithsonian National Air and Space Museum explains that World War I transformed airplanes from machines with limited capability into military aircraft performing specialized missions, and laid the foundation for the postwar aviation industry (2025).	War did not invent the airplane; it accelerated its performance, standardization, and mass production.

No.	Verified fact	Meaning for the debate
2	The U.S. Army explains that radar, demonstrated in 1937, developed rapidly during World War II and subsequently entered everyday applications such as air-traffic control and weather observation (2025).	Civilian diffusion of military research is possible, but it does not justify the human cost of war.

No.	Verified fact	Meaning for the debate
3	DARPA's four-node ARPANET became operational in 1969, and the U.S. Department of Defense launched the integrated NAVSTAR GPS program in 1973.	It is chronologically wrong to describe the internet and GPS as direct inventions of the world wars. They should be described as civilian transfers of Cold War defense research.

4	NATO lessons-learned material on Ukraine analyzes how low-cost one-way attack drones and electronic warfare are rapidly reshaping one another's methods of operation.	Do not cite battlefield results or production volumes from an ongoing war as settled figures. Use only the qualitative change—shorter technology iteration cycles—as evidence for judgment.

The first thing to read in this chapter is the **development timeline**. The airplane existed before World War I, and radar before World War II. ARPANET and GPS began after both world wars had ended. This chronology alone allows us to distinguish "inventions of war" from "development accelerated by war" and the "civilian transfer of Cold War defense research." Production volumes and interception rates in an ongoing war change constantly and are strongly shaped by the interests of those reporting them, so they are excluded from the quantitative evidence used in the sample response.

2.2.2. How to Read These Numbers

- Drone production volumes and numbers of strikes are operational figures that change with every reporting date; they do not demonstrate a technology's civilian benefit.
- The fact that a technology found civilian uses after a war does not prove the counterfactual that it would never have emerged without war.
- Interception rates, destruction rates, and claimed battlefield successes reported by belligerents reflect both measurement methods and institutional interests; they must be cross-checked against opposing-party data and independent investigations.
- The opportunities lost in medical, climate, and education technology when funding, talent, and semiconductors are concentrated on weapons do not appear in battlefield statistics.

2.3. Competing Answers

2.3.1. Answer A — War Drives Progress

War radically increases the cost of failure and the pressure of time, concentrating funding, talent, and frontline feedback in one place. Technologies that originated in or were accelerated by military demand—aviation and radio communications in World War I, radar and electronic computing in World War II, and ARPANET and GPS during the Cold War—became foundations for civil aviation, communications, logistics, and disaster response. The war in Ukraine may similarly diffuse advances in low-cost drones, electronic warfare, satellite communications, and rapid procurement into search and rescue, infrastructure inspection, and autonomous-mobility technologies.

The strength of this answer is that it does not ignore the real effect of war on the speed and scale of technological development. Yet the moment improved technical performance is labeled moral and social progress for humanity, the victims are reduced to a cost of innovation.

2.3.2. Answer B — War Drives Regression

The first thing war optimizes is not how to save people, but how to detect them sooner, strike them from farther away, and attack at lower cost. Drones and AI can reduce risks to soldiers, but they can also lower the political threshold for attack and increase automated killing with unclear lines of responsibility. Once infrastructure destruction, brain drain, interrupted education, normalized surveillance, and enormous opportunity costs are included, advances in selected technologies are not enough to claim that humanity has progressed.

The strength of this answer is its insistence that postwar civilian benefits do not justify war itself. Its weakness is that it may not adequately explain the value of defensive technology and deterrence, or of communications, interception, and mine-clearance technologies that saved lives during war.

2.3.3. Answer C — Progress Is Conditional

War may **advance technological capability, but it does not automatically advance humanity**. Progress must be judged by outcomes, not performance alone. We must examine the breadth of civilian transfer, the effect on protecting life, controllability, traceability of responsibility, the risk of misuse after diffusion, and the possibility of achieving the same results through nonmilitary public research. Semiconductor companies cannot evade responsibility merely by saying that they sold civilian components. They need end-use screening, human-rights due diligence, export-control compliance, and procedures to suspend supply when misuse is detected.

2.3.4. Decision Axes for Answer C

In the debate, evaluate "technological progress" and "human progress" on separate lines.

Decision axis	Question to verify
Technological capability	How much did performance, price, production speed, and reliability actually improve?
Life and safety	Did deaths, injuries, and misidentification harms among civilians and combatants decline?
Civilian diffusion	Who receives the postwar benefits transferred to medicine, transport, communications, and disaster response?

Decision axis	Question to verify
Control and accountability	Are there final human judgment, logs, audits, shutdown mechanisms, and identifiable accountable parties?
Alternatives and opportunity cost	Could the same technology have been developed through nonmilitary public research and international cooperation?

If only technological capability improves while life, accountability, and alternative-path indicators deteriorate, it is entirely coherent to answer: "The technology advanced, but humanity regressed." That is not a contradiction; it is the central distinction of this chapter.

2.4. AI Research Design

2.4.1. Prompt for the AI

> Research technologies whose development, mass production, or deployment accelerated during World Wars I and II and the Russia–Ukraine war. Distinguish among airplanes, radar, nuclear technology, drones, electronic warfare, satellite communications, AI, and semiconductors. For each technology, present in a table: □ whether it existed before the war, □ how its performance or operating method changed during the war, □ postwar civilian transfer, □ civilian harm and misuse, and □ the issuing institution, document title, year, and primary-source

URL. Do not describe ARPANET or GPS as a direct invention of either world war; verify the actual development dates. Exclude battlefield results, production volumes, and interception rates from an ongoing war as quantitative evidence for the conclusion, and distinguish verified facts, institutional claims, and the analyst's inferences. Finally, identify the evidence that most strongly supports the technology-acceleration thesis, the human-regression thesis, and the civilian-transfer thesis, along with the threshold that would change each conclusion.

2.4.2. Data Acquisition

Priority	Source	Items to verify
1	Primary documents from militaries, governments, and international organizations	Technology development date, operational scope, definitions
2	Technology histories from museums and public research institutions	Prewar existence, wartime improvements, timing of civilian transfer
3	Harm and humanitarian data	Civilian casualties, infrastructure destruction, losses to healthcare and education

Priority	Source	Items to verify
4	Company and research-institution materials	Actual functions of components and software, end-use controls, feasibility of postwar conversion

2.4.3. Assessing the Information

1. **Distinguish invention from acceleration.** If a technology existed before the war, describe accelerated performance improvement, mass production, or integration rather than "invention."
2. **Cross-check the timeline.** Do not misattribute Cold War programs such as ARPANET, launched in 1969, and integrated GPS development, launched in 1973, as direct products of the world wars.
3. **Separate capability from outcome.** Technical indicators such as detection range do not automatically prove improvements in life, freedom, or well-being.
4. **Exclude volatile figures from ongoing wars from the evidence used in the response.** Treat battlefield results, casualties, interception rates, and production volumes only as background until independently verified.
5. **Ask about the opportunity cost of alternatives.** Label as a counterfactual assumption any estimate of the results that might have been achieved had the same funding and talent gone to peaceful public research.

2.4.4. Core Questions for Debate

- The fact that war increases the **speed** of technology is different from the judgment that it improves the **direction** of humanity.
- Do not force the benefits of civilian transfer and wartime losses of life, freedom, and education into the same unit.
- Dual-use technologies such as semiconductors and AI require end-user screening, human control, and procedures to halt supply after misuse is discovered.
- A strong answer neither glorifies war nor rejects technology. It states the conditions under which technological capability becomes human progress.

2.5. Case Debate

The following is an **interview scenario**, not a real company case. You are an image-processing chip design engineer six months into your first job. Your team wants to deploy on a wildfire search-and-rescue drone an object tracker trained jointly on publicly available wartime footage and commercial drone data. Compared with the existing model, its missing-person detection rate is 12 percentage points higher and its computing power consumption is 18% lower, but its ability to track vehicles and people over long periods could be repurposed for military targeting. Retraining the system exclusively on civilian data would delay release by four months and is expected to reduce the detection rate by 6 percentage points. All of these figures are hypothetical.

1. The design team proposes which long-duration tracking, coordinate output, and external model-replacement functions should be restricted in the chip and firmware.
2. The rescue agency explains its need for detection in severe weather and re-identification of missing persons, while safety and ethics staff describe the risks of repurposing the system for surveillance and targeting.
3. The validation team compares the system with a control model trained solely on civilian data to test whether wartime material actually contributed to the higher detection rate.
4. The team chooses among **releasing the existing model unchanged, rebuilding it entirely with civilian data, and producing a civilian-only design with high-risk functions removed**.

The core question is not merely whether to screen the customer. It is whether engineers can materially reduce the risk of misuse at the product-function and data stages.

2.6. Sample 90-Second Response

"I choose **Answer C: a civilian-only design with high-risk functions removed**. The airplane existed before World War I, while ARPANET and GPS began in 1969 and 1973, respectively. This chronology refutes the claim that war invented every technology, but it does not deny that war accelerated development and deployment. In the scenario, the 12-percentage-point increase in detection, 18% reduction in power, and four-month delay are all hypothetical. I also accept the counterargument that using only civilian data would sacrifice rescue performance and time to market. I would therefore retain the model while blocking long-duration tracking of individuals, external transmission of precise coordinates, and unauthorized model replacement. I would compare it with the civilian control model to verify the contribution of wartime data, then disclose both missing-person detection and false-tracking rates. I would postpone release if the restrictions could be bypassed or false tracking exceeded the preset threshold. I would continue monitoring these criteria after the civilian transfer. Only when greater technological capability translates into protection of life and effective control should we call it human progress."

2.7. Follow-Up Questions

Cross-Examination and Rebuttal

- If the civilian benefits of airplanes, the internet, and GPS are so great, must we concede that war is a necessary engine of innovation?
- Could the same pace have been achieved by applying wartime levels of funding and regulatory flexibility to climate and infectious-disease research?
- What important information might the sample response miss by excluding current drone figures from an ongoing war?
- If an image-processing chip manufacturer could not know the end user, how far does its responsibility extend?

- Can a weapons system be said to have "human control" when a person approves the action but follows the AI recommendation almost automatically?
- By what criteria would you distinguish technologies that should be barred from postwar civilian transfer from those that should be actively transferred?

2.8. Sources

- Smithsonian National Air and Space Museum, *The Evolution of World War I Aircraft* (2025)
- U.S. Army, *Radar demonstration transforms the Army* (2025)
- DARPA, *ARPANET* (updated 2025)
- GPS.gov, *The History of GPS Technology: Government Collaboration and Technology Transfer* (2018)
- NATO JALLC, *Ukraine Insights: Electronic Warfare against One-Way Attack Drones* (2026)
- U.S. Congressional Research Service, *Defense Primer: Emerging Technologies* (2026)
- Joseon Dynasty Chaekmun Archive, King Seonjo's 1568 *War or Peace?*

Data note: Battlefield results, interception rates, and production volumes from an ongoing war were excluded from the quantitative evidence in the sample response because they change rapidly and the reporting parties have strong interests in the figures. The performance and schedule figures in the scenario are hypothetical assumptions for debate.

3. Should Reproducibility Take Priority over Peak Performance?

Today's Chaekmun

If a research sample's peak performance conflicts with repeatability, which result should advance to the next stage of development?

King Sejong's 1447 chaekmun on law and administration[1] required an answer not only about adding a new standard, but also about who would administer that standard and according to what principles. In R&D, one peak result may demonstrate possibility, but another person can use it only if the same procedure can be repeated on another day, wafer, and instrument.

3.1. Why This Question, and Why Now?

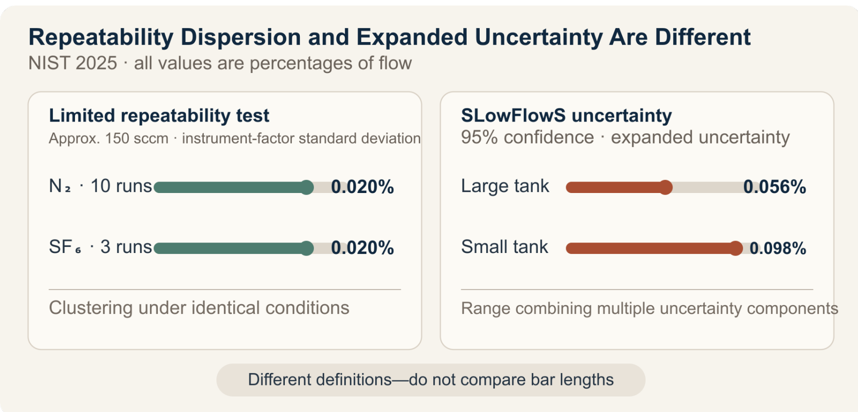
SLowFlowS, a low-flow standard for semiconductor process gases published by the U.S. National Institute of Standards and Technology in 2025, covers a flow range of 0.01–1,000 cm^3/min and reports 95% confidence-level expanded uncertainties of 0.056–0.098% of flow. In preliminary repeatability tests from the same research, the standard deviations of instrument factors for ten nitrogen runs and three sulfur hexafluoride runs at approximately 150 sccm were each 0.02%.

[1]The historical chaekmun (策問) connected to this chapter asked about the cycle in which a law creates abuses and another law is then enacted, and about where authority should reside. The Chaekmun Archive reconstructs the meaning of its published translation in the concise Classical Chinese "法立而弊生□弊生而法又立。其所以救之者何歟□政權當歸於何所歟□" This is not a literal transcription of the original. It asks: "When a law is established, abuses arise; when abuses arise, another law is established. How can this be remedied, and where should authority reside?" Today's question does not translate the legal institutions of that era into experimental regulations. It reinterprets the structure of the question—that even a sound standard cannot guarantee results unless its operators and verification procedures are sound—and connects it to research and development.

These figures cannot be collapsed into the statement “there was almost no error.” The 0.02% figure describes how tightly values clustered under specific repeatability conditions, while 0.056–0.098% is a 95% expanded uncertainty incorporating calibration, temperature, pressure, volume, and other uncertainty components. Even tightly clustered repeated values may be inaccurate if all are biased in the same direction.

Peak performance from a research sample should be read in the same way. A peak is valuable for exploring a technology’s upper bound. Yet if the best sample alone is selected and reported as though it were the mean, or a difference smaller than the measurement uncertainty is called an improvement, the manufacturing team inherits a target it cannot reproduce. A new researcher should place the peak, central tendency, spread, number of out-of-specification results, measurement uncertainty, and results after condition changes in a single table.

3.2. Data Lens



3.2.1. Evidence Table

Category	Figure and range in the primary data	Meaning for R&D decisions
Measurement range	SlowFlowS covers 0.01–1,000 cm^3/min at standard conditions of 273.15 K and 101.325 kPa.	Performance figures must be reported together with their applicable range and reference conditions.
Repeatability test	At approximately 150 sccm, the standard deviations of the instrument factors for ten nitrogen runs and three sulfur hexafluoride runs were each 0.02%.	This supports low dispersion under the same conditions, but does not establish reproducibility across all instruments and dates.
Expanded uncertainty	The large tank's value was 0.056% of flow and the small tank's was 0.098%. Both are 95% confidence-level expanded uncertainties.	Because these values differ in definition from standard deviation, they must not be placed in a single bar ranking.
Temperature validation	Over 15 hours, the 24 temperature sensors on the tank surface agreed within 63 mK of one another, and each sensor had a standard deviation below 2 mK.	Long-duration, multipoint validation provides stronger evidence than one favorable instantaneous measurement.

Category	Figure and range in the primary data	Meaning for R&D decisions
Species effect	When manufacturer correction factors were applied to measurements of gases other than the nitrogen used for calibration, errors of up to ±6% of flow remained.	Even with good repeatability, systematic error can grow when the material or method changes.

3.2.2. How to Read These Numbers

The figures 0.02% and 0.056–0.098% answer different questions. The first is a standard deviation from a limited repeatability test; the second is a 95% expanded uncertainty combining multiple uncertainty components. A small standard deviation alone does not justify saying that measurement accuracy is 0.02%.

Ten nitrogen runs and three sulfur hexafluoride runs are also not a sufficient sample for manufacturing validation. The NIST study validates a new national flow standard; it is not a yield-assurance test for a particular process recipe. It does, however, model a valuable research practice: disclosing how many measurements were taken, what variables changed, and which uncertainties were included alongside the performance result.

Peak performance and reproducibility are not enemies. A peak creates a hypothesis during exploration; repeatability and reproducibility determine whether that hypothesis is ready to advance. Do not erase the peak from the report. Instead, disclose whether it came from selecting only favorable samples without a predefined exclusion rule and whether it persists under independent remeasurement.

3.3. Competing Answers

3.3.1. Answer A — Explore the Peak First

The purpose of early research is to find performance possibilities not yet observed. Focusing only on making the mean stable can cause teams to discard high-risk hypotheses too soon and explore only around existing processes. Even a peak from one sample is important evidence that can redirect the next experiment, provided it passes a minimum check showing that it is not a measurement error.

The peak should not, however, be labeled a completed development result. The team must disclose the raw data, sample selection, and measurement conditions, label it an "exploratory result," and request funding for follow-up experiments that identify the cause and establish reproducibility.

3.3.2. Answer B — Transfer the Reproducible Result First

Manufacturing and product development use distributions that meet specification repeatedly, not a sample that happens to be good. A high peak accompanied by large sample-to-sample dispersion and repeated specification failures indicates a narrow process window or an uncontrolled confounding factor. Advancing a condition with a lower peak but stable central tendency and spread is the more responsible choice.

An overly conservative criterion, however, may eliminate opportunities to continue studying an innovative candidate. Even when the stable condition advances, the peak-performing candidate should remain on a separate research track so its mechanism can be identified.

3.3.3. Answer C — Two Gates: Exploration and Transfer

Approve peak performance as “technical possibility” and the repeated distribution as “readiness for transfer.” For the peak-performing candidate, design independent remeasurement and reproducibility tests across instruments, dates, and wafers. For the stable candidate, test process capability and performance loss in a limited pilot.

Do not rank both candidates using a single number. Gate the peak candidate on the improvement mechanism and reproducibility success rate; gate the stable candidate on its mean, spread, out-of-specification rate, and cost. If neither candidate passes its gate, record that the hypothesis and failure conditions have become clear, rather than declaring “no result.”

3.4. AI Research Design

3.4.1. Prompt for the AI

Evaluate which of two power-semiconductor research recipes should advance to a pilot process. Preserve raw histories of breakdown voltage, on-resistance, missing values, and remeasurements by sample, wafer, instrument, measurement date, and operator. Calculate the maximum, mean, median, standard deviation, and number of out-of-specification results, and show the measurement system's calibration history and 95% expanded uncertainty in separate columns. Distinguish values excluded under a predefined rule from those removed after the fact. Evaluate repeatability under identical conditions, reproducibility after changing instruments, dates, and operators, and the production process window separately. Separate facts, calculations, assumptions, and recommendations, and attach the document title, issuing body, year, and direct URL to every external source. Present the advantages and disadvantages of prioritizing the peak, prioritizing reproducibility, and applying a two-gate model, along with the numerical conditions that would change the conclusion.

3.4.2. Data Acquisition

Priority	Source	Items that must be recorded
1	Sample-level raw data	All measurements, sample and wafer locations, missing values, remeasurements, and reasons for exclusion
2	Experimental recipes and equipment logs	Material lot, process conditions, equipment, chamber state, measurement date, and operator
3	Measurement-system data	Calibration standard, resolution, repeatability and reproducibility, uncertainty budget, and environmental conditions
4	Specifications and product requirements	Minimum breakdown voltage; on-resistance, thermal, and lifetime limits; customer use conditions
5	Follow-on pilot plan	Sample size, randomization, independent reproducing institution, and success, stop, and retest criteria

3.4.3. Assessing the Information

1. **Selection bias:** Do not retain only the best samples or remove outliers using criteria defined after the results are known.
2. **Measurement distinctions:** Do not mix repeatability, reproducibility, accuracy, and expanded uncertainty as though they were the same metric.
3. **Effect size:** Determine whether the difference between candidates exceeds measurement uncertainty and sample dispersion.
4. **Process transfer:** Do not count multiple dies from one wafer as an equivalent number of independent wafers.
5. **Transfer gate:** A result must pass not only the mean criterion but also out-of-specification, spread, process-window, and independent-reproducibility criteria.

3.4.4. Core Questions for Debate

- Can a recipe advance if its peak greatly exceeds specification but one-third of its repeated samples fall short?
- Can a performance difference smaller than the measurement uncertainty be called a research result?
- If the mean is unchanged on a different instrument but the spread increases, has the result been reproduced?
- How much on-resistance penalty is acceptable for the stable recipe?

3.5. Case Debate

The following figures are **interview assumptions**, not actual results from a particular company or product. You are a new engineer on a power-semiconductor research team. Only one recipe can advance to the pilot process.

- For Recipe A, the highest breakdown voltage among 12 devices is 1,280 V, the mean is 1,195 V, and the standard deviation is 42 V; four devices fall below the 1,200 V specification.
- For Recipe B, the maximum is 1,250 V, the mean is 1,232 V, and the standard deviation is 11 V; all 12 devices meet or exceed 1,200 V.
- Assume that the measurement's 95% expanded uncertainty is ±8 V.
- B's on-resistance is 6% higher than A's, implying an efficiency penalty.

1. The research team tests whether A's peak represents a physical improvement rather than measurement or selection error.
2. The metrology team reviews the ±8 V uncertainty budget and remeasurement after changing the instrument and date.
3. The product team calculates the effect of B's 6% increase in on-resistance on thermal design and customer specifications.
4. You choose among **advancing A, advancing B, and holding both**.
5. You also propose a follow-up experiment and decision-reversal condition for the recipe not selected.

A strong answer does not end with selecting the candidate with the higher mean. It must explain how specification failures will be handled, whether uncertainty and dispersion have been distinguished, and how much performance loss is acceptable in exchange for stability.

3.6. Sample 90-Second Response

"I choose **Answer B: advance the more reproducible Recipe B to a limited pilot**, while retaining A on the research track. NIST's 2025 process-gas standard reports 95% expanded uncertainties of 0.056–0.098% across 0.01–1,000 cm^3/min. It is a real-world example showing that tightly clustered repeated values do not prove an absence of systematic error. Our figures, by contrast, are hypothetical. A peaks at 1,280 V, but its mean is 1,195 V and four of 12 devices fall below specification. B has a mean of 1,232 V, a standard deviation of 11 V, and no failures. I accept the counterargument that A's peak shows greater technical potential, but the next stage must receive a process window another engineer can reproduce. I would hold the transfer if B's 6% higher on-resistance violates the thermal specification. I would retest A on three wafers using independent equipment and 20 devices per wafer. If it simultaneously achieves a mean of at least 1,230 V, a standard deviation no greater than 15 V, and uncertainty no greater than $\pm$8 V, I would reopen the decision. After pilot transfer, I would continue disclosing lot-to-lot dispersion and specification failures transparently. I would record the peak as an exploratory result and approve development transfer only for a reproducible result."

3.7. Follow-Up Questions

Cross-Examination and Rebuttal

- Would your conclusion change if all four failures in Recipe A came from the wafer edge?
- If Recipe B's 6% increase in on-resistance raised system power by only 1%, on what basis would you accept it?
- If all 12 devices came from a single wafer, why should the sample size not simply be treated as 12?
- If the mean holds on another instrument but the standard deviation rises to 22 V, would you consider the result reproduced?
- If measurement uncertainty increases from ±8 V to ±20 V, how would you redesign the comparison of the two recipes?
- If asked to disclose the peak in a press release first, what wording and limitations would you attach?

3.8. Sources

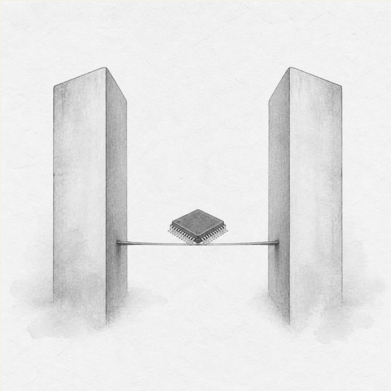
- U.S. NIST, Pope et al., *SLowFlowS: A Novel Flow Standard for Semiconductor Process Gases* (2025)
- U.S. National Library of Medicine PMC, full SLowFlowS paper and experimental data (2025)
- U.S. NIST, *Research Updates—Gas Flow for Semiconductor Manufacturing* (2025)
- BIPM/JCGM, *International Vocabulary of Metrology—Repeatability and Reproducibility* (2012, accessed 2026)
- BIPM/JCGM, *Guide to the Expression of Uncertainty in Measurement* (2008)

- U.S. NIST, *CHIPS for America Metrology Program* (accessed 2026)
- Joseon Dynasty Chaekmun Archive, King Sejong's 1447 *The Abuses of Law and the Seat of Authority*
- National Institute of Korean History, *Annals of King Sejong*, entry for the 27th day of the eighth month in Sejong's 29th year (1447)

Data note: The ranges and values 0.01–1,000 cm^3/min, 0.02%, 0.056–0.098%, 24 sensors, 63 mK, 2 mK, and up to ±6% come from the NIST primary data. The breakdown voltages of 1,280, 1,250, 1,195, and 1,232 V; standard deviations of 42 and 11 V; the 1,200 V specification; ±8 V uncertainty; 6% on-resistance increase; and follow-on gates are interview assumptions.

4. U.S.–China Export Controls and the Nationality of Technology

Today's Chaekmun



Under U.S. controls on advanced semiconductor exports to China, should a newly hired strategist at a Korean equipment maker with high China exposure suspend a Chinese contract and accelerate qualification in alternative markets?

A chaekmun preserved in an anthology from King Jungjong's reign[1] treated diplomacy as more than eloquence or short-term achievement. It asked who would take responsibility for conveying the state's intentions and maintaining trust, and how. Today's candidate must likewise go beyond choosing either "the U.S. side" or "the Chinese side." The answer must account for the time gained through controls, the markets Korean companies may lose, China's development of substitute technologies, and the distinctive technologies Korea must secure.

4.1. Why This Question, and Why Now?

U.S. controls on China cannot be summarized as a blanket "ban on semiconductor exports." Controls on advanced computing chips and semiconductor manufacturing equipment began in October 2022, while the performance thresholds and anti-circumvention scope were adjusted in 2023. Measures announced in December 2024 expanded the scope to 24 types of equipment, three types of

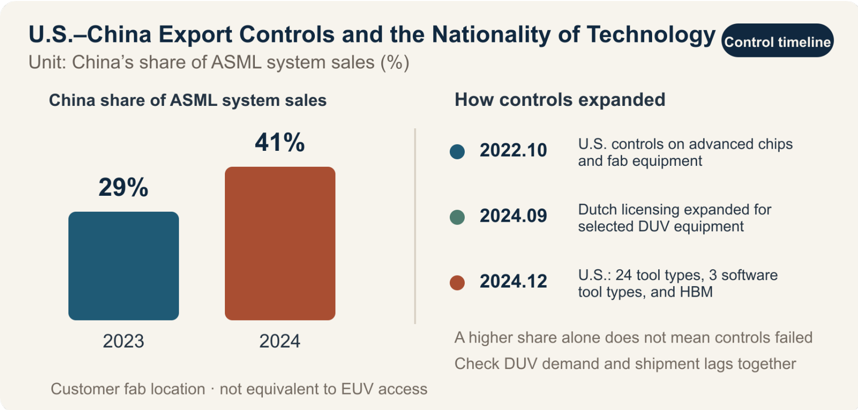
[1]The chaekmun (策問), or "royal policy question," that inspired this chapter asked: "What does an envoy charged with conveying the nation's intentions accurately and preserving its dignity need before eloquence and literary skill?" The Chaekmun Archive offers the Classical Chinese reconstruction "奉使傳國意□才辯與德行孰先□", which is not a literal transcription of the original. Because the exact examination year and ranking are not established in public sources, it is identified only as a "response preserved in an anthology from King Jungjong's reign." This chapter's question is likewise not a translation of the original. It reinterprets the structure that places long-term trust and responsibility before the techniques of diplomacy, and applies it to export controls.

software tools, HBM, and 140 China-related entities. The Netherlands also introduced licensing for selected advanced DUV lithography equipment in 2023 and expanded the scope in September 2024. This is not a blanket ban on all lithography systems, but a licensing regime based on the item, end user, and destination.

Korea faces conflicting effects at the same time. If advanced lithography equipment and AI chips enter China more slowly, Korean memory, foundry, and advanced-packaging companies gain time to leverage their technology lead and allied markets. Korean equipment and materials companies with high exposure to Chinese customers, however, may bear the costs of export licenses, field service, replacement parts, and declining revenue. The longer controls remain in place, the greater the incentive for Chinese firms to invest in domestic equipment, process automation, yield improvement, and higher productivity from legacy tools.

The central question, then, is not whether one supports the controls. It is **whether Korea can convert the time gained through controls into technologies that others cannot readily replace**, and how the benefits of the security alliance and the costs to industry should be allocated.

4.2. Data Lens



4.2.1. Evidence Table

No.	Figure and scope in the primary data	Meaning for the debate
1	The U.S. BIS implemented controls on advanced computing chips, supercomputers, and semiconductor manufacturing equipment bound for China on October 7, 2022, and revised performance thresholds and anti-circumvention provisions on October 17, 2023.	Export controls are not a one-time measure; they are rules that evolve with technology.

No.	Figure and scope in the primary data	Meaning for the debate
2	BIS measures announced on December 2, 2024, covered 24 types of semiconductor manufacturing equipment, three types of software tools, HBM, and 140 additions to the Entity List.	Korean equipment companies may also face licensing effects depending on U.S.-origin technology and software and the end user.

No.	Figure and scope in the primary data	Meaning for the debate
3	Beginning September 7, 2024, the Netherlands expanded the scope of national licensing for selected advanced DUV lithography equipment. The Dutch government explicitly described it as case-by-case review, not a comprehensive ban.	Saying that the measure "blocked all Chinese imports of photolithography equipment" overstates its scope.

No.	Figure and scope in the primary data	Meaning for the debate
4	China’s share of ASML system sales rose from 29% in 2023 to 41% in 2024. In the same data, EUV accounted for 38% of 2024 system sales and ArF immersion DUV for 44%.	China’s revenue share can rise even as controls tighten because of permitted DUV demand and shipment lags. The effectiveness of controls cannot be judged from a single revenue figure.
5	ASML said Chinese DUV demand was stronger than expected in 2025 and forecast that revenue from China would fall substantially in 2026 from 2024–2025 levels.	Regulatory effects may not appear immediately. Orders, shipments, and revenue recognition must be separated.

No.	Figure and scope in the primary data	Meaning for the debate
6	In their supply-chain and commercial dialogue, Korea and the United States affirmed the principles of protecting national security while minimizing disruption to the global semiconductor supply chain and sustaining industrial viability and technological development (2023).	The alliance's own objectives explicitly balance security and industrial costs. Korea must negotiate cost sharing and predictability.

4.2.2. How to Read These Numbers

The chart's **29%→41%** is not proof that "the controls failed." It represents the share of ASML's total system sales shipped to customer fabs located in China; it does not mean that China secured advanced EUV systems. Nor is it evidence that the controls automatically benefited Korean companies. Its significance is that **technology controls and market transactions do not move immediately in the same direction**.

When analyzing a Korean company, collect the following four sets of figures separately:

- Share of revenue from China and share of revenue from controlled items
- Share of U.S.-origin components and software in product cost and functionality
- Time from license application to approval or denial, and days of service interruption
- Orders from alternative markets, customer-qualification lead times, and performance and yield gaps for critical equipment

China's technological self-reliance should likewise not be judged by announced investment or patent counts alone. It must be verified through equipment adoption in actual production lines, wafer throughput, defect density, utilization, customer qualification, and repeat orders.

4.3. Competing Answers

4.3.1. Answer A — Leverage the Alliance Opportunity

U.S. market and technology controls buy Korea time to protect its technology lead and margins in advanced semiconductors. If China cannot secure sufficient EUV and selected advanced DUV lithography equipment, advanced AI chips, and HBM, the barriers to leading-edge processes and AI-accelerator production rise. If Korea's memory, HBM, advanced-packaging, materials, and equipment companies establish themselves as trusted suppliers within the Korea–U.S. alliance, they may also benefit from joint research, subsidies, and long-term contracts with customers in the United States, Japan, and Europe.

This answer's strength is its recognition that a technology lead is not permanent and that the time gained must be converted into investment. The controls themselves, however, do not guarantee Korean profits. Competitors in the United States, Japan, and Europe receive the same opportunity, while Korean companies' Chinese fabs and equipment customers also bear regulatory costs.

4.3.2. Answer B — Controls Will Backfire

Broad, frequently changing controls can constrain Chinese exports, installation, parts replacement, and maintenance by Korean equipment and materials companies and raise compliance costs. The more important long-term risk is that China will invest in domestic equipment, process automation, multipatterning, yield control, and productivity improvements for legacy tools rather than waiting for blocked equipment. BIS's inclusion in its 2024 controls of "software that improves the productivity of less advanced equipment" indicates that the United States also sees this avenue for workaround innovation as a threat.

A strategy of permanently excluding a market rival becomes ever more expensive as technology changes and transactions move through third countries. If Korea merely follows U.S. rules passively, it may lose the Chinese market while retaining a double vulnerability: dependence on the United States, the Netherlands, and Japan for key foundational technologies.

4.3.3. Answer C — Build a Technology Lead and Diversify Markets

Korea should not seek loopholes between two great powers. It should secure technologies that neither side can easily replace. Manufacturers need distinctive performance in HBM, advanced packaging, low-power memory, and high-yield processes; equipment makers need it in inspection, metrology, etching, deposition, process control, and maintenance data. At the same time, companies must lower their concentration of Chinese revenue, understand their dependence on U.S.-origin technology, and accelerate customer qualification in allied markets.

The conditional position does not mean "trade moderately with both sides." It means **invest the time gained through controls in R&D and customer diversification, trade only within the licensing rules, and change strategy when revenue concentration, technology gaps, or licensing delays cross predefined limits**.

4.4. AI Research Design

4.4.1. Prompt for the AI

> Research the opportunities and risks that U.S. semiconductor export controls on China create for Korean semiconductor manufacturers and equipment makers. Search in this order: □ primary BIS and Federal Register rules, □ the Dutch government's licensing scope, □ annual reports from ASML and Korean companies, and □ Korean government trade statistics. For each claim, provide a table with the institution, document title, publication date, statistical base year, unit, and primary-source URL. Distinguish "prohibited" from "requires a license," "order" from "shipment and revenue," and "China revenue" from "revenue from controlled items." Put verified facts, company and government claims, and the analyst's inferences in separate columns, and do not use unsourced or conflicting figures in the conclusion. Finally, construct the strongest possible cases for an active-opportunity position, a cautionary position, and a conditional position, then propose indicators Korean companies should review quarterly.

4.4.2. Data Acquisition

Priority	Source	Items that must be recorded
1	BIS rules, Federal Register, and EAR	Effective date, ECCN, performance thresholds, end user, licensing policy, exceptions
2	Dutch and Korean government materials	Equipment requiring licenses, case-by-case review, exceptions and consultations concerning Korean companies
3	Company annual reports and earnings materials	Definition of regional revenue, timing of order, shipment, and revenue recognition, product mix
4	Customs and trade statistics	HS code, value and quantity, nominal and real terms, distinction between China and Hong Kong
5	Patents, news reports, and analytical material	Link to primary data, conflicts of interest, production validation, contrary evidence from other sources

4.4.3. Assessing the Information

1. **Legal scope:** Verify from the rule's language whether the measure is a "blanket ban" or a "licensing requirement."
2. **Timing alignment:** Check whether an order placed before a rule's effective date was recognized as revenue only later.

3. **Denominator alignment:** Determine whether China's revenue share refers to total revenue or system sales, and whether it measures units or value.
4. **Causal verification:** Do not assume that controls were the sole cause simply because Chinese investment rose after controls were introduced.
5. **Company-specific exposure:** Evaluate Chinese dependence, alternative customers, and cash capacity separately for large manufacturers and smaller equipment makers.

4.4.4. Core Questions for Debate

- How many years of **technology time** have the controls bought Korea, and into what R&D should that time be converted?
- Who should bear, and how should they share, the alliance's security benefits and the revenue losses of Korean equipment makers?
- What indicators show Chinese localization through manufacturing results rather than announcements?
- Should an "irreplaceable technology" that prevents Korean companies from being pushed around be measured by performance, yield, or customer-qualification time?

4.5. Case Debate

In March 2027, a midsized Korean inspection-equipment company receives a KRW 120 billion order from a Chinese foundry. China generated 34% of its revenue in the previous year, and the equipment uses a U.S.-origin laser-control module and design software. The contractual shipment date is 45 days after a new BIS rule takes effect. A substitute order from a U.S. customer is worth KRW 70 billion, but qualification will take nine months, and the company has enough cash to absorb losses for 14 months. The Chinese customer asks the company not to disclose the end user or process node. All figures are hypothetical conditions created for the debate.

1. The compliance team presents the ECCN, end user, licensing requirement, and sanctions risk if the license is denied.
2. The sales team presents the profit from the Chinese contract, the risk of losing the customer, and the U.S. customer's qualification timeline.
3. The technology team presents the time required and the performance penalty involved in legally replacing the U.S.-origin module.
4. The strategy team chooses among **suspending the contract while applying for a license, signing a conditional contract, and abandoning the contract while shifting to an alternative market**.
5. The final proposal includes available cash flow, a cap on China revenue, R&D line items, a customer-qualification schedule, and stop conditions if the rules change.

In this situation, "diversion through a third country" is not an option. A strong answer presents a quantified, scheduled path that keeps the company alive and builds technology while complying with the law.

4.6. Sample 90-Second Response

"I choose **Answer C: suspend the contract while verifying the license and end user**. BIS expanded its 2024 control scope to 24 types of equipment, three types of software, and HBM, while the share of ASML's system sales shipped to China rose from 29% in 2023 to 41% in 2024. This means that even if controls widen a technology gap, they do not automatically guarantee profits for Korean companies. The KRW 120 billion contract, 34% China revenue exposure, KRW 70 billion substitute order, and nine-month qualification period in the scenario are all hypothetical. The strongest counterargument is that abandoning the contract would lose the Chinese market before the substitute order is secure. I would therefore verify the ECCN and scope of the U.S.-origin module and software, and contract only if the license is approved or the item is proven not to require a license, with clauses barring re-export and allowing suspension if the rules change. If the customer continues to conceal the end user or the license is denied, I would abandon the contract and accelerate alternative-market qualification and development of a Korean module. I would recheck licensing conditions every quarter. Controls become an opportunity only when the time they create is converted into technology and customer diversification."

4.7. Follow-Up Questions

Cross-Examination and Rebuttal

- How can you argue that controls are effective when ASML's China revenue share rose to 41%?

- If China implements advanced processes through DUV multipatterning without EUV, how much does the time gained through controls shrink?
- If a Korean equipment maker's China revenue falls from 34% to 15%, should the government and large manufacturers share the cost?
- How would you determine whether U.S. rules apply when U.S.-origin software is used in only part of product development?
- If the United States extends controls more broadly to Korean HBM, would you maintain the opportunity thesis?
- If you had to choose one measure of distinctive technology, which would it be: throughput, defect density, utilization, or customer-qualification time?

4.8. Sources

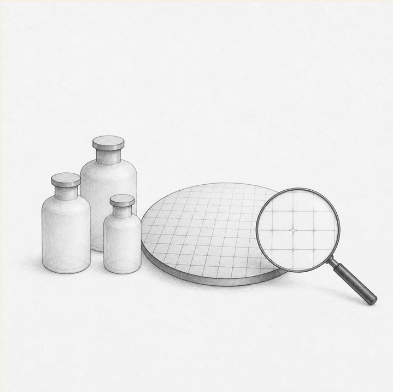
- U.S. BIS, guidance on 2022 and 2023 advanced computing and semiconductor manufacturing controls
- U.S. BIS, December 2024 announcement on semiconductor manufacturing equipment and HBM controls
- U.S. BIS, *Export Administration Regulations §744.23* (accessed 2026)
- Government of the Netherlands, expansion of export licensing for advanced DUV lithography equipment (2024)
- ASML, *Q4 and Full Year 2024 Investor Presentation* (2025)
- ASML, *2025 Annual Report—Financials* (2026)
- ASML, *Q3 2025 Financial Results* (2025)

- Korea–U.S. Supply Chain and Commercial Dialogue Joint Statement, principles for export-control cooperation (2023)
- Joseon Dynasty Chaekmun Archive, Jungjong-era *What Qualities Should a Diplomat Possess?* (accessed 2026)

Data note: Because export controls and their exceptions change frequently by item, performance, end user, and destination, recheck the primary texts from BIS, the Federal Register, the Dutch government, and company disclosures immediately before commercial publication.

5. Does Lower Import Value Prove Successful Localization?

Today's Chaekmun



If imports of Japanese hydrogen fluoride have fallen, can a newly hired procurement specialist report this as successful localization?

A Joseon-era chaekmun[1] distinguished praise for abundant resources from the realities experienced by the people. Today's procurement specialist must likewise do more than repeat the favorable

[1]This chapter connects to the historical chaekmun *Seven Abuses in Honam and Their Remedies*, in which King Jeongjo asked about Honam's conditions and possible remedies in 1798. The transmitted text begins, "王若曰▫湖南一路卽我朝興王之地也。" Following the declaration "The king speaks: the province of Honam is the land where our dynasty arose," the question demanded a diagnosis encompassing both the region's resources and talent and its accumulated abuses.

phrase "imports from Japan have declined." The report must also verify the yield and price of domestic substitutes, the origin of their raw materials, and their customer-qualification status.

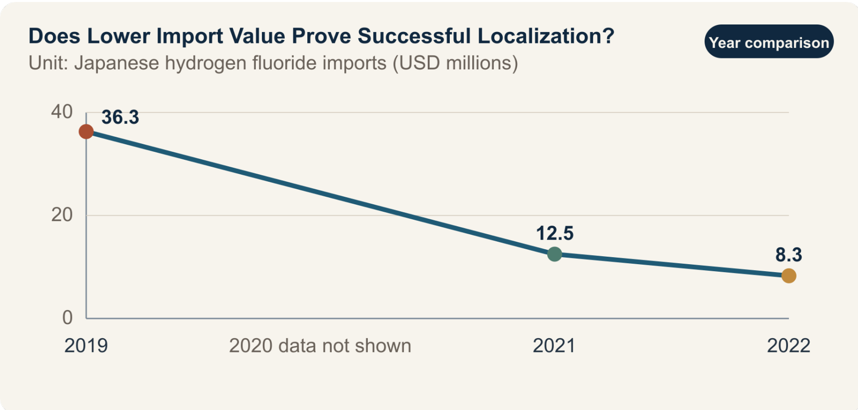
5.1. Why This Question, and Why Now?

According to data from the Ministry of Trade, Industry and Energy, the value of hydrogen fluoride imports from Japan fell 66%, from USD 36.3 million in 2019 to USD 12.5 million in 2021. Extending the trade-statistics series, imports of Japanese hydrogen fluoride for semiconductor use are reported at USD 8.3 million in 2022. Immediately before the controls, Japan's share of high-purity hydrogen fluoride was 43.9%. This trend demonstrates a major shift in sourcing.

A decline in import value, however, can contain several different stories. Domestic production may have increased, imports may have shifted to third countries, unit prices or exchange rates may have changed, or production itself may have fallen. In semiconductor processing, "the same chemical formula" does not automatically mean "the same material performance." Differences in metallic impurities, moisture, particles, containers, and transport controls can affect etch uniformity and yield.

A new procurement specialist's report is not a document for judging sentiment between nations. It should connect import value, volume, and unit price by HS code with approved items by supplier, defect rates and yields before and after introducing the domestic material, raw-material origins, emergency inventory, and customer requalification lead times. "Localization complete" is a much stronger claim than saying that the source of supply has changed.

5.2. Data Lens



5.2.1. Evidence Table

Category	Figure and scope in the primary data	Limitation when used in a report
Policy data	The value of hydrogen fluoride imports from Japan was USD 36.3 million in 2019.	This is a pre-control baseline, but the corresponding volume and purity mix must also be disclosed.

Category	Figure and scope in the primary data	Limitation when used in a report
Policy data	Import value was USD 12.5 million in 2021, 66% lower than in 2019.	The decline signals a change in sourcing, but does not prove the manufacturing performance of domestic material.
Trade-statistics-based data	The value of Japanese hydrogen fluoride imports for semiconductor use was USD 8.3 million in 2022.	The HS-code scope and aggregation methodology must be recorded before comparison with other years.
Pre-control data	Japan's share of high-purity hydrogen fluoride was reported at 43.9%.	The figure cannot be used without knowing whether the denominator is value, volume, or a particular grade.
Calculation from primary data	Imports fell by approximately 77%, from USD 36.3 million in 2019 to USD 8.3 million in 2022.	Even the calculated decline cannot distinguish among increased domestic production, substitution through third countries, and lower demand.

5.2.2. How to Read These Numbers

First, import value is the product of `volume × unit price × exchange rate`. A decline in value does not show whether volume was unchanged or whether the price of high-purity product shifted. Always verify weight and unit price for the same HS code and period.

Second, distinguish "lower dependence on Japan" from "self-reliance across the domestic value chain." Even when final purification takes place domestically, the producer may depend on a specific foreign supplier for feedstock, containers, analytical equipment, or process know-how. If only the country name changes while single-source dependence remains, resilience has not improved enough.

Third, the procurement team is not the sole arbiter of localization. The production team must verify long-run yield and defects, the quality team must approve change control, and customers must complete requalification where required. Prototype success, deployment on one line, approval across all product families, and stable repeat delivery must not be described as the same stage.

5.3. Competing Answers

5.3.1. Answer A — Recognize the Achievement

If import value has fallen substantially and domestic supply has actually entered mass production, the process represents genuine learning. More suppliers create purchasing leverage and more options for continuing production during an unexpected licensing delay. If every partial achievement is dismissed while waiting for perfect self-reliance, the next round of investment and shop-floor improvement may lose momentum.

The basis for recognizing success, however, must be delivery and performance rather than nationality. It is more accurate to report a “supply-chain diversification achievement” once repeated delivery, customer approval within the same product family, and verified emergency scale-up capacity are in place.

5.3.2. Answer B — Beware a Premature Declaration of Success

Declaring localization successful on the basis of import value alone conceals price increases, production declines, rerouting through third countries, and quality costs. If the substitute is more expensive and delivers lower yield, the total cost per wafer may rise even while reported imports fall. A protected supplier may also have less incentive to improve quality.

In particular, if feedstock is concentrated in another single country, the result merely transfers dependence on Japan to another source. The procurement team’s job is not to produce a promotional slogan, but to locate where the single point of failure has moved.

5.3.3. Answer C — Assess Localization in Stages

Do not force the conclusion into a binary of "complete" or "failed." Divide it into development, line validation, customer qualification, repeat mass production, and raw-material diversification. Treat the decline in import value as an initial observation, then advance to the next stage only when domestic-use share, yield, total cost of ownership, delivery, and raw-material origin meet the defined criteria.

Disclose the conditions that would change this conclusion. The status can be upgraded to "complete" if the substitute's yield gap disappears, customer requalification is finished, and emergency scale-up capacity is verified. Conversely, if quality costs and the price premium persist while dependence on a new country increases, the diversification strategy must be redesigned.

5.4. AI Research Design

5.4.1. Prompt for the AI

> Create an investigation worksheet that a newly hired procurement specialist can use to verify whether declining Japanese imports of semiconductor-grade hydrogen fluoride represent a localization achievement. Check, in order, statistics from the Korea Customs Service, Ministry of Trade, Industry and Energy, and Korea International Trade Association; Japanese Ministry of Economy, Trade and Industry export-control documents; company disclosures; and customer-qualification materials. Record

the HS code, period, country, value, weight, unit price, and purity and end-use scope, and determine whether the 2019, 2021, and 2022 figures are comparable. In a separate table, show the domestic supplier's raw-material origin, capacity, actual delivery volume, yield by product family, and customer-approval status. Distinguish verified statistics, government and corporate claims, calculated values, and internal assumptions, and attach a direct URL to every item. Finally, state the conditions under which the phrase "localization complete" should be retained, revised, or deleted.

5.4.2. Data Acquisition

Priority	Source	Items that must be recorded
1	Primary customs and trade data	HS code, country of import, value, weight, unit price, period, and revision status
2	Export-control rules	Items requiring licenses, effective date, bulk and individual licenses, timing of rule changes
3	Internal procurement and quality records	Purchase volume by supplier, lot failures, yield, returns, and emergency-procurement lead time

Priority	Source	Items that must be recorded
4	Customer change approvals	Requalification scope by product family, approval date, reasons for holds, regular-audit results
5	Domestic supplier due diligence	Feedstock origin, purification and analysis capability, scale-up limit, foreign dependence of critical equipment

5.4.3. Assessing the Information

1. **Product match:** Distinguish all industrial hydrogen fluoride from high-purity semiconductor grades.
2. **Decompose value:** Separate changes in import value into changes in volume, unit price, and exchange rate.
3. **Confirm the stage:** Distinguish development announcements from customer approval and temporary deliveries from repeat mass production.
4. **Compare performance:** Compare yield with a control group on the same product, line, and period, recording other process variables.
5. **Transferred dependence:** Determine whether the domestic product's feedstock, containers, or analytical equipment has become concentrated in a new single country.

5.4.4. Core Questions for Debate

- What portion of the 77% decline in import value is explained by expanded domestic production?
- Under what conditions could a 0.4-percentage-point yield loss be accepted as the insurance premium for supply security?
- How should the price premium and emergency-procurement risk be combined into total cost per wafer?
- How does the report's scope of responsibility change when it says "supply-chain diversification" instead of "localization"?

5.5. Case Debate

All figures below are **interview assumptions**, not data from an actual company. You are a procurement specialist three months into your first job. Import value for Japanese material has fallen 77% from the base year, but the domestic substitute is 15% more expensive and delivers yield 0.4 percentage points below the incumbent material in one product family. Current inventory covers 90 days, and customer requalification takes nine months. The country of origin of the domestic supplier's purification feedstock has not yet been fully verified.

1. The procurement team recalculates import value, volume, unit price, and total cost of ownership by supplier.
2. The production and quality teams use a controlled test under identical conditions to determine whether the 0.4-percentage-point gap is attributable to the material.

3. The sales team identifies which products and customers can be served during the nine-month requalification period.
4. You choose among **localization complete, partial localization, and supply-chain diversification in progress** as the report's conclusion.
5. Before the 90-day inventory is depleted, you propose corrective actions covering feedstock-origin due diligence, quality improvement, and the dual-sourcing ratio.

The interviewer is not looking for patriotic rhetoric. The interviewer wants to hear which quality and cost risks the company is accepting, who will verify them, and when the conclusion will change.

5.6. Sample 90-Second Response

"I choose **Answer C and would report 'supply-chain diversification in progress,' not 'localization complete.'** Public data show that the value of Japanese hydrogen fluoride imports fell approximately 77%, from USD 36.3 million in 2019 to USD 8.3 million in 2022. But import value is the product of volume, unit price, and exchange rate, so it does not directly prove the manufacturing performance of domestic material. The 15% price premium, 0.4-percentage-point yield gap, 90 days of inventory, and nine-month requalification period in the scenario are all hypothetical. I accept the counterargument that reduced dependence on Japan improves supply security. Yet if feedstock is concentrated in another single country and the yield penalty persists, only the address of dependence has changed. Within 90 days, I would therefore audit feedstock origin and scale-up capacity, retest the yield gap on the same product and line, and maintain dual sourcing throughout customer qualification. I would upgrade the status to 'complete' once the yield gap disappears and customer approval and repeat deliveries are confirmed. Conversely, if the total-cost disadvantage including quality costs and the new single-country dependence persist, I would add a third supplier. Localization should be judged by approved quality and delivery capability, not declining import value."

5.7. Follow-Up Questions

Cross-Examination and Rebuttal

- How would you revise the report if import value fell but import volume rose?
- Which product price and time horizon would you use to calculate the monetary cost of a 0.4-percentage-point yield loss?
- If the Japanese inventory runs short during the nine-month customer requalification, which products would receive priority?
- Can the material be recognized as localized if the domestic supplier buys critical feedstock from only one country?
- How would you set the maximum period for accepting a 15% price premium as the cost of supply security?
- If a government announcement conflicts with internal quality data, how would you record the conflict in the final report?

5.8. Sources

- Ministry of Trade, Industry and Energy, *Achievements in Strengthening the Competitiveness of Materials, Parts, and Equipment* policy material (2022)
- Korea International Trade Association, Institute for International Trade, *Economic Effects and Implications of Lifting Japan's Export-Control Measures*, Trade Brief No. 09 (2023)
- Japan Ministry of Economy, Trade and Industry, *Security Export Control Policy* (accessed 2026)
- Joseon Dynasty Chaekmun Archive, King Jeongjo's 1798 *Seven Abuses in Honam and Their Remedies*
- Academy of Korean Studies, Archives of Old Documents, search records for royal examination responses from 1798

Data note: Public import statistics and case assumptions are kept separate; 15%, 0.4 percentage points, 90 days, and nine months are educational assumptions.

Part II.

Judgment in Research, Design, and Manufacturing

6. Should Reliability Testing Be Cut to Meet a Deadline?

Today's Chaekmun

Should a product ship before some reliability tests are complete in order to meet the customer's deadline?

King Jungjong's 1515 chaekmun on realizing ideal government in the present[1] demanded more than good intentions; it required a

[1]The historical chaekmun (策問) connected to this chapter asked what must come first to realize the government of the sage-kings Yao and Shun in that age, and how Confucius would govern the country. The Chaekmun Archive reconstructs the meaning of its published translation in the concise Classical Chinese "欲致堯舜之治於今日□何者爲先□子若孔子□將何以治國歟□" This is not a literal transcription of the original. It means: "To realize the government of Yao and Shun today, what must come first? If you were Confucius, how would you govern the country?" Today's question does not equate a historical ideal of government with a quality standard. It reinterprets the structure of the question—which demands an order of execution and conditions for verification rather than merely a worthy goal—and connects it to reliability decisions.

statement of what to do first and how to execute it. "We will protect quality" and "We will meet the deadline" are likewise not answers when they remain declarations. The team must decide which tests detect which failure modes, how much risk remains unseen, and what may ship to whom and within what limits.

6.1. Why This Question, and Why Now?

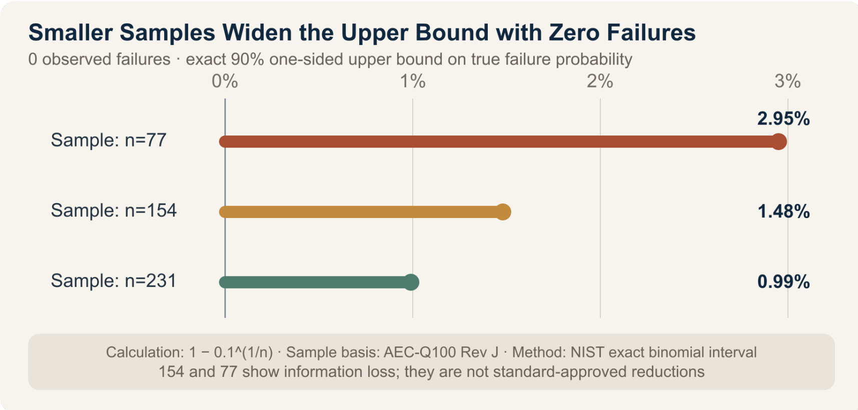
In an advanced package, the different coefficients of thermal expansion of the die, bumps, interposer, underfill, and substrate act within one system. When a material or thickness changes, dominant failure modes may also change, including interfacial delamination, cracking, and intermetallic-compound growth. Legacy reliability data cannot simply be borrowed because the previous generation had the same process name and external form.

For representative stress-test groups, AEC-Q100 Rev J (2023) requires 77 devices per lot across three lots—231 in total—with zero failures. This sample structure is associated with LTPD 1 at a 90% confidence level, but the standard states that it should not be used to estimate process-control performance or shipment PPM directly. Passing a test is different from claiming precise knowledge of the failure rate in actual use.

Accelerated testing compresses time, but it is not a license to raise any stress indiscriminately. NIST explains that random samples should be tested across multiple stress cells; the stress must accelerate the same failure mechanism seen under use conditions; and excessively high stress must not create a new failure mechanism.

When a deadline is tight, teams should validate the failure mode and acceleration model before simply halving the test time.

6.2. Data Lens



6.2.1. Evidence Table

Category	Figure and scope in the primary data	Meaning for process and quality decisions
Sample structure	A representative stress test in AEC-Q100 Rev J uses 77 devices per lot × three lots, for a total of 231 devices with zero failures.	The lot count is structured to include at least some process and material variation. Replacing it with 77 devices from one lot does not provide equivalent evidence.

Category	Figure and scope in the primary data	Meaning for process and quality decisions
Statistical meaning	The standard associates LTPD 1 with a 90% confidence level and states that this sample alone is insufficient to estimate process-control performance or PPM.	"Zero failures" does not mean that the failure probability is zero.
Temperature-humidity stress	The document specifies 1,000 hours at 85°C/85% RH, or accelerated conditions including 96 hours at 130°C/85% RH and 264 hours at 110°C/85% RH.	Do not compare time alone; verify the applied bias, package structure, and target failure mode together.
Reduced-sample calculation	With zero observed failures, the exact 90% one-sided upper bounds are 0.99% for 231 devices, 1.48% for 154, and 2.95% for 77.	Reducing the sample by two-thirds makes the upper bound on "unseen failures" roughly three times wider.
Accelerated-test principles	NIST calls for stress cells by failure mode, random samples, acceleration models, and extrapolation to use conditions.	If excessive stress creates a new failure mode, the result is not a faster version of the same test, but a different test.

Category	Figure and scope in the primary data	Meaning for process and quality decisions
Current military standard	MIL-STD-883 Rev L Change 1, published by the U.S. Defense Logistics Agency, is the 2025 edition.	The applicable standard and revision must be checked for each contract and product family; standards cannot substitute for one another.

6.2.2. How to Read These Numbers

The figures 0.99%, 1.48%, and 2.95% are not predictions of the actual field failure rate. Assuming identical Bernoulli trials in which no failures are observed, they are the exact 90% one-sided upper bounds on the true failure probability, calculated as (1-0.1^{1/n}). They illustrate why even zero failures among 231 devices cannot be described as a "0% failure rate."

Samples of 154 and 77 are not reduced-sample plans allowed by AEC-Q100. They are hypothetical cases in which only two or one of three lots have been completed because of a delivery deadline, used to compare the resulting loss of statistical information. Actual qualification must follow the applicable standard, product family, scope of change, and customer approval.

Accelerated-test time and sample size are not fully interchangeable. A shorter test at a higher temperature requires a physical model and validation data showing that it accelerates the same failure mechanism. A larger sample with insufficient test time can miss late-emerging failure modes, while completing the time requirement on only one lot can miss lot-to-lot variation.

6.3. Competing Answers

6.3.1. Answer A — Complete Every Planned Test

For a product with changed materials or structure, legacy data cannot erase potential failure modes. A mass-production shipment is difficult to reverse, and if an early failure appears in a customer system, the costs of recall, analysis, and lost trust can exceed the cost of a delayed delivery. Hold shipment until the planned number of lots, stress duration, and post-stress electrical tests are all complete.

"All tests complete," however, does not mean that every risk has been eliminated. The team must also review whether the test plan adequately reflects the real failure modes and use conditions.

6.3.2. Answer B — Reduce Testing Based on Risk

For unchanged structures and validated failure modes, legacy lot data may be used while samples and analytical resources are concentrated on stresses associated with the change. This is an engineering choice that can meet the customer's deadline while reducing redundant tests. It applies only when equivalence of materials, suppliers, process windows, and acceleration models is documented and the standard and customer permit it.

Yet "test only the changes" may miss interactions that are not yet understood. If both the underfill and interposer thickness have changed, data from the previous package become less representative.

6.3.3. Answer C — Hold Production Shipments and Release Only Limited Engineering Samples

Do not shorten the full qualification test, but provide only nonsale engineering samples for the customer's assembly and firmware validation under restrictions on quantity, use, and return. Distinguish them from production and field shipments, and document the incomplete tests, known risks, and prohibited uses by agreement.

This choice preserves part of the customer's schedule, but engineering samples can effectively be used as production units. It is unavailable unless traceable serial numbers, the permitted location of use, a ban on onward transfer, and an immediate right of recall after a test failure are in place.

6.4. AI Research Design

6.4.1. Prompt for the AI

> Create a reliability release-decision table for a new advanced package before its delivery deadline. Compare changes in the underfill, interposer thickness, bumps, substrate, and process conditions with the previous

product, and connect each change to its failure mechanisms. Verify the applicable contractual standard and revision, and record each test's temperature, humidity, bias, cycles, duration, number of lots, sample size per lot, and allowable failures. Distinguish completed, in-progress, not-started, failed, and right-censored data. For zero observed failures, calculate the exact 90% one-sided upper bound for each sample size, but do not interpret it as the field failure rate. To reuse legacy data, require evidence that the material, supplier, process window, failure mode, and acceleration model are equivalent. Separate facts, calculations, interview assumptions, and recommendations, and attach the institution, document, year, and direct URL to every external source. Present the advantages and disadvantages of a full hold, risk-based reduction, and a limited engineering-sample release, along with the conditions that would change the conclusion.

6.4.2. Data Acquisition

Priority	Source	Items that must be recorded
1	Change list	Materials, dimensions, suppliers, equipment, process windows, and differences from the previous product

Priority	Source	Items that must be recorded
2	Raw test data	Sample and lot, stress conditions and duration, censoring, pre- and post-stress electrical tests, failure location
3	Failure-analysis data	Cross sections, SAM, X-ray, electrical analysis, failure mode, and root cause
4	Standards and customer requirements	Applicable document and revision, sample and allowable failures, change notification and deviation approval
5	Shipment controls	Engineering or production status, quantity, place of use, traceability, recall, and no-onward-transfer conditions

6.4.3. Assessing the Information

1. **Equivalent standards:** Do not apply test conditions from another product family merely because their names are similar.
2. **Sample independence:** Do not treat multiple dies from one lot as representing the variation of multiple lots.
3. **Zero-failure interpretation:** Do not express zero observed failures as a true failure probability of zero.

4. **Acceleration validity:** Do not use stress that creates a failure mechanism absent under use conditions for lifetime extrapolation.
5. **Data reuse:** Unless equivalence of the change and failure modes is demonstrated, do not substitute legacy data for a pass result on the current product.

6.4.4. Core Questions for Debate

- If all 77 devices in one test lot pass, what is the strongest claim you can make to the customer?
- If a 72-hour delivery deadline conflicts with a 96-hour HAST, can the test be shortened to 48 hours by raising the stress?
- Who must control 20 engineering samples, and how, to prevent their use in field equipment?
- What values are omitted when the cost of a schedule delay and the cost of a latent defect are both converted into Korean won?

6.5. Case Debate

The following figures are **interview assumptions**, not actual results from a particular company or product. You are a new engineer on the process and quality team for a new advanced-package product.

- Both the underfill material and interposer thickness have changed in this product.
- Using the AEC-Q100 sample structure as an internal benchmark, the company planned a HAST of 96 hours at 130°C/85%

RH on three lots × 77 devices, along with 1,000 temperature cycles.

- The customer's assembly-validation milestone is 72 hours away.
- So far, only one HAST lot of 77 devices is complete, with zero failures; the other two lots remain in progress.
- Temperature cycling has reached 300 of 1,000 cycles with zero failures.
- Legacy-product data are available, but equivalence has not been established because the underfill and interposer combination is different.

1. The reliability team presents the incomplete tests and failure mechanisms associated with each change.
2. The process team reviews material and process-window equivalence between the legacy data and the current product.
3. The customer team determines whether the units needed in 72 hours are for assembly validation or field sale.
4. You choose among **shipping the full production quantity, holding all units until testing is complete, and providing only 20 nonsale engineering samples under restrictions**.
5. You state the action after a test failure and the numerical and documentary conditions for switching to production shipment.

A strong answer does not conceal "zero failures" and "testing incomplete" in the same sentence. It separates the minimum support that can be offered to the customer's schedule from the quality gates that cannot be reduced.

6.6. Sample 90-Second Response

"I choose **Answer A: hold all units until the planned testing is complete**. A representative AEC-Q100 Rev J sample consists of 77 devices per lot across three lots—231 total—with zero allowable failures. When zero failures are observed, the 90% one-sided upper bound is approximately 0.99% for 231 devices and 2.95% for 77. These are not predicted field failure rates, but they show how reducing the sample increases uncertainty. In the scenario, both the underfill and interposer have changed, while only one HAST lot and 300 temperature cycles have been completed. These figures and the 72-hour deadline are hypothetical. I accept the counterargument that the delay may disrupt the customer's schedule and cost a market opportunity. I would instead disclose the remaining tests and new ship date immediately. Shipment requires completion of HAST on all three lots and 231 devices, 1,000 temperature cycles, electrical testing, and failure-mode review. I would show test progress and causes of any failure to the customer and the accountable quality executive in the same table. If even one device fails, the hold remains until root-cause analysis and a revised test plan are approved. Deadlines can be renegotiated; unverified risk cannot be transferred to the customer."

6.7. Follow-Up Questions

Cross-Examination and Rebuttal

- If the customer says it will install the 20 engineering samples in the final product, how would your choice change?

- If one of 77 devices fails HAST but analysis identifies handling damage, how would you redesign the test?
- If all three lots use the same batch of raw material, can the full sample of 231 be treated as independent evidence?
- What physical evidence is required to replace a 96-hour HAST with 48 hours at a higher temperature?
- How would you qualify the statement that there were zero failures through 300 temperature cycles when reporting it to the customer?
- Can information that a competitor already mass-produces the same structure justify releasing the entire shipment?

6.8. Sources

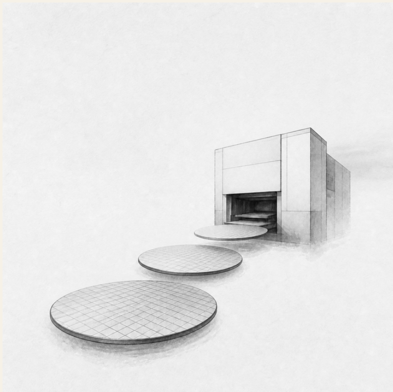
- Automotive Electronics Council, official reliability-document list (accessed 2026)
- Automotive Electronics Council, *AEC-Q100 Rev J: Failure Mechanism Based Stress Test Qualification for Integrated Circuits* (2023)
- U.S. NIST/SEMATECH, *Confidence Intervals for a Proportion*—exact binomial intervals (accessed 2026)
- U.S. NIST/SEMATECH, *Accelerated Life Tests* (accessed 2026)
- U.S. NIST/SEMATECH, *Physical Acceleration Models* (accessed 2026)
- U.S. Defense Logistics Agency, official MIL-STD-883 document list (accessed 2026)
- U.S. Defense Logistics Agency, *MIL-STD-883 Rev L Change 1* (2025)

- Joseon Dynasty Chaekmun Archive, King Jungjong's 1515 *The Path to the Government of Yao and Shun Today*

  Data note: The sample structure of 77 devices per lot × three lots, 231 total, with zero allowable failures; LTPD 1 and a 90% confidence level; and the temperature, humidity, and duration of the damp-heat tests come from the AEC-Q100 Rev J primary source. The 90% one-sided upper bounds of 0.99%, 1.48%, and 2.95% for zero observed failures in samples of 231, 154, and 77 are calculations based on the exact binomial method presented by NIST; samples of 154 and 77 are not reduced-sample plans permitted by the standard. The changes to underfill and interposer, 72-hour deadline, completion of one HAST lot, 300 of 1,000 temperature cycles, 20 engineering samples, and switching conditions are interview assumptions.

7. How Far Should HBM Capacity Expansion Be Brought Forward?

Today's Chaekmun



If a customer places a five-year order but the equipment takes seven years to pay back, should a new analyst recommend a full capacity expansion?

A Joseon-era chaekmun[1] demanded institutions and conduct that would work to the end, not merely momentum at the start. During an HBM boom, a good start may mean ordering equipment quickly;

[1]This chapter follows the historical 1507 chaekmun *Government Consistent from Beginning to End*, in which King Jungjong asked how the purpose of an undertaking could be preserved to the finish. The Chaekmun Archive reconstructs its meaning in the concise Classical Chinese “善始者未必善終□何也□欲始終惟一□其道何由□”, which is not a literal transcription of the original. It asks: “Why does one who begins well not necessarily end well, and what path allows beginning and end to remain consistent?”

a good finish means protecting cash flow if demand falls and converting the equipment to the next product generation.

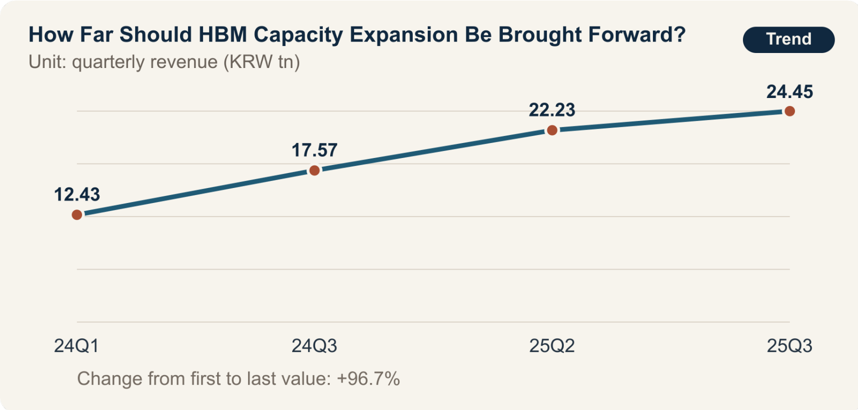
7.1. Why This Question, and Why Now?

SK hynix revenue rose from KRW 12.43 trillion in the first quarter of 2024 to KRW 17.57 trillion in the third quarter of 2024, KRW 22.23 trillion in the second quarter of 2025, and KRW 24.45 trillion in the third quarter of 2025. A simple comparison of the first quarter of 2024 with the third quarter of 2025 shows an increase of approximately 97%. Operating profit in the third quarter of 2025 was KRW 11.38 trillion.

The product mix also changed. In company disclosures, HBM represented 30% of DRAM revenue in the third quarter of 2024 and was forecast to reach 40% in the fourth quarter. These figures show the importance of AI memory, but they do not mean that HBM shipments, contract duration, or future demand were secured. Revenue growth reflects not only volume, but also pricing and a shift toward higher-value products.

When a new analyst on the capital-investment team turns the boom narrative into an investment proposal, the timelines must be aligned. The analysis must place customer order duration, cancellation rights and prepayments, equipment order and installation, yield ramp, packaging bottlenecks, depreciation, and the payback period on one axis. Even if a customer states five years of demand, a seven-year payback makes the final two years of cash flow dependent on demand outside the contract.

7.2. Data Lens



7.2.1. Evidence Table

Category	Figure and scope in the primary data	Meaning for the investment decision
Company disclosure	Revenue in the first quarter of 2024 was KRW 12.43 trillion.	This is the starting point for comparison, not HBM-only revenue.
Company disclosure	Revenue was KRW 17.57 trillion in the third quarter of 2024 and KRW 22.23 trillion in the second quarter of 2025.	This is an upward path in quarterly revenue, but price, volume, and product mix must be decomposed.

Category	Figure and scope in the primary data	Meaning for the investment decision
Company disclosure	Third-quarter 2025 revenue was KRW 24.45 trillion and operating profit was KRW 11.38 trillion.	The figures demonstrate strong profitability, but do not guarantee seven years of cash flow from a new line.
Company announcement	HBM represented 30% of DRAM revenue in the third quarter of 2024 and was forecast to reach 40% in the fourth quarter.	The denominator is DRAM revenue, not total revenue, and 40% was a forecast at the time.
Calculation from primary data	Revenue increased approximately 97% from the first quarter of 2024 to the third quarter of 2025.	This is a simple comparison of two nonadjacent quarters, not a long-term growth rate or a measure of downside demand.

7.2.2. How to Read These Numbers

KRW 24.45 trillion is total company quarterly revenue. It is not HBM unit volume or contract backlog by customer. The KRW 11.38 trillion operating profit likewise combines the effects of existing equipment and product pricing, so it cannot be inserted directly into the internal rate of return for a new line.

The denominator for the 30% HBM figure is DRAM revenue. It is not unit share, and higher-priced products appear larger when measured by revenue. Because 40% was a fourth-quarter forecast at the time of the announcement, it must be recorded separately from the actual result. A new analyst should use different colors for disclosed actuals, management forecasts, and the analyst's own demand assumptions.

A long-term order is not committed demand until the contract has been read. The analyst must examine minimum purchases, price renegotiation, cancellation fees, customer design changes, quality shortfalls, and rights to change suppliers. The downside scenario must include not only the cost of canceling equipment, but also the supporting investment in packaging, testing, power, and skilled employees.

7.3. Competing Answers

7.3.1. Answer A — Expand Fully and Early

HBM equipment procurement and process and packaging qualification take time. If a company waits until demand is fully confirmed, it may miss the customer's product cycle. Securing equipment when long-term orders and a technology lead have been established can bind customers and standards to the supplier and slow market entry by followers.

The opportunity cost of delayed expansion is also large. If equipment queues and cleanroom and power construction stretch out, volume missed in one quarter may not be recovered in the next. The ability to meet a customer's supply-security requirement becomes evidence supporting the next contract.

7.3.2. Answer B — Guard Against Overinvestment

If the seven-year payback exceeds a five-year order, the company bears demand and pricing risk after the contract ends. If the pace of AI investment slows or customers change their in-house accelerator architectures, the fixed-cost and depreciation shock will grow. A full equipment order that ignores the memory industry's history of overexpansion can deepen the next downturn.

Nor does increasing HBM wafer output alone create a saleable product if packaging, testing, substrates, and power do not keep pace. Concentrating the budget on the largest equipment while missing the real bottleneck is even more dangerous.

7.3.3. Answer C — Use a Three-Stage Expansion Linked to Demand

Divide the build-out into cleanroom and infrastructure, critical equipment, and follow-on equipment, opening the next order only when customer prepayments, minimum purchases, qualification, and packaging utilization are confirmed. This secures part of the equipment lead time while preserving cancelable capacity if downside demand emerges.

The switching conditions are contract quality and cash flow. Stop the next stage if cancelable orders increase, prepayments decline, or the downside scenario shows a cash shortfall during the investment period. Conversely, bring expansion forward when the customer shares risk and the packaging bottleneck and yield constraints have been resolved.

7.4. AI Research Design

7.4.1. Prompt for the AI

> Prepare a recommendation comparing full expansion with a three-stage build-out for a five-year HBM order and a new line with a seven-year payback. Investigate, in order, company disclosures and earnings releases, customer contracts and demand forecasts, equipment purchase orders, monthly process and packaging capacity, and the financial model. Distinguish revenue, shipment volume, ASP, and HBM's share of DRAM revenue, and place actual results, company forecasts, and internal assumptions in separate columns. Present by customer the minimum purchase, cancellation rights, prepayments, and price adjustments; equipment lead times and cancellation fees; wafer, stacking, and testing bottlenecks; and the yield ramp. Calculate cash flow and payback under base, downside, and upside cases, disclosing the source and sensitivity of every assumption. Finally, propose numerical thresholds and accountable decision makers for opening or stopping the next order stage.

7.4.2. Data Acquisition

Priority	Source	Items that must be recorded
1	Customer contracts and demand agreements	Duration, minimum purchases, cancellation rights, price adjustments, prepayments, quality and delivery terms
2	Equipment and construction contracts	Order and installation lead times, payment schedule, cancellation fees, conversion and used-equipment resale options
3	Process and packaging operating data	Wafer starts, stacking and test capacity, yield, utilization, bottlenecks, and staffing
4	Company disclosures and market data	Distinction between actuals and forecasts, denominator for HBM share, changes in price, volume, and mix
5	Financial model	Depreciation, working capital, power and materials costs, base and downside cash flow, and payback period

7.4.3. Assessing the Information

1. **Actuals and forecasts:** Do not report a figure that was a forecast at the time, such as 40% for the fourth quarter of 2024, as an actual result.
2. **Contract quality:** Minimum purchases, cancellation and price-adjustment rights, and prepayments determine risk sharing more than the order term does.
3. **Bottleneck alignment:** Convert wafer, stacking, testing, and substrate capacity into the same finished-product units.
4. **Timeline alignment:** Put the customer's contract term and the equipment life, depreciation, and payback period on the same yearly axis.
5. **Downside survival:** Examine cancellation costs, fixed costs, and cash balances first under low demand.

7.4.4. Core Questions for Debate

- How can the two years of payback risk remaining after the five-year order be shared with the customer?
- How would you calculate whether the opportunity cost of securing equipment or a 12% cancellation fee is larger?
- If packaging rather than wafers is the bottleneck, which equipment should be ordered first?
- What advance investment answers the objection that a phased build-out may move too slowly and lose the customer?

7.5. Case Debate

The figures below are **hypothetical interview conditions**, not SK hynix's actual investment plan. You are a new analyst on the capital-investment team. A major customer has proposed five years of volume, but the new line takes seven years to pay back. In the downside scenario, demand is 50% of the base case, while the cancellation fee on equipment already ordered is 12% of the contract value. The customer's minimum purchase and prepayment and the packaging bottleneck remain under negotiation.

1. The sales team discloses the cancelable portion of the five-year volume and the price-adjustment clauses.
2. The production team presents monthly wafer, stacking, and test capacity and the expected date for reaching target yield.
3. The procurement team summarizes lead times by tool, the 12% cancellation cost, and the feasibility of staged ordering.
4. The finance team compares seven-year cash flow under the base case and a 50% downside case.
5. You choose between **a single full expansion and a three-stage expansion**, and write the conditions for opening and stopping the next stage.

Even if you choose full expansion, the statement "AI demand will continue" is insufficient. Even if you choose three stages, you must explain whether the first stage moves fast enough to secure a place in the equipment queue.

7.6. Sample 90-Second Response

"I recommend **Answer C: a three-stage expansion**. Disclosed quarterly revenue rose approximately 97%, from KRW 12.43 trillion in the first quarter of 2024 to KRW 24.45 trillion in the third quarter of 2025, while HBM represented 30% of DRAM revenue in the third quarter of 2024. Yet total company revenue and HBM revenue share do not guarantee a seven-year return on investment. The five-year customer contract, seven-year payback, 50% downside demand, and 12% cancellation fee in the scenario are all hypothetical. The full-expansion counterargument—that delayed equipment and qualification could cause us to miss the boom—is valid, so I would secure the cleanroom and bottleneck equipment first. But committing the full order is risky when the payback runs two years beyond the contract. I would open the next stage only after confirming customer minimum purchases and prepayments and packaging yield. If the 50% downside case shows a cash shortfall during the investment period, I would absorb the cancellation fee and stop follow-on orders. Conversely, if the customer shares the risk and the bottleneck is resolved, I would accelerate the schedule. At every stage, I would recalculate cash flow and yield and report them to the board. Capacity speed should be set by contracts and process data, not the mood of the boom."

7.7. Follow-Up Questions

Cross-Examination and Rebuttal

- If the customer's five-year volume is a nonbinding forecast, should even the first stage be held?
- How would you compare a 12% equipment cancellation fee with seven years of carrying cost for excess capacity?
- If a competitor expands even in the 50% downside scenario, would you prioritize defending market share?
- What difference does it make to the investment proposal that HBM's reported share is a percentage of DRAM revenue?
- If packaging yield is low but wafer-equipment lead times are long, which order would you place first?
- If the customer prepays enough, does the concern about a seven-year payback disappear?

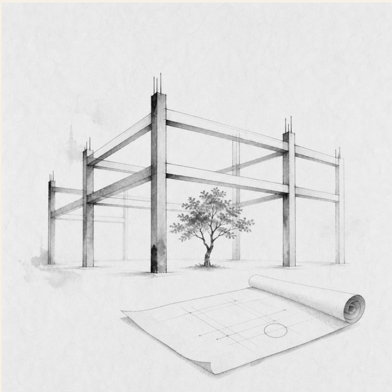
7.8. Sources

- SK hynix, *1Q24 Financial Results* (2024)
- SK hynix, *3Q24 Financial Results* (2024)
- SK hynix, *3Q25 Financial Results* (2025)
- Joseon Dynasty Chaekmun Archive, King Jungjong's 1507 *Government Consistent from Beginning to End*

Data note: Company results and announced figures are kept separate from case assumptions; five years, seven years, 50%, and 12% are educational conditions.

8. Should We Continue a Project That Missed Its Target?

Today's Chaekmun



Should a publicly funded fab project be extended when it missed its production target but still retains skilled people and patents?

The Joseon chaekmun—an examination question through which the king asked candidates to reason through an urgent problem of state[1]—does not allow a worthy original purpose to excuse a failed outcome. Nor does it say that missing the original numbers requires discarding every capability created along the way. A junior policy evaluator must separate target attainment, residual capabilities, and the opportunity cost of additional funding before recommending extension, redesign, or termination.

8.1. Why This Question Matters Now

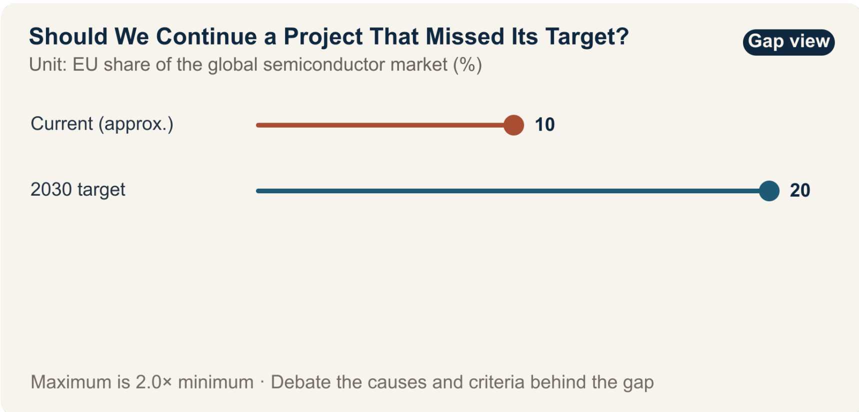
The EU set a goal of raising its share of the global semiconductor market from about 10% to 20% by 2030. The European Chips Act entered into force on September 21, 2023. In 2026, the European Commission stated that existing policies had mobilized more than €52 billion in public and private investment and created about 46,000 direct and indirect jobs.

The move from 10% to 20% expresses direction and ambition; it is not an outcome. "Mobilized investment" may differ from actual EU budget expenditure and may combine announcements, plans, construction starts, and executed spending. The estimate of 46,000 jobs combines direct and indirect employment, so net employment, job duration, and occupational mix must be checked.

[1] This chapter connects the 1507 chaekmun posed by King Jungjong, "Governance Consistent from Beginning to End," to the problem of ending or carrying forward an unsuccessful project. The Chaekmun Archive condenses and reconstructs the question in Classical Chinese as "善始者未必善終□何也□欲始終惟一□其道何由?"; this is not a literal transcription of the original. It asks, "Why does someone who begins well not necessarily end well, and what path allows one to remain consistent from beginning to end?"

Project-level evaluation must be narrower still. Examine actual incremental capacity, customer-qualified products, the additionality of private investment, the reemployment of skilled workers, the use of patents in volume production, and repeat orders won by certified suppliers. Assessing only the original production target misses learning; assessing only the abstraction of “capability” conceals target shortfalls and sunk costs.

8.2. Data Lens



8.2.1. Evidence Table

Category	Figure or Scope in the Source	Meaning for Project Evaluation
Policy baseline	EU policy materials put its share of the global semiconductor market at about 10%.	This is a starting point that may vary with the definition and baseline year.
Policy target	The 2030 target is a 20% share of the global market.	This is a legal and policy objective, not achieved production output.
Entry into force	The European Chips Act entered into force on September 21, 2023.	This is the institutional start date, not the construction, execution, or volume-production date of an individual project.
Commission statement	The Commission announced that more than €52 billion in public and private investment had been mobilized by 2026.	The definition of "mobilized," double counting, and the amount actually disbursed must be checked.
Commission statement	The Commission stated that about 46,000 direct and indirect jobs had been created.	The estimated range, net employment, duration, and job quality require separate verification.

8.2.2. How to Read These Numbers

If the 20% global-share target is calculated by production value, the scale of production the EU must add will depend on global market growth. You must also establish whether the calculation uses nominal revenue, production capacity, or shipment volume. A statement that the target is twice the baseline does not, by itself, reveal how many fabs are needed.

The €52 billion "mobilized" may not mean that €52 billion in public funds was paid out. Separate existing private plans from investment induced by subsidies, and announced amounts from actual execution. Likewise, construction-period indirect employment and long-term volume-production jobs should not be treated as equivalent components of the 46,000-job estimate.

To evaluate an unsuccessful individual project, prepare three ledgers. The first records production, financial, and employment commitments; the second records reusable talent, patents, and supplier networks; the third records the opportunity cost of directing additional funding away from other projects. Money already spent is not a reason to extend a project. Compare only the value that can still be recovered.

8.3. Opposing Answers

8.3.1. Answer A — Extend to Preserve Capabilities

Short-term output does not capture all the learning generated in a strategic industry. If skilled workers, process patents, and qualified suppliers remain—and can reduce the risk of failure and shorten qualification for the next line—immediate termination may disperse the knowledge assets created by public investment.

Where infrastructure and workforce development are complete and additional funding can remove a defined bottleneck, a limited extension may be more efficient than restarting the entire original plan. The remaining assets, however, must have an identified user and a contract that puts them to work in a follow-on project.

8.3.2. Answer B — Sunset and Terminate

"Long-term capability" can become a convenient explanation for missing targets. Patents without production capacity or customers, trained workers with nowhere to move, and supplier certifications without orders are achievements only on paper. Without a termination threshold, an unsuccessful project continues to occupy funding and talent, blocking better alternatives.

Continuing because the money already spent feels too valuable to lose is a sunk-cost fallacy. If the future benefit created by additional funding is smaller than the benefit of another project, terminate the project and transfer its people and patents separately.

8.3.3. Answer C — Conditional Redesign with a Narrower Scope

Do not extend the original plan unchanged. Retain only the components whose failure mechanisms are clear and whose reuse value is high. Set 12-month milestones for production, customer qualification, actual transfer of people and patents, and new private investment, with a sunset clause that triggers automatic termination if they are missed.

Evaluate the redesign by the marginal benefit of new funding, not by past expenditure. Disclose whether the new budget removes a bottleneck and creates marketable capacity, and whether it produces more value than allocating the funds to another project. An independent evaluator should verify results using evidence that does not come exclusively from the original project team.

8.4. AI Research Design

8.4.1. Prompt for the AI

> Write an evaluation of whether a publicly funded semiconductor fab that missed its production target should be extended, redesigned, or terminated. Investigate, in order, the relevant laws and funding agreements; the original and revised business plans; actual spending, production, and employment data; customer qualifications; records showing the use of patents, training, and suppliers; and external audits. Distinguish targets and

announced figures—such as the EU's roughly 10% baseline, 20% goal for 2030, €52 billion mobilized, and 46,000 jobs—from actual execution and net effects. For the individual project, tabulate capacity versus plan, actual sales, the additionality of private investment, redeployment of skilled workers, use of patents in volume production, and repeat supplier orders. Separate verified facts, stakeholder claims, estimates, and scenario assumptions, and attach the institution, document title, publication date, and direct URL to every external source. Include the opportunity cost of additional funding and a 12-month sunset clause.

8.4.2. Data Acquisition

Priority	Evidence	Required Fields
1	Original plan and funding agreement	Production, investment, and employment targets; schedule; change procedure; clawback and sunset provisions; accountable party
2	Actual execution and operating data	Amount paid and spent; equipment and operating capacity; yield; sales; customer qualifications; causes of delay

Priority	Evidence	Required Fields
3	Workforce and patent records	Retention and movement of skilled workers; patent registration, use, and revenue; reuse in follow-on projects
4	Supplier and private-investment records	Repeat orders after qualification; new private capital; additionality created by subsidies
5	Alternative-project evidence	Production, R&D, and workforce outcomes possible with the same additional funding, and transition costs

8.4.3. Evidence Assessment

1. **Targets versus results:** Mark policy targets, announcements, commitments, construction starts, volume production, and sales as distinct stages.
2. **Exclude sunk costs:** Compare future costs with recoverable future benefits, not with money already spent.
3. **Reality of capabilities:** Identify the follow-on organization, schedule, and contract that will use the talent, patents, and supplier network.
4. **Additionality:** Determine whether private investment was newly induced by subsidies or merely moved from an existing plan.

5. **Enforcing the sunset:** Specify who triggers automatic termination after missed milestones, including exceptions and authority to amend them.

8.4.4. Core Issues for Debate

- Can 62% attainment of a production target and the residual value of people and patents be combined into a single score?
- How do you distinguish whether additional funding equal to 25% of the original budget is rescuing sunk costs or removing a future bottleneck?
- What is the minimum evidence that 2,000 skilled workers and 40 qualified suppliers were actually reused?
- How do you give an independent evaluator enough autonomy to enforce a 12-month sunset clause despite political pressure?

8.5. Case Discussion

The following is an **interview scenario, not an actual EU project**. You are a junior researcher on an industrial-policy evaluation team. A supported fab achieved only 62% of its production target, but 2,000 skilled workers, 300 patents, and 40 qualified suppliers remain. Continuing the project requires additional funding equal to 25% of the original budget. The actual reuse rate of the remaining capabilities and customers' repeat purchases have not yet been verified.

1. The operations team attributes the 62% result across demand, yield, equipment, and infrastructure.

2. The workforce team verifies the roles, retention, movement, and follow-on placement of the 2,000 workers.
3. The technology team verifies the use of the 300 patents in volume production and repeat orders won by the 40 suppliers.
4. The finance team compares the additional 25% with the benefit of directing those funds to other projects.
5. You recommend one of three options: **extend the original plan, redesign it for 12 months, or terminate it and transfer the assets**.

If you choose redesign, do not merely say, "Give it one more chance." Specify numerical thresholds and the person responsible for automatic termination after 12 months. If you choose termination, explain where the workers, patents, and suppliers will go.

8.6. Sample 90-Second Answer

"I recommend **Answer C: a narrower 12-month redesign, not an extension of the original plan**. The EU set a goal of raising its share of the global semiconductor market from about 10% to 20% by 2030, and the Commission says existing policy has mobilized more than €52 billion in investment. But a target and a mobilized sum do not prove the productivity of an individual fab. In this case, the 62% target attainment, 2,000 skilled workers, 300 patents, and 25% additional budget are all assumptions. I accept the argument for extension that abandoning people and patents useful to follow-on production would destroy value. The opposite risk is using the language of 'strategic capability' to conceal sunk costs. I would therefore spend additional funds only on verified bottlenecks and asset transfer, while publishing gates for customer-qualified capacity, workforce placement, use of patents in volume production, and repeat supplier orders. If the milestones are missed, or another project offers a higher marginal benefit, the project should terminate automatically. I would approve the next stage only after actual sales and additional private investment are verified. The purpose of redesign is not to hide failure, but to convert what remains into assets for the next decision."

8.7. Follow-Up Questions

Cross-Examination and Rebuttal

- If weaker market demand caused the project to reach only 62% of its production target, could the project still be considered a technical success?
- If the 2,000 skilled workers find jobs at other companies, is that evidence of the original project's success or a reason to terminate it?
- Would you redesign the project if the 300 patents had little value but the certifications of the 40 suppliers remained useful?
- How would you compare fairly the additional 25% with the option of spending it on another workforce program?
- If the project narrowly misses its targets after 12 months, would you enforce the sunset clause as written?

- Can you report separately on the political target of a 20% market share and the economics of an individual project?

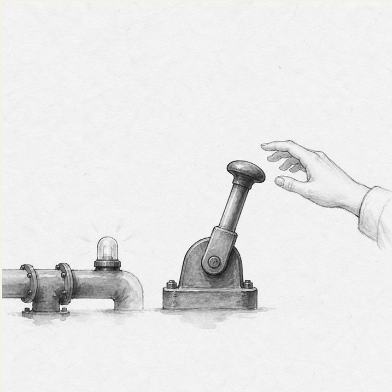
8.8. Sources

- European Commission, *European Chips Act* (entered into force in 2023; accessed 2026)
- European Commission, *Chips Act 2.0* (2026)
- European Commission, *Proposal for the Chips Act 2.0* (2026)
- Joseon Dynasty Chaekmun Archive, "Governance Consistent from Beginning to End" (King Jungjong, 1507)

Data note: EU policy and announcement figures are separated from the scenario assumptions. The 62%, 2,000 workers, 300 patents, 40 suppliers, 25%, and 12 months are educational evaluation conditions or decision criteria proposed in this chapter.

9. Should One Gas Alarm Stop the Line?

Today's Chaekmun



Should a junior engineer request a line shutdown based solely on an unconfirmed gas-leak alarm, even if delivery will be delayed by two days?

The Joseon chaekmun—an examination question through which the king asked candidates to reason through an urgent problem of state[1]—shows that setting priorities requires considering both the

[1]The passage linked to this chapter comes from King Gwanghaegun's 1611 chaekmun, "The Most Urgent Task after the Imjin War": "今之國事▫何者最急▫用人、賦役、田政、戶籍▫孰先孰後?" It asks, "What is most urgent in the affairs of state today? Of appointing capable people, managing corvée labor, administering land, and maintaining household registers, which should come first and which later?" It is not an old statement about gas alarms translated into modern language. This chapter instead connects it to the judgment required to identify an irreversible danger when several losses compete for priority. See the individual chaekmun and commentary and the Academy of Korean Studies record of the 1611 examination.

nature of the harm and whether it can be reversed. In a fab, a delivery delay can be recovered; loss of life or health from hazardous-gas exposure cannot.

9.1. Why This Question Matters Now

Semiconductor fabs use toxic, flammable, and corrosive substances while operating continuously for long periods. Stopping the entire line for every alarm raises the cost of false positives and creates alarm fatigue. Waiting for confirmation when a leak is real, however, sacrifices evacuation time. Junior staff in particular may overstate production losses and discount their own sensor interpretation. The answer is not simply to demand courage, but to define shutdown stages and reporting lines in advance.

The ILO estimates that work-related factors cause 2.93 million deaths and 395 million nonfatal occupational injuries each year. These global totals show why prevention matters; they do not tell us whether one alarm in a particular fab indicates a real leak. An operational decision requires the substance's toxicity, concentration, alarm duration, agreement between independent sensors, ventilation and evacuation time, and the sensor's recent failure history.

9.2. Data Lens

9.2.1. Evidence Table

Evidence	Figure or Scope in the Source	Meaning for Fab Safety Decisions
1	The ILO estimates that work-related accidents and diseases cause 2.93 million deaths each year.	Safety losses include irreversible harm that is difficult to express as an average delivery cost.
2	Nonfatal occupational injuries are estimated at 395 million per year.	Counting deaths alone misses treatment, lost work, and long-term health effects.

Evidence	Figure or Scope in the Source	Meaning for Fab Safety Decisions
3	ILO data attribute about 63% of worldwide work-related deaths to Asia and the Pacific.	A regional total shaped by workforce size and industrial structure must not be used as the risk probability for one factory.

9.2.2. How to Read These Numbers

The 2.93 million figure is a global estimate combining work-related **accidents and diseases**, not the number of fab gas incidents in one year. The 395 million figure counts injuries and may not equal the number of individual people injured. Nor is the 63% regional share causal evidence that Asian factories are inherently more dangerous; it reflects the distribution of workers and industries.

These numbers do not command, “Shut down the entire line for every alarm.” They provide the background for keeping the harm from a missed detection in the decision table. Site criteria must separately define exposure limits for each gas, sensor reliability, agreement among alarms, available evacuation time, and the effectiveness of partial isolation.

9.3. Opposing Answers

9.3.1. Answer A — Immediate Full Shutdown

Because a single missed detection of hazardous gas can cause irreversible harm, even one alarm reported by a junior employee should first move the system to a safe state. The cost of stopping and then using an independent measurement to identify a false alarm is bounded; the cost of missing actual exposure may not be. The reporting employee must face no retaliation if alarms are to retain their protective function.

9.3.2. Answer B — Verify the False Alarm First

In a process with repeated sensor false alarms, stopping the entire line for every short, isolated signal can damage delivery performance and equipment stability and may teach employees to ignore alarms. A procedure is needed for partial shutdown and remeasurement based on the material's hazard and the status of adjacent sensors and ventilation.

9.3.3. Answer C — Isolate the Zone and Escalate in Stages

Define automatic actions by toxicity, concentration, duration, and agreement between two sensors. If lethality is high or little evacuation time is available, even a single alarm triggers a full shutdown. When the event can be localized, evacuate and isolate the zone before taking an independent measurement. A safety officer, not the production organization, authorizes restart.

9.4. AI Research Design

9.4.1. Prompt for the AI

> Review the ILO's primary occupational-safety sources separately from the site's approved gas-safety procedures. For global work-related deaths, nonfatal injuries, and the Asia-Pacific share, record the issuing organization, baseline year, estimate range, unit, and direct URL. Then tabulate the safety data sheet for the gas, sensor calibration records, recent alarm and false-positive history, ventilation zones, evacuation time, and safe-equipment-shutdown procedure. Do not allow AI to decide independently whether to stop the line; distinguish statutory thresholds from internal company thresholds. Put established facts, sensor observations, worker statements, and inferences in separate columns, and label missing information "unverified."

9.4.2. Data Acquisition

Order	Evidence	What to Verify
1	Site emergency-response procedure and safety data sheet	Hazard by substance; automatic-shutdown threshold; evacuation and first aid

Order	Evidence	What to Verify
2	Raw sensor logs and calibration records	Time, concentration, duration, adjacent sensors, calibration expiry
3	Equipment and ventilation drawings and worker locations	Isolatable zones, airflow, evacuation route, people remaining
4	Alarm, shutdown, and restart history	Cause of false alarm, downtime, approver, recurrence

9.4.3. Evidence Assessment

1. Use raw logs to distinguish a communication fault from an actual sensor measurement.
2. Determine whether normal adjacent sensors are truly exculpatory evidence, given airflow and detection limits.
3. Do not treat the harm of a missed detection and the cost of a false alarm as equal; reflect severity and reversibility.
4. Record any retaliation against a stop-work request or delay in reporting it.
5. Require all three conditions for restart: cause removed, independent measurement completed, and safety officer approval.

9.4.4. Core Issues for Debate

- For which substances, concentrations, and evacuation conditions should a single sensor trigger an immediate full shutdown?
- Can a recent history of false alarms justify disregarding the present alarm?
- If production and safety teams disagree, who should hold final authority to restart the line?

9.5. Case Discussion

You are a junior process engineer on the night shift. One sensor exceeds the gas threshold for 12 seconds while adjacent sensors remain normal. The same sensor produced two false alarms in the past month, and a full shutdown will delay customer delivery by two days. Three workers are in the affected zone. **The times, staffing, and delivery impact in this section are hypothetical conditions created for discussion.**

1. Choose among immediate evacuation and partial shutdown, full shutdown, or remeasurement while operating.
2. Explain the conditions under which normal readings from adjacent sensors would change your conclusion.
3. Set the sequence and timing for reporting to the safety team, facilities team, and production manager.
4. Decide who authorizes restart after portable independent detection and sensor calibration.

5. Propose record fields that protect the junior employee from retaliation even if the event proves to be a false alarm.

9.6. Sample 90-Second Answer

"I choose **Answer A: evacuate the three workers immediately and request a full line shutdown**. The ILO estimates that work-related accidents and diseases cause 2.93 million deaths each year, but this global statistic does not establish whether the present alarm is genuine. The 12-second alarm, two recent false positives, normal adjacent sensors, and two-day delivery delay in this case are all assumptions. The objection that a false alarm is plausible and a full shutdown is costly is valid. But because the harm from missing hazardous gas in an occupied zone may be irreversible, I would use that information to decide when to restart—not to delay the first evacuation. Immediately after shutdown, I would begin portable independent detection, sensor calibration, and a ventilation check in parallel. I would restart only when the safety officer confirms all three of the following: two independent measurements are normal, the sensor-fault mechanism has been identified, and no one remains in the zone. If any condition is unmet, I would maintain the shutdown regardless of delivery. Even if the event proves to be a false alarm, I would record the request as appropriate to the information available at the time and protect the requester from retaliation."

9.7. Follow-Up Questions

- Why do worldwide fatality statistics fail to prove whether this sensor alarm is genuine?
- Under what conditions can a leak remain possible even when adjacent sensors read normal?
- How would your conclusion change if a partial shutdown would itself obstruct evacuation?
- When false alarms recur, should the site first raise the alarm threshold or replace the sensor?
- Which metric would show whether protection for employees who request a shutdown works in practice?
- What evidence would make you escalate immediately from a partial to a full shutdown?

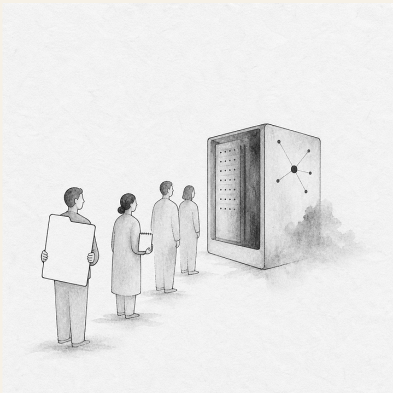
9.8. Sources

- International Labour Organization, *Safety and Health at Work* (2023; accessed 2026)
- International Labour Organization, *Nearly 3 Million People Die of Work-Related Accidents and Diseases* (2023)
- International Labour Organization, *Safety and Health at Work in Asia and the Pacific* (2022)
- Joseon Dynasty Chaekmun Archive, King Gwanghaegun's "The Most Urgent Task after the Imjin War"
- Academy of Korean Studies, 1611 Examination Record

Data note: The ILO figures are global estimates. The 12 seconds, two false alarms, two-day delivery delay, and three workers are scenario assumptions.

10. Who Should Get Priority in the GPU Queue?

Today's Chaekmun



When shared corporate GPUs are scarce, who should receive them first: a revenue team or a junior experimental team?

The Joseon chaekmun—an examination question through which the king asked candidates to reason through an urgent problem of state[1]—did not ask only how to use people whose talent was already visible. It also asked how to cultivate and recognize ability that had

[1] This chapter adapts the actual chaekmun "How Should We Find Talent?," associated with a 1447 record from the reign of King Sejong, to today's problem of allocating compute. The Chaekmun Archive condenses and reconstructs it in Classical Chinese as "人才國之寶也。何以養之、辨之、用之□君臣相遇之難□其故何歟?"; this is not a literal transcription of the original. It asks, "Talent is a treasure of the state: how should it be cultivated, recognized, and used, and why is it difficult for a ruler and a subject to find each other?"

not yet emerged. Allocating GPUs likewise goes beyond ranking teams by current revenue. It asks how to guarantee new teams a minimum opportunity to prove what they can do.

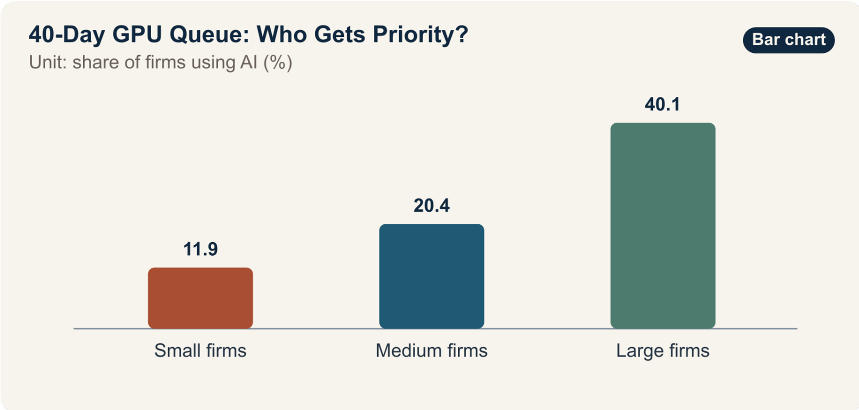
10.1. Why This Question Matters Now

In an OECD comparison compiled from official national statistics, AI adoption rates were 11.9% for small firms, 20.4% for medium-sized firms, and 40.1% for large firms. The large-firm rate was about 3.4 times the small-firm rate. Even when the same technology is publicly available, actual opportunity can differ sharply when firms have unequal funding, data, talent, training, and time to recover from failure.

These adoption rates cannot be converted directly into an internal GPU allocation ratio. They measure the share of firms using AI, not work quality, GPU hours, or return on investment. They also combine different countries and industries. An internal decision requires data on waiting time by team, actual utilization versus reservations, experimental reproducibility, customer deadlines, data readiness, and the downstream use of outputs.

Inside the company, distinguish holding an account from receiving a real opportunity to experiment. Giving everyone an account does not equalize the conditions for producing results if teams lack data cleaning, mentoring, review time, and a chance to retry after an initial failure. Allocation rules must therefore address not only GPU time, but also preparation support and a path to retry after the first experiment fails.

10.2. Data Lens



10.2.1. Evidence Table

Category	Figure or Scope in the Source	Meaning for Internal Allocation
OECD aggregate	AI adoption among small firms was 11.9%.	This is firm-level use and does not directly measure the GPU demand or ability of small teams.
OECD aggregate	AI adoption among medium-sized firms was 20.4%.	It suggests that barriers in resources and technical support may be easier to overcome as scale increases.

Category	Figure or Scope in the Source	Meaning for Internal Allocation
OECD aggregate	AI adoption among large firms was 40.1%.	Higher adoption does not guarantee the performance or efficiency of every deployment.
Calculation from source data	The large-firm adoption rate was about 3.4 times the small-firm rate.	This signals an access gap; it does not mean GPU scarcity is the sole cause.

10.2.2. How to Read These Numbers

The 11.9%, 20.4%, and 40.1% figures are OECD member-country aggregates of whether firms use AI. Consult the document's definition to determine whether they refer to generative AI alone or to all AI. An average weighted by the number of firms differs from one weighted by employees or hours of use.

Correlation between firm size and adoption does not establish causation. Large firms may have more data infrastructure, legal and security support, dedicated staff, and training, in addition to GPUs. The effectiveness of a GPU allocation policy should therefore be evaluated by reduced waiting times, completed experiments, reproducible results, operational adoption, and gaps between teams—not by the number of accounts alone.

Fair allocation is not the same as giving every team equal time. Work such as a customer demonstration deadline has urgent delay costs; junior experimentation without a track record requires an opportunity. Both objectives can be served by separating an urgent customer pool from a baseline exploration pool, reclaiming unused reservations quickly, and granting larger allocations to teams that pass an initial experiment.

10.3. Opposing Answers

10.3.1. Answer A — Prioritize Revenue and Deadlines

Allocating first to teams with clear customer schedules and cash-flow implications can produce business results faster from the same GPUs. Teams whose data and code are ready are also more likely to use the resources they reserve. Tying up scarce capacity in experiments that are not ready increases the opportunity cost to the whole company.

A demonstration with an imminent deadline may be irrecoverable once missed. It is reasonable to create a separate urgent queue and evaluate expected revenue, contract probability, and readiness.

10.3.2. Answer B — Protect Equal Opportunity

If rankings depend only on current performance, teams that already have customers, staff, and data will remain at the front of the queue, while junior and research teams cannot run even the first experiment needed to demonstrate ability. Exploration without short-term revenue can still create the next product, reveal failure mechanisms, and produce reusable data. Equity is not charity; it is an investment in future options.

Performance-based allocation favors teams that produce easily measurable outcomes and may systematically crowd out basic research and long-term improvement. Every new team should receive a baseline allocation and mentoring sufficient to run one small validation.

10.3.3. Answer C — Two Pools with Staged Promotion

Separate an urgent customer pool from a baseline research pool, then assess readiness, expected effect, and waiting time within each. Guarantee a minimum experiment to new teams, while requiring them to preregister their data, code, and evaluation metrics. Promote a team to a larger allocation when its first result is reproducible and has demonstrated follow-on value.

If actual utilization is low relative to the reservation or no output is submitted, reclaim the remaining allocation and transfer it to the next team. Urgent customer teams must also report downstream revenue and customer outcomes. Repeated failure to deliver expected results lowers their priority. Publish both the allocation rule and an appeals path so managerial discretion is not perceived as favoritism.

10.4. AI Research Design

10.4.1. Prompt for the AI

> Design an operating policy that allocates scarce shared corporate GPUs fairly between revenue teams and junior experimental teams. Use OECD AI adoption rates by firm size only as external context. Collect internal scheduler logs, reservations, actual utilization and waiting time; data readiness by team; experiment plans and reproducibility; customer deadlines; and actual business outcomes. Minimize sensitive personal and team information and test for discrimination related to rank or department. Distinguish "access," "actual use," "completed output," and "operational adoption." For every external number, provide the institution, document title, year, unit, denominator, and direct URL. Compare fixed quotas, performance ranking, and a lottery-review hybrid, and propose an urgent customer pool, a baseline research pool, reclamation of unused allocations, an appeal process, and quarterly audit criteria.

10.4.2. Data Acquisition

Priority	Evidence	Required Fields
1	GPU scheduler logs	Request, approval, start, end, waiting time, utilization versus reservation, reason for failure or termination
2	Experiment preregistration	Data readiness, code, expected GPU time, evaluation metric, security and privacy approval
3	Output repository	Reproducibility, model, code, failure record, reuse by follow-on teams, operational adoption
4	Customer and business records	Deadline, contract stage, delay cost, demonstration result, actual revenue contribution
5	Team-attribute audit data	Waiting-time gaps by rank, department, and tenure; appeals; exceptions; managerial discretion

10.4.3. Evidence Assessment

1. **Separate the metrics:** Count account ownership, GPU allocation, actual use, completion, and business performance as different stages.
2. **Verify readiness:** Do not automatically prioritize an urgent request if its data, code, and evaluation metrics are not ready.

3. **Correct opportunity gaps:** Give new teams without past performance a small validation through which they can create evidence of performance.
4. **Check reproducibility:** Recognize a learning outcome only when failure conditions, code, and logs—not just successful results—can be reused.
5. **Audit disparities:** Disclose waiting-time distributions and exception approvals by team size and rank, not only the overall average.

10.4.4. Core Issues for Debate

- Which is more urgent: a customer demonstration in two weeks or a 40-day average wait for junior teams?
- When only 60% of GPU demand can be served, what principle protects a "baseline research allocation"?
- Does evaluating the potential of teams without past performance simply reintroduce evaluator bias?
- What records must be reproducible for a failed experiment to count as an output?

10.5. Case Discussion

The following values are **internal allocation assumptions for an interview**, separate from the OECD statistics. You are a junior employee on the shared-compute operations team. The revenue team must give a customer demonstration within two weeks; 12 junior research teams have waited an average of 40 days. Current GPU capacity can serve only 60% of total requested demand. Actual utilization versus reservations and the downstream reuse of failed experiments have not yet been disclosed by team.

1. The revenue team submits its customer deadline, prepared data and code, and the cost of delay.
2. Each junior research team preregisters the purpose, required time, and reproducibility plan for a small validation.
3. The operations team examines the distribution hidden behind the 40-day team average and identifies unused reservations.
4. You design one of three approaches: **a fixed quota, performance ranking, or a lottery-review hybrid**.
5. The policy must include a two-week urgent path, a baseline opportunity to experiment, reclamation of unused time, promotion and demotion, and an appeal process.

Do not allow unprepared work to occupy resources for extended periods merely because a junior team deserves consideration. Nor should the same department monopolize exceptions in the name of revenue. Everyone must be able to predict the rule and verify the outcome.

10.6. Sample 90-Second Answer

"I propose **Answer C: two pools combining review and a lottery**. In the OECD aggregate, AI adoption was 11.9% among small firms, 20.4% among medium-sized firms, and 40.1% among large firms—making the large-firm rate about 3.4 times the small-firm rate. This gap shows that simply making a tool available does not equalize opportunity, but it does not directly set an internal allocation ratio. The customer demonstration in two weeks, 12 junior teams, 40-day average wait, and 60% capacity in this case are all assumptions. The objection that delaying a ready customer project may lose real revenue is valid. I would therefore assess the urgent customer pool by deadline, readiness, and expected impact, while using a lottery to give preregistered new teams in the baseline research pool their first experiment. Reproducible results earn promotion to the next stage; unused reservations or missing outputs trigger immediate reclamation. Each quarter, I would publish waiting time, actual utilization, reproducibility, and departmental exceptions. If junior teams remain blocked from obtaining a first result, or if unused capacity in the customer pool exceeds the threshold, I would rebalance the pools. Fair access does not mean dividing time equally. It means opening a path through which ability can be demonstrated."

10.7. Follow-Up Questions

Cross-Examination and Rebuttal

- If the customer demonstration in two weeks reflects a possibility rather than a signed contract, should it enter the urgent pool?

- How would you reduce departmental and educational-background bias when evaluating readiness among the 12 junior teams?
- If a few long projects distort the 40-day average, which quantiles should be disclosed?
- Which should come first: investing to raise the 60% capacity or improving the allocation rule?
- If the logs from a failed experiment are reused by a follow-on team, should that receive the same reward as a revenue outcome?
- How would you answer a high-performing team that has no unused allocation but must still wait to protect the baseline research pool?

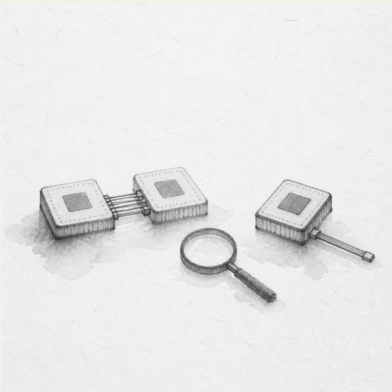
10.8. Sources

- OECD, *Generative AI and the SME Workforce: New Survey Evidence* (2025)
- OECD, *Introduction: Generative AI and the SME Workforce*—AI Adoption by Firm Size (2025)
- Joseon Dynasty Chaekmun Archive, "How Should We Find Talent?" (record from the reign of King Sejong; accessed 2026)
- National Institute of Korean History, *Gang Hui-maeng* (accessed 2026)

Data note: The OECD figures by firm size are external evidence. The two weeks, 12 teams, 40 days, and 60% are educational assumptions for internal resource allocation.

11. Chiplet Standards and Proprietary Optimization

Today's Chaekmun



Should the next AI accelerator use UCIe for every chiplet connection, or use a proprietary interface in selected segments to improve performance and power?

King Sejong's 1447 chaekmun on law and authority[1] did not assume that establishing a standard ended the problem. An interface standard can broaden the supplier base and provide a common language for verification, but it does not guarantee optimal performance in every package. A proprietary link can be tailored to a specific workload, but it can also lock design knowledge and the supply chain to one source.

11.1. Why This Question Matters Now

AI accelerators combine compute dies, I/O dies, caches, and memory interfaces inside one package. Compared with a single large monolithic die, this makes it easier to assign functions to different process nodes. It also requires bandwidth, latency, power, thermals, and testing between dies to be reconciled again at the system level. UCIe provides a common foundation spanning the physical layer, protocols, software model, and compliance testing.

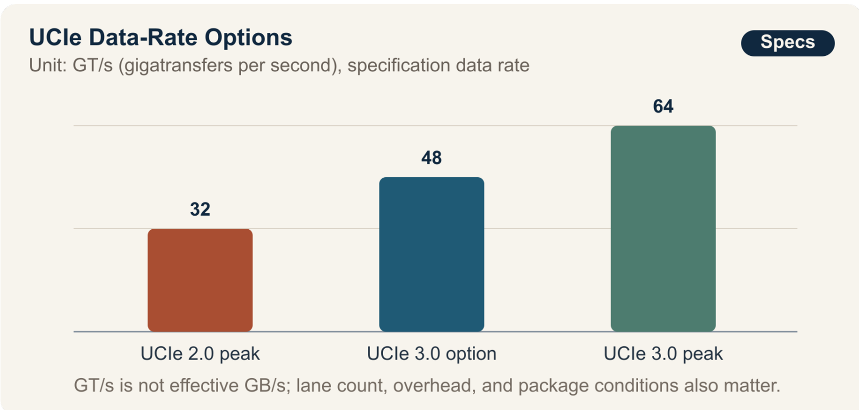
In August 2025, the UCIe Consortium announced UCIe 3.0 with support for 48 GT/s and 64 GT/s, doubling the peak data rate from

[1]The actual chaekmun （策問） connected to this chapter asked about the cycle in which rules create abuses and new rules are then established, as well as where responsibility should reside. The Chaekmun Archive condenses and reconstructs the question in Classical Chinese as "法立而弊生□弊生而法又立。其所以救之者何歟□政權當歸於何所歟□"; this is not a literal translation of the complete original. It asks, "When laws are established, abuses arise, and when abuses arise, new laws are established. How can this be remedied, and where should authority reside?" Today's question does not translate Joseon's governing institutions into chiplet specifications. It borrows the question's structure to ask who verifies and reverses a design when shared rules and engineered exceptions each create new costs. See the individual entry in the Joseon Dynasty Chaekmun Archive.

UCIe 2.0's 32 GT/s. NIST's report on 2024 standards activity, however, concluded that no single universal standard could yet be considered widely adopted and that relatively few real integrations combined chiplets from different companies. The goals of the standard and the ecosystem's current maturity must be considered together.

A junior design engineer cannot replace the ultimate decision-maker. The engineer can, however, quantify which boundaries should use the standard, whether the performance advantage of a proprietary link exceeds its verification, redesign, and single-source risks, and which failed test would trigger a return to the standards-based design.

11.2. Data Lens



11.2.1. Evidence

Evidence	Figure or Scope in Official Sources	Meaning for the Design Decision
1	The UCIe Consortium announced that UCIe 3.0 supports 48 GT/s and 64 GT/s, doubling the peak data rate from UCIe 2.0's 32 GT/s.	High-speed link options remain available when the standard is selected. GT/s, however, is not the same as the GB/s available to an application.
2	UCIe 3.0 adds an extended sideband with a reach of up to 100 mm, early firmware download, priority packets, and emergency-shutdown notification, while specifying backward compatibility with earlier versions.	Architecture review must cover initialization, management, and fault isolation, not only the data path.
3	UCIe 2.0 introduced UCIe-3D for bump pitches ranging from roughly 10–25 μm to below 1 μm, as well as test, telemetry, and debug structures for multiple chiplets.	Fine pitch alone does not guarantee yield. Bonding, thermal stress, known-good-die screening, and repair strategy are also required.

Evidence	Figure or Scope in Official Sources	Meaning for the Design Decision
4	NIST's report on 2024 activity concluded that no single universal chiplet standard was yet widely adopted and that multisupplier integration examples remained limited.	Treating "standards-compliant" as synonymous with "immediately replaceable chiplet" overstates ecosystem maturity.
5	The U.S. NAPMP announced an investment of about $3 billion in six priority areas for advanced packaging, pilot facilities, and workforce development.	Beyond the interface, bottlenecks remain in substrates, process equipment, power and thermals, optical connections, codesign, and volume-production validation.
6	A NIST chiplet-standards workshop identified gaps in package assembly design kits (PADKs), thermal and mechanical constraints, functional tests for locating defects, and fail-fast procedures.	A test plan that verifies only the link specification cannot close the boundaries of responsibility in a multisupplier package.

11.2.2. How to Read These Numbers

64 GT/s is not 64 GB/s. GT/s is the link's symbol-transfer rate. Effective bandwidth depends on lane count, encoding and error-correction overhead, protocol efficiency, directionality, package routing, and thermal constraints. The chart therefore shows the specification's speed options; it does not predict the throughput of a particular product.

The ranges of 10–25 μm and below 1 μm are bump pitches considered by UCIe-3D, not evidence that a supplier achieved a desired yield at those pitches. The sideband reach of up to 100 mm must not be confused with the reach of the primary data path. The roughly $3 billion figure is the scale of a policy program, not proof of our project's return or delivery schedule.

At a minimum, a standards-based design and a proprietary design must be measured under the same workload and package conditions using the following metrics:

- Effective bandwidth and both the mean and tail of round-trip latency
- Energy per transfer, link power, package temperature, and time spent throttling
- Traceability of PHY, protocol, and DFx requirements, and fault-injection pass rate
- Known-good-die screening rate, assembly yield, and probability of a first-silicon redesign
- Weeks required to replace an IP vendor, foundry, or OSAT, and the scope of qualification that must be repeated

11.3. Opposing Answers

11.3.1. Answer A — Standards First

Standardizing every supplier boundary on UCIe 3.0 allows common PHY and protocol requirements, compliance tests, and management and debug procedures to be reused. It also provides a basis for negotiation when evaluating a second IP source or a different chiplet. A standards-first approach is reasonable when replaceability and verification assets across a product family matter more than peak performance in the first product.

Specification compatibility is necessary for package compatibility, but not sufficient. Integration can fail even with a standard PHY if teams cannot reconcile bump maps, power-delivery networks, thermal design, firmware, and error-handling policies. The area and protocol overhead of a standard block may also exceed the power limit of a particular data path.

11.3.2. Answer B — Proprietary Optimization

When the workload is fixed and one company designs both dies and the package, a proprietary link can remove unnecessary features. Adjusting lanes, buffers, and voltage for short routing and dedicated flow control may improve performance and power. If standards-based IP validation is running late at product launch, an established internal link may also reduce schedule risk.

The price is paid in system verification and the supply chain. Dependence on one vendor's PHY, packaging process, and tools makes it harder to agree whether a defect belongs to the die, link, or assembly. The next generation or a move to external chiplets may invalidate existing verification assets. A simulation showing that the link is “faster” does not by itself offset these costs.

11.3.3. Answer C — Conditional Hybrid by Boundary

Fix UCIe at boundaries with external suppliers and on management and debug paths. Consider a proprietary link only at one performance bottleneck where the same accountable organization controls both dies. Adopt it only if it passes tests for real-workload benefit, power and thermal margin, fault isolation, redesign schedule, and a fallback path.

A hybrid does not automatically eliminate the weaknesses of the two approaches. Additional interface types can enlarge the verification matrix and complicate incident response. Before the architecture freezes, document exactly where the replaceable standards boundary ends and which failed test will cause the proprietary segment to be abandoned.

11.4. AI Research Design

11.4.1. Prompt for the AI

> Compare three approaches to die-to-die connectivity in a next-generation AI accelerator: standardizing on UCIe 3.0, using a proprietary interface, and a hybrid that uses UCIe only at supplier boundaries. Separate UCIe

Consortium specifications and announcements, NIST standards reports and CHIPS R&D materials, actual IP-vendor data sheets, and our simulations and verification logs. For each claim, provide the organization, document title, version or publication date, section and page, unit, and direct URL. When converting 32, 48, or 64 GT/s to GB/s, do not invent lane counts or overhead assumptions. Put specification support, vendor claims, independent measurements, and internal assumptions in separate columns. Divide interoperability into PHY compliance, protocol, package and PADK, thermal and power, initialization and firmware, fault isolation, and vendor replacement. Conclude with the pretests and failure thresholds that distinguish the three options, and the last point at which the project can return to the standards-based design.

11.4.2. Data Acquisition

Order	Evidence	Required Fields
1	UCIe specifications and official announcements	Version, data rate, package type, sideband, DFx, backward compatibility, compliance clauses

Order	Evidence	Required Fields
2	NIST and CHIPS standards materials	Gaps in multisupplier integration; PADKs; thermals and mechanics; test and repair; supply-chain issues
3	PHY and packaging vendor materials	Supported process and bump pitch, PPA conditions, BER test conditions, IP change and support periods
4	Internal design data	Effective bandwidth and latency by workload, link power, temperature, area, schedule, number of verification items
5	Assembly and test logs	Known-good-die, bonding yield, fault injection, defect-isolation time, rework feasibility

11.4.3. Evidence Assessment

1. **Separate specification from implementation.** Distinguish the maximum a specification permits from what a particular IP implementation achieved on a particular process.
2. **Restore the units.** Do not mix GT/s, Gb/s, GB/s, and TB/s, or unidirectional and bidirectional values, or per-lane and package-wide values.

3. **Match test conditions.** Do not rank PPA results unless they use the same workload, voltage, temperature, package routing, and error rate.
4. **State the scope of interoperability.** Record whether only PHY compliance passed, or whether firmware, DFx, packaging, and thermals were also jointly verified.
5. **Turn assumptions into decision gates.** Express the proprietary design's benefit and redesign cost as ranges, and change the decision when a range is breached.

11.4.4. Core Issues for Debate

- Which verification costs does a standard reduce, and which nonstandard problems remain in our package?
- Does the proprietary link's advantage survive in the final AI workload rather than only in a link benchmark?
- How do we measure vendor replaceability in actual weeks of revalidation rather than by counting vendor names?
- Which test allows a junior engineer to request a stop, and who holds final approval authority?

11.5. Case Discussion

You are a junior die-to-die design engineer on an AI inference-accelerator project that combines six chiplets. Product launch is 14 months away; package power is capped at 700 W, including a 70 W link-power budget. Every value below is an internal assumption for discussion, not a vendor guarantee or measured volume-production result.

- **All-UCIe design:** Uses a 64 GT/s UCIe 3.0 PHY. It is estimated to deliver 96% of target performance across three representative workloads, draw 68 W for links, require 12 months for integration, and involve 900 verification requirements. Adding an alternative PHY takes 10 weeks.
- **All-proprietary design:** Uses an internal link targeting 80 GT/s. It is estimated to deliver 104% of target performance, draw 56 W for links, and require 10 months for integration, but it has 1,350 verification requirements and only one PHY source. A replacement design would take 24 weeks.
- **Hybrid design:** Uses UCIe at external I/O and management boundaries, with a proprietary link only at the bottleneck between two compute chiplets. It is estimated to deliver 100% of target performance, draw 62 W for links, require 11 months for integration, and involve 1,100 verification requirements. The proprietary segment is single-sourced.
- The budget permits only one first-silicon respin, and the architecture freezes in eight weeks.

You must normalize the simulation conditions for all options and submit a design review with the packaging, verification, and procurement engineers. Choose among **all-UCIe, all-proprietary, and a boundary-based hybrid**. Your final design must specify where each interface is adopted, the verification to complete within eight weeks, mitigation for the single source, and the numerical condition that causes you to abandon the proprietary design.

11.6. Sample 90-Second Answer

"I choose **Answer C: fix UCIe 3.0 at external supplier boundaries and retain a proprietary link at only one bottleneck**. UCIe 3.0 supports 48 and 64 GT/s, doubling the specification's peak data rate from UCIe 2.0's 32 GT/s, but NIST finds that multisupplier integrations remain limited. The name of a standard alone cannot guarantee interoperability. In the scenario, the proprietary design is strongest at 104% performance and 56 W of link power, but it is also assumed to require 1,350 verification items and 24 weeks to replace. I accept the argument that proprietary optimization protects performance and power, so for eight weeks I would compare the options under the same workload and package conditions. I would retain the proprietary segment only if it raises performance by at least 5% or reduces power by at least 10 W, while passing fault-injection, initialization, and thermal tests. If it fails any one of these tests or slips by two weeks or more, I would return to the all-UCIe design. I would preserve the test logs and reasons for abandoning an option for reuse in the next generation. The deciding criterion is not peak speed, but preserving vendor replaceability and accountable verification."

11.7. Follow-Up Questions

- What additional information is required to convert 64 GT/s into effective bandwidth?
- What are three reasons a multisupplier package can fail even after passing UCIe compliance testing?

- If a proprietary link improves performance by 4% and saves 12 W, which criterion takes priority?
- If the sole PHY supplier promises to shorten the replacement period from 24 to 12 weeks, how would you verify the promise?
- If intermittent errors appear in first silicon, which logs distinguish responsibility among the die, package, and protocol?
- If the next generation is increasingly likely to add an external chiplet, when should today's hybrid become an all-standards design?

11.8. Sources

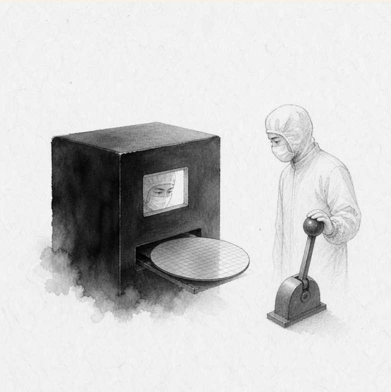
- UCIe Consortium, *UCIe 3.0 Specification With 64 GT/s Performance and Enhanced Manageability* (2025)
- UCIe Consortium, *Specifications*—Overview of UCIe 1.0–3.0 (accessed 2026)
- UCIe Consortium, *UCIe 2.0 Specification: Continuing Innovation to Drive an Open Chiplet Ecosystem* (2024)
- NIST, *Semiconductors and Microelectronics Standards Working Group Annual Report for 2024*, NIST IR 8577 (2025)
- NIST CHIPS R&D, *Chiplets Interfaces & Digital Twin Technical Standards Workshops Summary Report* (2024)
- NIST, *National Advanced Packaging Manufacturing Program Announcement* (2023)
- NIST, *Nanoscale Mechanical Characterization for Hybrid-Bonding-Ready Structures in Advanced Packaging* (2026)

- Joseon Dynasty Chaekmun Archive, King Sejong's 1447 "The Abuses of Law and the Seat of Power"
- National Institute of Korean History, *Seong Sam-mun*

Data note: The chart's 32, 48, and 64 GT/s values are specification data rates announced by the UCIe Consortium. NIST's interoperability assessment appears in a 2025 report covering 2024 activity. The scenario's 80 GT/s, performance, power, schedule, verification-item count, temperature, and switching criteria are all assumptions for discussion, not actual company or product figures.

12. Should a Black Box That Improved Yield Run the Process?

Today's Chaekmun



Should an AI be allowed to operate a process autonomously if it raises yield by 1 percentage point but cannot explain why?

The Joseon chaekmun—an examination question through which the king asked candidates to reason through an urgent problem of state[1]—asks us to distinguish an AI's improved average yield from evidence that it will operate safely on an unfamiliar product.

12.1. Why This Question Matters Now

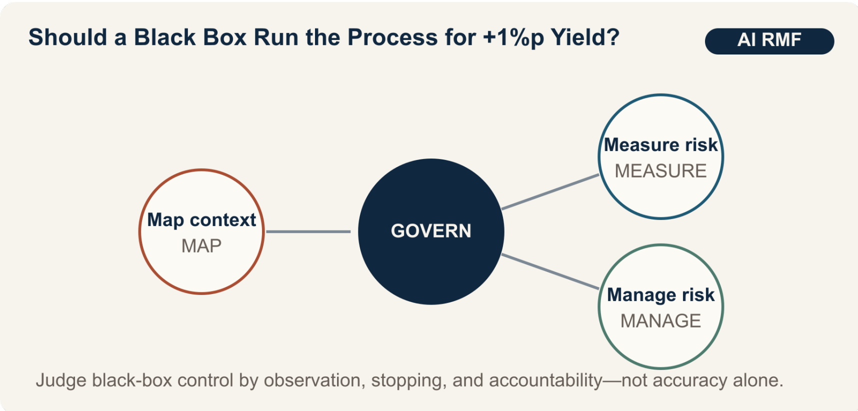
Hundreds of interacting variables and equipment states make it difficult for people to explain every semiconductor-process interaction. AI may improve recipe adjustment and anomaly detection. But when the data distribution changes after a product transition, sensor replacement, or equipment maintenance, yesterday's optimization can create today's defects. A junior validation engineer must move beyond the abstraction of "explainability" and design the conditions that stop operation and determine who receives the handoff.

The NIST AI Risk Management Framework organizes risk management around four functions: GOVERN, MAP, MEASURE, and MANAGE. GOVERN is a cross-cutting function that permeates the others. It calls for defined roles and responsibilities, policies, organizational culture, and continuing review. The essence of autonomous process control is not a statement that a human is sitting in front of a screen. It is a structure in which independent observations and

[1]The passage linked to this chapter comes from King Myeongjong's 1558 chaekmun, "The Way of Heaven and Human Governance": "天道難知亦難言也。日月麗乎天□一晝一夜□有遲有速者□孰使之然歟?" It asks whether observing a visible regularity entitles us to claim too quickly that we know its cause. It was not a contemporary question about process AI; the connection between phenomenon, cause, and responsibility is a modern one. The individual chaekmun and commentary and the Academy of Korean Studies' Annals Wiki entry on "Cheondochek" caution against confidently asserting principles that are difficult to know.

execution logs allow a person to report, stop, and reverse an actual anomaly.

12.2. Data Lens



12.2.1. Evidence Table

Evidence	Scope in the Source	Operational Meaning
1	NIST AI RMF 1.0 organizes the core of risk management into four functions: GOVERN, MAP, MEASURE, and MANAGE.	Do not decide deployment on yield alone; consider roles, context, measurement, and response together.

Evidence	Scope in the Source	Operational Meaning
2	GOVERN is a cross-cutting function that permeates MAP, MEASURE, and MANAGE.	Define the accountable operator, change approval, authority to stop, and record retention alongside model performance.
3	In 2026, NIST released a concept paper for an AI RMF profile on trustworthy AI in critical infrastructure.	Treat process AI across the full life cycle of equipment availability, safety, and supply chains, not only by laboratory accuracy.

12.2.2. How to Read These Numbers

The public evidence in this chapter does not provide a single “safety score.” The four functions form an iterative structure, not a certification checklist completed once in sequence. GOVERN does not eliminate risk, and MEASURE does not mean average yield alone. Measure false positives and false negatives by product, operator-intervention rates, drift-detection time, and time to restore the previous recipe.

The scenario's 1-percentage-point yield gain and false-positive rates are therefore validation conditions, not industry averages. Percent and percentage point must also remain distinct. If the false-positive rate rises from 2% to 6%, it has increased by 4 percentage points and tripled.

12.3. Opposing Answers

12.3.1. Answer A — Performance-First Autonomous Operation

Waiting for a complete causal explanation can forfeit a validated improvement in a complex process. If independent offline testing and a limited pilot show performance more stable than human control, autonomous operation can begin with low-risk, recoverable product families. Scope limits and real-time monitoring can compensate for gaps in explanation.

12.3.2. Answer B — Explanation and Recovery First

An average yield improvement of 1 percentage point can conceal rising false positives after a product transition or rare, severe losses. Without understanding causes, it is harder to reproduce failures after equipment maintenance or recipe changes, and easier to hide responsibility behind the model. Until diagnosis and safe rollback are validated, AI should recommend actions and a human should execute them.

12.3.3. Answer C — Limited Operation by Product Family

Do not reduce autonomous operation to two boxes labeled "explainable" and "unexplainable." Define operating envelopes by product family, dual observation paths, operator stop authority, drift alarms, and time to restore the previous recipe. If false positives, false negatives, or recovery time breach a threshold, downgrade immediately to recommendation mode and resume autonomous operation only after independent validation.

12.4. AI Research Design

12.4.1. Prompt for the AI

> Using NIST AI RMF 1.0 and the 2026 concept paper for a critical-infrastructure profile, design a validation plan for autonomous semiconductor-process operation. For each of GOVERN, MAP, MEASURE, and MANAGE, tabulate the accountable party, required data, test, stopping condition, and evidentiary log. In addition to average yield, require false-positive and false-negative rates by product, operator-intervention rate, distribution-shift detection time, and safe-rollback time. Separate claims by the model-development team from independent validation results. Cite each source with the NIST document title, year, section, and direct URL. Put equipment failure modes and data gaps unknown to the AI in separate columns.

12.4.2. Data Acquisition

Order	Evidence	Fields to Verify
1	NIST AI RMF and Playbook	Definition by function, life cycle, independent review, risk prioritization
2	Process history and model versions	Product, equipment, recipe and sensor version, training period, deployment point
3	Shadow operation and limited pilot	Differences from human decisions, false positives and false negatives, yield, reason for intervention
4	Incident and recovery logs	Detection time, stop time, restoration of previous recipe, lost wafers

12.4.3. Evidence Assessment

1. Separate training lots from test lots and evaluate product-transition periods independently.
2. Examine distributions by product to determine whether an average improvement conceals losses in a specific product.
3. Verify that execution logs fully connect model recommendations to commands actually issued to equipment.
4. Use a live drill to verify that an operator can stop the system without retaliation.

5. Require joint process, quality, and safety signatures for restart, rather than approval by the development team.

12.4.4. Core Issues for Debate

- How much harm and recoverability can be tolerated when explanation remains incomplete?
- Which takes precedence as a stop signal: average yield or false positives in the worst-performing product family?
- What logs and response times—not merely the presence of a button—prove that a human is "in the loop"?

12.5. Case Discussion

You are a junior engineer on a process-AI validation team. In a limited pilot, average yield rose by 1 percentage point, but false positives increased from 2% to 6% after a product transition. The average time from anomaly detection to human intervention is 18 minutes; restoring the previous recipe takes 25 minutes. **Every number in this section is a hypothetical condition for discussion, not an actual fab result.**

1. Recommend one of three options: continue autonomous operation, permit it only for existing product families, or stop it entirely.
2. Decide whether to downgrade new products to recommendation mode and reclassify false positives by cause.
3. Set a stopping threshold and the number of consecutive validation lots required for restart.

4. Assign stop and restart authority among operators, developers, and the process owner.
5. Write the procedure for returning to the previous model when safety or quality losses exceed the yield benefit.

12.6. Sample 90-Second Answer

"I choose **Answer C: allow autonomous operation only for existing product families and downgrade new products to recommendation mode.** NIST AI RMF calls for four functions—GOVERN, MAP, MEASURE, and MANAGE—and makes GOVERN the accountability structure across them. That is why higher average yield alone cannot authorize autonomous operation. The 1-percentage-point yield gain, increase in false positives from 2% to 6%, 18-minute intervention time, and 25-minute rollback time in this case are all assumptions. I accept the objection that waiting for a complete explanation in a complex process can forfeit an improvement. Just as Yi I distinguished observed phenomena from the principles behind them, I would validate average performance separately from the failure conditions that follow a product transition.[2] I would give operators immediate stop authority and independently validate drift alarms and rollback drills. I would expand autonomous operation to new products once their false-positive rate returns to roughly 2%; if it

[2]This is a modern paraphrase of Yi I's (李珥) "Cheondochek," the winning response in the first stage of a special civil-service examination in 1558. His answer distinguishes changing *gi* (氣) from the underlying principle, *li* (理), while arguing that real-world abuses must still be corrected. See the individual entry in the Joseon Dynasty Chaekmun Archive and the Academy of Korean Studies' Annals Wiki entry on "Cheondochek".

approaches 6% again or recovery time grows, I would move existing products to manual mode as well. I would also protect anyone who requests a stop from retaliation. Human control must be demonstrated by actual stop-and-recovery logs, not by an approval stamp."

12.7. Follow-Up Questions

- How would you compare a 1-percentage-point yield gain and a 4-percentage-point rise in false positives in the same cost unit?
- If performance deteriorates only after a product transition, what would you check first in the training data?
- Is a model safer merely because it explains itself well, even if its performance is lower?
- How would you measure organizational pressure that prevents an operator from pressing the stop button?
- If the independent validation team relies on the same data, is it sufficiently independent?
- What conditions would move your conditional position to full autonomous operation?

12.8. Sources

- U.S. National Institute of Standards and Technology, *Artificial Intelligence Risk Management Framework* (*AI RMF 1.0*) (2023)
- NIST AI Resource Center, *AI RMF Core* (2023; accessed 2026)

- NIST, *Concept Note: AI RMF Profile on Trustworthy AI in Critical Infrastructure* (2026)
- Joseon Dynasty Chaekmun Archive, King Myeongjong's "The Way of Heaven and Human Governance"

Data note: NIST's four functions are grounded in public documents. The yield, false-positive, intervention, and recovery figures are assumptions created for the scenario.

Part III.

Organizations, Security, and Talent

13. Critical Technologies and Engineer Mobility

Today's Chaekmun

How far should engineers' career moves be restricted and monitored to protect national core technologies?

The Joseon chaekmun—an examination question through which the king asked candidates to reason through an urgent problem of state[1]—did not begin by choosing either war or peace. It asked what

[1]The Chaekmun Archive condenses and reconstructs King Seonjo's 1568 special-examination question in Classical Chinese as 「禦外之道□戰與和而已。義、力、勢□何所先後□」 It asks, "When responding to an external threat, in what order should principle, capability, and circumstances be considered?" It is neither a literal transcription of the surviving examination text nor a translation of this chapter's modern question. The connection here is only the mode of reasoning: protection must be assessed through its purpose, actual capacity for control, and specific circumstances rather than by invoking principle alone. See the individual entry in the Chaekmun Archive and the KRpia bibliographic entry for *Park Se-dang's Chaekmun*.

should be corrected first when principle, capability, and circumstances diverged. Technology protection presents the same problem. Even when the principle of “national security” is weighty, treating every departing employee as a suspect undermines the ability to compete for talent. Speaking only of “freedom of occupation,” while ignoring access privileges and evidence of removal, misses real leakage risks.

This chapter is not a test of personal loyalty. It distinguishes **the technology to protect, the evidence to verify, the measure to use, and the point at which that measure ends**. Nationality and the country of a new employer must not become proxies for risk; the same conduct must trigger the same procedure.

13.1. Why This Question Matters Now

The Notice on the Designation of National Core Technologies, effective in October 2025, expanded and revised the list to 79 technologies across 13 fields. That does not mean 79 people or 79 companies should be monitored. It defines the **scope of technologies** subject to protection.

The Guidelines for the Protection of Industrial Technology, effective in June 2026, require organizations holding national core technologies to manage access rights, block and log unauthorized transfers, identify anomalous activity by employees preparing to leave, and manage the return of company assets. The Guidelines also provide reference forms for confidentiality and noncompete agreements. Nothing in them, however, grants a company a blanket right to copy the entire contents of a departing employee's personal phone and laptop as a matter of course.

The Special Act on National High-Tech Strategic Industries allows contracts with designated specialist personnel to include restrictions on joining the same type of industry overseas and the duration of those restrictions, prevention of confidential-information leakage, and the provision of post-employment information. The system depends on designation and contract; it is not an automatic ban on international travel or job changes for every semiconductor engineer. The government's introduction of a "Top-Tier Visa" in 2025 to recruit global talent in semiconductors, AI, and other fields also demonstrates that international talent mobility is a source of competitiveness.

Enforcement results show that the risk is real. The Korean National Police Agency announced that in 2025 it had apprehended 378 people in 179 technology-leakage cases, eight of which involved national core technologies. "Apprehension," however, is not a final conviction, and the 179 cases do not establish an incidence rate that includes unreported cases. The figures support the need for protective controls; they do not justify presuming that a particular employee changing jobs is a criminal.

13.2. Data Lens

13.2.1. Evidence

Evidence	Fact Established in the Official Text	Proper Use in the Debate
1	The Korean National Police Agency announced that in 2025 it apprehended 378 people in 179 technology-leakage cases, eight of which involved national core technologies.	This shows the scale of enforcement, not the crime rate, the number of final convictions, or the risk posed by every departing employee.
2	The October 2025 Notice expanded and revised the national core technology list to 79 technologies across 13 fields.	First determine whether a specific technology falls within the Notice; the object of protection is not a person's job title.

Evidence	Fact Established in the Official Text	Proper Use in the Debate
3	The 2026 Guidelines for the Protection of Industrial Technology require management of access rights and records of viewing, storage, and removal, as well as review of anomalous activity by employees preparing to leave and the return of company assets.	This supports baseline controls over company systems and company assets. It does not support a complete search of personal devices.
4	The Personal Information Protection Commission explains that work-network logs are personal information and must not be used for employee surveillance, but may be managed to verify access outside an employee's authority.	The answer is not "logs exist, so we can inspect anything," but a targeted review with defined purpose, fields, and retention period.

Evidence	Fact Established in the Official Text	Proper Use in the Debate
5	In Supreme Court case 2009Da82244, the Court held that the validity of a noncompete agreement depends on a comprehensive assessment of the interest to be protected; the employee's predeparture position; duration, geography, and occupation; compensation; circumstances of departure; and public interest.	Separate confidentiality from a noncompete that restricts the occupation itself; the latter requires a narrow scope and compensation.
6	Article 15 of the Constitution guarantees freedom of occupation, while Article 3 of the Personal Information Protection Act requires the minimum information necessary for a stated purpose and minimal intrusion into private life.	The more intrusive the measure—such as copying an entire personal device—the stronger the need for a specific basis and consideration of less intrusive alternatives.

13.2.2. Distinguishing the Control Measures

The six levels in the illustration are not official statutory grades. They are an **intrusiveness ladder** constructed for this debate.

Level	Measure	Scope Examined	Condition for Escalation
1	Role-based access	Work-required folders in company repositories	Role change, excessive privilege, change in protection classification
2	Download and access logs	Company-system metadata such as time, account, filename, volume, and external-media connection	Access far outside the normal baseline combined with a business explanation that does not fit

Level	Measure	Scope Examined	Condition for Escalation
3	Targeted predeparture review	Specific period, account, files, and company assets	A concrete connection between national-core-technology files and an unauthorized removal path
4	Confirmation of confidentiality, return, and deletion	Nonpublic information and company assets defined by contract	Missing returns, refusal to confirm deletion, or evidence of public disclosure
5	Compensated, limited noncompete	Specific technology, role, industry, and period	Designation and contract, protectable interest, reasonable scope, and compensation are all established

Level	Measure	Scope Examined	Condition for Escalation
6	Targeted forensics on a personal device	Files, period, and area specified by agreement or lawful procedure	Concrete evidence supporting external copying or transmission and independent legal review

13.2.3. How to Read These Numbers

Do not interpret the eight national-core-technology cases among 179 cases as a "4.5% probability of national core technology leakage." The two numbers describe different categories of cases in which apprehensions occurred in 2025; there is no denominator representing all companies or departing employees. Apprehensions must also be distinguished from final convictions.

An anomaly such as a 4.8 GB download is a starting point, not a conclusion. The same volume means different things depending on whether it was approved handover material, how many times it exceeded the employee's ordinary baseline, whether the files were actually written to external media, and whether a transfer succeeded. A gap in the logs proves neither "no leakage" nor "leakage."

Noncompetition and confidentiality are different tools. Confidentiality prohibits using or disclosing defined information; a noncompete prohibits performing defined work for a period. If a narrower measure can protect the technology, the entire occupation should not be restricted first.

13.3. Opposing Answers

13.3.1. Answer A — Proactive Protection

Once a national core technology has been copied outside the organization, it is difficult to retrieve. Design drawings and process recipes fit on a small storage device, so litigation after departure may not reverse the harm. Organizations holding these technologies should therefore maintain role-based least privilege, removal approval, download and external-media logs, immediate withdrawal of high-risk privileges at departure, and return of company assets as ordinary controls.

Confidentiality alone may not suffice where designated specialist personnel handle genuinely irreplaceable, leading-edge technology. It may be reasonable to agree in advance on a noncompete tailored to a specific competing role and the useful life of the technology, with adequate compensation, and to seek a prompt court decision when evidence of breach appears. This position does not call for more surveillance. It calls for strengthening the company systems that can be controlled before harm occurs.

The scope of protection must not, however, be drafted broadly or imposed identically on every departing employee. Overbroad restrictions increase control costs and false positives. They can drive strong employees away from critical work and ultimately weaken the capacity to create the very technology being protected.

13.3.2. Answer B — Protect Mobility and Privacy

An engineer's general experience, public knowledge, and problem-solving ability are not company files. Copying personal devices and banning employment throughout the same industry worldwide merely because someone changes jobs or joins a foreign company can unduly restrict occupational freedom and privacy. A country should also avoid the contradiction of recruiting foreign talent while immobilizing its own.

Security should protect the technology at its source instead of binding the person. Segmented repositories, least privilege, approved transfer channels, auditability of company devices, watermarks and hashes for important files, and confirmation of return and deletion at departure can reduce risk without looking deeply into private life. Without concrete evidence of transmission, the principle should be to reaffirm confidentiality duties while respecting the job change.

Ignoring log anomalies and external-media connections altogether in the name of privacy, however, would make it difficult for a company to perform legally required protective measures. Protecting legitimate mobility therefore requires a targeted review with a clearly defined scope and termination condition.

13.3.3. Answer C — Risk-Based Conditional Controls

Control intensity should depend on four facts, not on the person's nationality or destination: **the technology's protection classification, the person's actual access privileges, conduct outside the ordinary baseline, and evidence supporting external transmission**. When all four are low, least privilege and the ordinary departure process are sufficient. If some are elevated, conduct a targeted review of company logs and company assets. Escalate to independent legal review, limited forensics, an injunction, or referral to investigators only when evidence of external copying or transmission becomes concrete.

Treat a noncompete as a separate, last-resort measure. Narrow it to roles that directly overlap the protected technology, its useful life, and the relevant industry, and pay compensation. The person under review must receive the basis for the review, an opportunity to respond, and a process for appealing to an independent reviewer. Delete investigation data when the purpose-specific retention period ends.

The core principle is not "inspect everything whenever there is suspicion," but **raise the control by only one level when the evidence rises by one level**. If an approved handover explains the activity, return immediately to a lower level. If external transmission is verified, escalate under the same procedure regardless of the person's background.

13.4. AI Research Design

13.4.1. Prompt for the AI

As of August 2026, investigate the case of an engineer at a Korean semiconductor company who handles national core technology and leaves for a similar role in Korea or abroad. Use as primary sources only the Act on Prevention of Divulgence and Protection of Industrial Technology, its Enforcement Decree, the Guidelines for the Protection of Industrial Technology, and the Notice on the Designation of National Core Technologies in the Korean Law Information Center; the Special Act on Strengthening and Protecting Competitiveness of National High-Tech Strategic Industries and its Enforcement Decree; Articles 3 and 15 of the Personal Information Protection Act; Supreme Court case 2009Da82244; the Personal Information Protection Commission's employment guidance; and the Korean National Police Agency's 2025 enforcement results on technology leakage. Classify access privileges, download logs, predeparture review, confidentiality, noncompetes, and inspection of personal devices as "mandatory," "potentially permissible," "contract required," "court determination required," or "no basis." For every conclusion, attach the institution, document title, provision, effective, decision, or publication date, and direct URL. Do not confuse apprehension with conviction, a log anomaly with actual transmission, company assets with personal devices, or national core technology with general know-how. Conclude with a decision table that changes control intensity according to technology sensitivity, access privileges, evidence of anomalous conduct, and evidence of transmission.

13.4.2. Data Acquisition

Priority	Evidence	Required Fields
1	Current statutes, notices, and guidelines in the Korean Law Information Center	Effective date, exact provision, covered party, language establishing duty versus discretion
2	Full Supreme Court judgment	Case number, decision date, factors governing a noncompete, outcome of the case
3	Personal Information Protection Commission guidance	Status of logs as personal information, permissible purpose, ban on out-of-purpose surveillance, data-minimization principle
4	Korean National Police Agency enforcement data	Baseline year, cases versus people, apprehension versus conviction, national-core-technology category

Priority	Evidence	Required Fields
5	Internal company records	Protection classification, privilege matrix, approval records, logging scope and retention period, contracts and compensation

13.4.3. Evidence Assessment

1. **Classify the subject matter:** Verify that the file and process actually fall under the current Notice on national core technology or qualify as national high-tech strategic technology.
2. **Classify the authorization:** Separate authorization to view from authorization to download or remove.
3. **Classify the conduct:** Quantify departures from the ordinary baseline while checking alternative explanations such as handover, incident response, or copying for approved training.
4. **Classify the transmission:** Record external-media connection, file writing, email or cloud upload, and confirmation of receipt as different pieces of evidence.
5. **Classify the measure:** Determine whether a less intrusive measure achieves the same purpose and whether scope, duration, approver, deletion date, and appeal procedure are defined.
6. **Classify the conclusion:** Put “verified fact,” “reasonable inference,” “cannot be verified,” and “hypothetical scenario assumption” in separate fields.

13.4.4. Core Issues for Debate

- Which specific file, process, and privilege does "protecting the technology" refer to?
- What additional evidence is needed between a log anomaly and actual copying or transmission?
- If reviewing company systems achieves the purpose, why inspect the entirety of a personal device?
- Who sets the role, duration, geography, and compensation of a noncompete, and by which criteria?
- When do the end of the investigation, deletion of data, and the person's response and appeal procedures take effect?

13.5. Case Discussion

You are a first-year process-technology engineer at **Gaon Memory**, a fictional semiconductor company, and a junior member of a task force improving departure-security procedures. Every number below is a **hypothetical condition** created for discussion; none refers to an actual company or case.

A process-integration engineer with seven years of experience will leave in 14 days for an R&D role at a foreign company in the same industry. The engineer was authorized to view an advanced-process recipe that the company is assumed to manage as national core technology. There is no noncompete agreement supported by separate compensation; the employee signed only confidentiality and company-asset-return agreements.

Downloads during the seven days before notice of departure totaled 4.8 GB—16 times the 0.3 GB weekly median over the previous 90 days among employees in the same role. Of that total, however, 4.0 GB, or 83.3%, came from a handover folder approved by the team manager. The remaining 0.8 GB consisted of 12 files outside the handover list, three of which contained critical recipes. All were within the employee's viewing privileges, but removal had not been approved.

The logs also show that a personal external storage device was connected to the company laptop for 11 minutes. An error in the security agent means no record remains of whether any file was successfully written. The company wants to copy the entire contents of the employee's personal laptop and phone and impose a worldwide, industry-wide employment ban for 24 months. The departing employee agrees to return company assets, permit an audit of company accounts, and allow an independent forensic firm to compare hashes for the 12 disputed files, but refuses a complete copy of personal devices.

Choose one of the following:

1. Because most downloads were approved handover material, proceed only with the ordinary departure process.
2. Revoke access to core technology immediately; conduct a targeted review limited to company logs, company assets, and the 12 files; and propose a narrow, compensated noncompete.
3. Seize the entirety of the personal devices and immediately pursue a 24-month blanket noncompete and referral to investigators.

After choosing, specify the scope of the review, the employee's opportunity to respond, the role, duration, and compensation covered by a noncompete, the conditions for escalation and release, and the deletion date for investigation data.

13.6. Sample 90-Second Answer

"I choose **Answer B: respect the freedom to change jobs while reviewing only verified risks**. The Korean National Police Agency announced that in 2025 it apprehended 378 people in 179 technology-leakage cases, eight of which involved national core technologies. That is a serious risk, but not a basis for suspecting everyone who changes jobs. The 4.8 GB download, 16-times-normal volume, three critical recipes outside the handover list, and 24-month employment ban in this case are all assumptions. The counterargument is important: 83.3% of the material was approved, and connecting an external device does not prove intent to leak. I would first preserve the access records and company logs, then have an independent reviewer compare only the hashes of the 12 disputed files and give the employee an opportunity to explain. I would not copy the entirety of any personal device. I would propose a nine-month noncompete covering only roles that directly overlap the technology, with compensation equal to 80% of base salary. Verified external transmission would trigger legal action; confirmation that the activity was approved work would end the restriction and require deletion of investigation data. Protecting technology cannot turn a legitimate job change into a crime."

13.7. Follow-Up Questions

- Even if the 4.8 GB total is 16 times the normal baseline, what additional metric would you examine given that 83.3% was approved material?
- When the external-device write log is missing, should the record say "no leakage" or "cannot be verified"?
- If the departing employee also refuses a hash comparison of the 12 files, what additional evidence would cause you to seek a court order?
- How would you calculate whether the hypothetical nine-month restriction and compensation equal to 80% of base salary are excessive or insufficient?
- How would you explain any difference between joining a domestic competitor and a foreign company in the same industry in terms of technical risk and legal authority?
- If the investigation ends with evidence supporting innocence, who must verify deletion of logs and forensic data, and when?

13.8. Sources

- Korean National Police Agency, "2025 Technology-Leakage Enforcement Results at a Glance" (2026)
- Ministry of Trade, Industry and Energy, *Notice on the Designation of National Core Technologies*, Notice No. 2025-2 (2025)
- Ministry of Trade, Industry and Energy, Reasons for Enactment and Amendment of the *Notice on the Designation of National Core Technologies*: 79 Technologies in 13 Fields (2025)

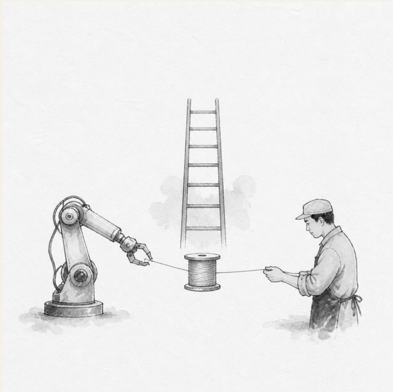
- Ministry of Trade, Industry and Energy, *Guidelines for the Protection of Industrial Technology*, Notice No. 2026-76 (2026)
- Korean Law Information Center, *Act on Prevention of Divulgence and Protection of Industrial Technology* (version effective 2026)
- Korean Law Information Center, Article 14 of the *Enforcement Decree of the Industrial Technology Protection Act*: Measures to Protect National Core Technologies (2026)
- Korean Law Information Center, *Special Act on Strengthening and Protecting Competitiveness of National High-Tech Strategic Industries* (version effective 2026)
- Korean Law Information Center, Article 24 of the *Enforcement Decree of the National High-Tech Strategic Industries Act*: Application for Designation of Specialist Personnel (2026)
- Personal Information Protection Commission, Q&A in the *Personal Information Protection Guidelines for Employment and Labor* (2023)
- Korean Law Information Center, Article 3 of the *Personal Information Protection Act*: Principles for Protecting Personal Information (version effective 2025)
- Korean Law Information Center, Article 15 of the *Personal Information Protection Act*: Collection and Use of Personal Information (version effective 2025)
- Korean Law Information Center, Articles 15 and 17 of the *Constitution of the Republic of Korea* (version effective 1988)
- Supreme Court of Korea, Decision 2009Da82244, March 11, 2010
- Korea Policy Briefing, Introduction of the "Top-Tier Visa" for Global Talent in Semiconductors, AI, and Other Fields (2025)

- Joseon Dynasty Chaekmun Archive, King Seonjo's Special-Examination Chaekmun (accessed 2026)
- KRpia, Bibliographic Entry for *Park Se-dang's Chaekmun* (accessed 2026)

Data note: The legal analysis uses the versions in force in the Korean Law Information Center as of August 20, 2026. The police figures describe enforcement outcomes. The company name, positions, data volumes, periods, ratios, and compensation rate in the case are all hypothetical. The control ladder in the illustration is an analytical framework that orders several official measures by intrusiveness; it is not a statutory classification. An actual investigation, noncompete, or forensic examination should not be decided solely under internal company rules, but must receive case-specific legal review by specialists in privacy, labor, and trade-secret law.

14. Who Owns the Time Saved by Automation?

Today's Chaekmun



If AI raises productivity by 20%, how should the company, the team, and employees share the resulting gains and reduction in working time?

The Joseon chaekmun—a royal policy examination question—[1] treats education not simply as the transfer of knowledge, but as the work of developing people and roles. The gains from automation

[1]The original passage most closely related to this chapter comes from the Myeongjong-era chaekmun "Where Should Education Go?": "教所以成材也。其本末次第與作人之方何如□" It asks, "Education is what develops capable people; what should be the order of its foundations and particulars, and what methods should be used to form people?" The original did not directly address automation or corporate allocation. This chapter interprets it in relation to preparing people whose work has been reduced for their next role. See the individual chaekmun and commentary.

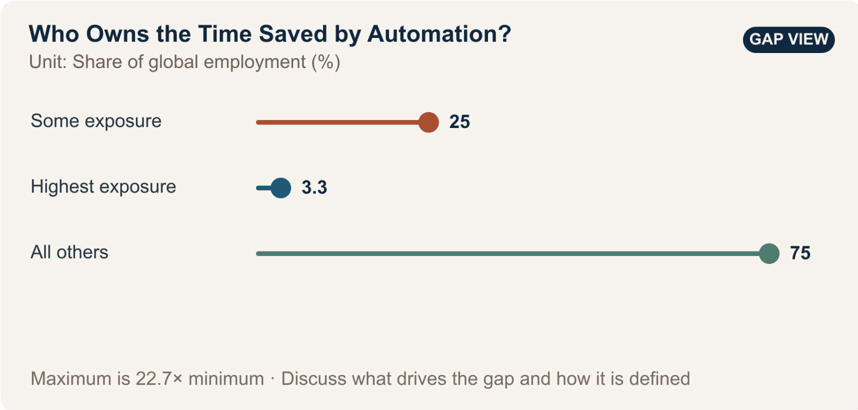
should likewise be judged by whether they merely reduce the remaining workforce or create new skills.

14.1. Why This Question Matters Now

The ILO estimates that one in four workers worldwide is in an occupation with some degree of exposure to generative AI. Exposure, however, is not a layoff rate. If the likelihood that some tasks will change is read as the percentage of entire jobs that will disappear, both automation investment and workforce planning become distorted. The company supplied the capital for equipment and models; employees supplied process data, exception-handling knowledge, and shop-floor expertise. If the productivity gain has shared origins, its allocation rules should be set before implementation rather than bargained over afterward.

Because the ILO sees job transformation as more likely than wholesale replacement, a new employee's practical question should be more specific than "Will AI eliminate jobs?" Before automation is introduced, the company should decide who owns the time it saves and who will pay for the paid learning time and on-the-job mentoring needed to move people into new roles.

14.2. Data Lens



14.2.1. Evidence Table

Evidence	Figure or range in the primary source	Significance for the debate
1	The ILO estimates that 25% of workers worldwide—one in four—are in occupations with some degree of exposure to generative AI.	Because task change is likely to be widespread, each role needs a transition plan.
2	The highest-exposure category accounts for 3.3% of global employment.	Claiming that “all 25% of exposed workers will be replaced” wrongly conflates distinct categories.

Evidence	Figure or range in the primary source	Significance for the debate
3	The ILO judges job transformation to be more likely than wholesale replacement.	Performance measures should extend beyond headcount reductions to changes in tasks, pay, working time, and redeployment.

14.2.2. How to Read These Numbers

The 25% figure is the share of employment in occupations that include tasks generative AI can perform—not the share of jobs that will disappear. The 3.3% figure is also not an observed layoff rate; it is the employment share in the highest-exposure category. Exposure also varies by country, gender, and occupational structure. These figures cannot be converted directly into a company's workforce-reduction target.

Inside the company, compare labor hours, errors, rework, and output before and after automation for the same product under the same quality standard. Also examine whether the saved hours reduced overtime or were absorbed only through headcount cuts, and whether paid training led to actual placement while preserving employment and pay.

14.3. Competing Answers

14.3.1. Answer A — Reinvestment First

Because the company bore the development cost and the risk of failure, a substantial share of the productivity gain should be reinvested in equipment upgrades and R&D. Distributing all gains immediately through shorter hours and bonuses could reduce the capacity for the next round of automation. Reinvestment that protects long-term employment is also a benefit returned to employees.

14.3.2. Answer B — Return the Gains to Labor

AI works because shop-floor employees supplied data and exception-handling knowledge. If the company cuts headcount, increases workloads, and monopolizes the productivity gain, trust and the transfer of expertise will break down. Part of the verified net gain should fund shorter working hours, bonuses, and paid retraining.

14.3.3. Answer C — Agree on an Allocation Formula in Advance

Settle the allocation formula before automation begins. Count only net productivity gains achieved while maintaining quality and safety, then divide them among reinvestment, team rewards, paid training, and shorter hours. If post-training placement and retention rates fall below their targets, improve training and job design before making additional workforce cuts.

14.4. AI Research Design

14.4.1. Prompt for the AI

Consult the original 2025 ILO report *Generative AI and Jobs: A Refined Global Index of Occupational Exposure* and distinguish its definitions of exposure, highest exposure, replacement, and job transformation. Build before-and-after task lists for semiconductor production roles, comparing output, quality, rework, labor hours, pay, headcount, paid training hours, and redeployment rates over the same period and product family. Present an allocation table for productivity gains that distinguishes the company's capital investment from employees' contributions of data and expertise. Put verified facts, company assumptions, worker views, and analyst inferences in separate columns. Attach the unit, period, role scope, and direct URL to every figure.

14.4.2. Data Acquisition

Step	Source	What to verify
1	ILO report and methodology	Definition of exposure, occupation and task units, highest exposure, and limits of interpretation
2	Process MES and quality records	Hourly output, defects, rework, and downtime for the same product
3	HR, payroll, and work records	Headcount, pay, overtime, attrition, and redeployment by role
4	Training and placement records	Whether training was paid, training hours, and actual placement and retention rather than completion alone

14.4.3. Evaluating the Information

1. Remove the effects of product mix and demand growth from any increase in output.
2. Fix the denominator: determine whether the reported 20% productivity gain is per person, per hour, or per piece of equipment.
3. Look beyond training completion to placement, post-placement pay, and continued employment.
4. Deduct the cost of any increase in errors or safety incidents from the productivity gain.
5. Do not rely on an average of employee views; examine the distribution by role, shift, and tenure.

14.4.4. Core Questions for Debate

- By what standard should the company distinguish recovery of its automation investment from employees' contributions of data and expertise?
- If saved time is filled with new tasks, can the productivity gain be said to have been returned to employees?
- Should retraining be judged by completion, placement, or preservation of pay?

14.5. Case Discussion

You are a new employee on the production innovation team. After automation, hourly productivity rose 20%, but headcount in one role fell 30%, and only 55% of retrained employees were placed in other roles. The company must allocate the next year's net productivity gain among equipment reinvestment, bonuses, a four-hour reduction in the workweek, and paid training. **All figures are hypothetical conditions for discussion and are not data from an actual company.**

1. Ask for the net gain to be recalculated after deducting quality losses, rework, and depreciation.
2. Propose an allocation among the four categories and explain your rationale.
3. Decide whether to maintain the 30% headcount reduction or pause further cuts during retraining.
4. Specify changes to roles, mentors, and paid learning time that would raise the 55% placement rate.
5. Define the reversal conditions that would reopen the allocation formula after six months.

14.6. Sample 90-Second Response

"I choose **Answer C: divide productivity gains between the company and employees according to criteria agreed in advance.** The ILO estimates that 25% of workers worldwide are in occupations with some degree of exposure to generative AI, but the highest-exposure category accounts for 3.3%, and job transformation is more likely than wholesale replacement. Exposure rates must not be read as layoff rates. The case figures—a 20% productivity increase, a 30% headcount reduction, and a 55% post-training placement rate—are all hypothetical. I accept the counterargument that the company must recover its investment if it is to fund the next round of automation. But if the allocation excludes the shop floor's contributions of data and expertise, the next transition will fail. I would divide the net gain after quality and rework costs as follows: 40% for equipment reinvestment, 25% for paid training, 20% for bonuses, and 15% for a four-hour reduction in the workweek. These proportions are also for discussion only. If placement remains below 70%, I would pause further workforce cuts and increase the training share. If placement and pay retention are confirmed, I would raise the reinvestment share. I would publish the results quarterly. Automation should be judged not by how many positions it eliminates, but by how many people move into safer, more skilled roles."

14.7. Follow-Up Questions

- How would your allocation change depending on whether the 20% productivity gain is per hour or per person?
- Should bonuses be paid before the automation development cost has been recovered?
- If workload intensity rises after a four-hour reduction in the workweek, has time truly been returned to employees?
- How should employees and the company share responsibility for a 55% retraining placement rate?

- If automation removes a new hire's repetitive work, what should replace that rung on the skills ladder?
- Under what conditions would you reduce the employee share and increase equipment reinvestment?

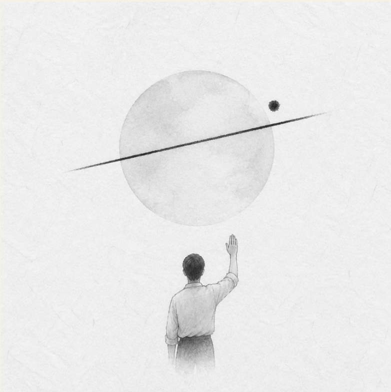
14.8. Sources

- International Labour Organization, *Generative AI and Jobs: A Refined Global Index of Occupational Exposure*, Working Paper 140 (2025)
- International Labour Organization, *Generative AI and Jobs: A 2025 Update* (2025)
- Joseon Dynasty Chaekmun Archive, Myeongjong-era "Where Should Education Go?"

Data note: The 25% and 3.3% figures are ILO estimates of occupational exposure. The productivity, workforce-reduction, placement, and allocation figures are either hypothetical case conditions or proposed values.

15. Yield Reporting and Organizational Voice

Today's Chaekmun



If a new employee discovers yield data that contradicts a manager's conclusion just before the quarterly results announcement, should the employee report it immediately?

Gwanghaegun's 1611 chaekmun[1] did not ask candidates to list every worthwhile action. It asked them to choose what should be fixed

[1] The historical chaekmun (策問) connected to this chapter asked how to prioritize the tasks of restoring government after war. The Chaekmun Archive condenses its meaning in the reconstructed Classical Chinese passage "今之國事□何者最急□用人、賦役、田政、戸籍□孰先孰後□." This is not a literal transcription of the original. It asks, "What is most urgent in the affairs of state today? Among appointing capable people, corvée labor, land administration, and household registration, which should come first and which later?" Today's question is not a direct translation of that sentence into a modern organization. It reinterprets the same structure—setting priorities among uncomfortable facts during a crisis—and applies it to yield reporting.

first and explain the sequence. An organization approaching a quarterly announcement faces the same difficulty: keeping the disclosure timetable, rechecking a possible analytical error, and preventing incorrect figures from reaching customers and investors are all urgent at once.

A new employee's low rank does not determine whether the data are true. Nor does that mean an initial calculation should immediately be circulated as established fact. The issue in this chapter is not simply the "courage to speak." It is how to turn **reproducible evidence, limited verification time, an independent reporting line, and protection after raising a concern** into a single procedure.

The OECD Guidelines for Responsible Business Conduct call on companies to maintain risk-based due diligence and grievance mechanisms. In practice, a minority view is not automatically correct. It is an early warning that should undergo reproduction and independent verification before an irreversible announcement.

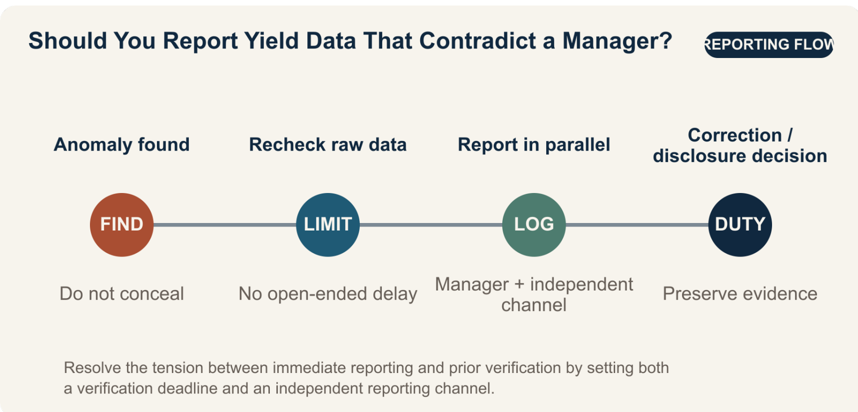
15.1. Why This Question Matters Now

Semiconductor yield affects revenue, cost, delivery schedules, and customer quality at the same time. Yet the same wafer records can produce different yield figures depending on rules for excluding product families, the treatment of reworked units, inspection stage, the definition of a good unit, and the aggregation cutoff. A difference between two figures is not enough to prove false reporting, but neither can the discrepancy be buried merely by saying that "the sample definitions differ."

The OECD's 2023 *Guidelines for Multinational Enterprises on Responsible Business Conduct* cover disclosure, human rights, employment and industrial relations, the environment, consumer interests, and science and technology, among other areas. Responsible business conduct does not end with installing a reporting mailbox. It requires an ongoing process: identify and assess adverse impacts; prevent or mitigate them; track outcomes; communicate how they were addressed; and provide remediation when needed.

A new yield analyst cannot instantly transform an entire corporate culture. What the analyst can do is preserve the raw data, freeze the computing environment, flag the rows that produce the discrepancy, and record both the verifier and the reporting deadline. Organizational voice is better measured not by the impression that "people can speak freely in meetings," but by the elapsed time across **discovery → request for verification → notification of the decision-maker → action → recurrence check**.

15.2. Data Lens



15.2.1. Evidence Table

Evidence	Primary source or calculation	Status	Significance for the debate
1	In its 2023 Guidelines, the OECD treats disclosure, human rights, employment, the environment, consumer interests, and science and technology as major areas of responsible business conduct.	Fact based on the OECD source	A concealed yield discrepancy is not merely a matter of personal etiquette; it sits at the intersection of disclosure, customer impact, and technological responsibility.

Evidence	Primary source or calculation	Status	Significance for the debate
2	This chapter's evidence framework proposes measuring organizational voice through retaliation after reporting, time to corrective action, and recurrence rates rather than merely the existence of an anonymous reporting channel.	Project analytical framework—not a statistic	Count not only reports, but whether reports lead to action and organizational learning.

Evidence	Primary source or calculation	Status	Significance for the debate
3	Separating the time allowed to review a dissenting view from the time allotted for the final decision makes it possible to design for both deliberation and speed.	Project analytical framework—not a statistic	It prevents disagreement from being erased without allowing verification to continue indefinitely.
4	In this chapter's case, the proposed announcement of 92% and the recalculation of 89.8% differ by 2.2 percentage points. Expressed as defect rates, the change is from 8% to 10.2%, a relative increase of 27.5%.	**Arithmetic using hypothetical figures for discussion**	A yield gap may look small while implying a much larger change in defect volume and cost. These are not actual company figures.

15.2.2. How to Read These Numbers

First, distinguish **percent from percentage points**. The difference between 92% and 89.8% is 2.2 percentage points, not 2.2%. Relative to a 92% baseline, the decrease is about 2.4%.

Second, the defect rate may be closer to the decision that needs to be made than the yield rate. Under the same assumptions, the defect rate rises from 8% to 10.2%, a relative increase of 27.5%. That calculation alone, however, cannot establish the financial loss. It also requires wafer input, die size, saleable grades, rework rates, and customer compensation terms.

Third, the aggregation definitions must match. If the two figures cover different product families, process stages, periods, or treatment of rework, the comparison is invalid. The first report should therefore say, "Applying two definitions to the same raw data reproduces a 2.2-percentage-point gap," rather than "The yield figure is wrong."

Fourth, the OECD material is not a statistical study of the effectiveness of whistleblowing at a specific company. This chapter applies the Guidelines' accountability process to a yield case. Reporting delay, retaliation, action-completion rate, and recurrence of the same error are **internal operating indicators proposed by this chapter**, not industry averages published by the OECD.

15.3. Competing Answers

15.3.1. Answer A — Report Immediately

Any material discrepancy discovered before an announcement should be escalated immediately, regardless of rank. Correcting an inaccurate yield figure after it has gone public can damage customer production plans, cost estimates, and investor trust at once. If the new employee remains silent, the manager may finalize a conclusion without ever seeing the conflicting data. Alerting the audit, quality, and disclosure teams with the raw data and reproduction steps attached is not disloyalty to the organization; it is a concrete form of loyalty.

This approach also has weaknesses. If one unverified analysis circulates widely, preparations for the announcement may stop, sensitive information may move beyond authorized access, and a legitimate difference in definitions may become an allegation of misconduct. “Immediately” should mean submitting the concern without delay through the designated independent reporting line—not emailing the entire company.

15.3.2. Answer B — Verify First

Because yield is highly sensitive to definitions, the finding should undergo at least a minimal second check before it is reported. Unless the raw-data hash, query version, exclusion rules, and sample period are aligned, 92% and 89.8% may describe different populations. Rather than spend the limited time before the announcement on rumor and rebuttal, two analysts should reproduce the calculation independently and summarize the source of the discrepancy on one page.

But “more verification is needed” can become the most common instrument of delay. Without a named verifier and a deadline—and if the direct manager alone controls the conclusion—a request to postpone the issue until the next quarter can function as concealment. A verify-first approach must include a time limit and explicit escalation conditions.

15.3.3. Answer C — Time-Boxed Independent Verification

I would freeze the raw data immediately, request independent reproduction within 24 hours, and at the same time notify the disclosure officer on a need-to-know basis that a “2.2-percentage-point discrepancy under verification” exists. If the gap survives an independent calculation or exceeds a customer or financial materiality threshold, I would escalate it to the audit, quality, and disclosure committee without requiring the direct manager’s consent. If the discrepancy disappears when the definitions are aligned, I would record the correction to the analysis and keep the proposed announcement.

This is not a midpoint between immediate reporting and prior verification. It is a design that **separates in advance the audience for the report, the required level of evidence, the verification deadline, and the escalation conditions**. It protects time for dissent while fixing the time for the final decision.

15.4. AI Research Design

15.4.1. Prompt for the AI

Investigate how a semiconductor manufacturer should resolve differences in yield calculations and internal disagreement. Check sources in this order: □ the company's data dictionary and yield formulas, □ raw data and query or code versions, □ approval authority across process engineering, quality, and production, □ audit and disclosure rules, and □ the OECD Guidelines for Responsible Business Conduct. Compare the populations, inspection stages, periods, and treatment of reworked and test units behind 92% and 89.8% in a table. Calculate the discrepancy in percentage points, relative yield change, and defect-rate change. Build an authority matrix showing each department's claim, evidence, and interest; the neutral mediator; stop and restart conditions; and the final approver. Put verified facts, responsible employees' claims, AI inferences, and hypothetical discussion assumptions in separate columns. Do not enter personal names, a reporter's identity, or undisclosed customer information. The conclusion should include 24-hour independent verification; conditions for pausing or maintaining the announcement; protection for dissent; decision-log fields; and a post-decision recurrence check.

15.4.2. Data Acquisition

Step	Source	Required record
1	Raw data from the MES, inspection equipment, and data warehouse	Extraction time; lot and product family; process stage; missing and duplicate values; raw-data hash
2	Yield queries and analytical code	Repository version; numerator and denominator definitions; exclusion rules; treatment of rework; execution environment
3	Financial and customer-impact data	Saleable grades; scrap and rework costs; customer compensation; revenue recognition; disclosure deadline

Step	Source	Required record
4	Internal quality, audit, and reporting policies	Recipient; independent verifier; escalation path; confidentiality; anti-retaliation protection; response deadline
5	Original OECD and regulatory documents	Document title; publication year; scope; direct URL; differences from company policy

15.4.3. Evaluating the Information

1. **Reproducibility:** Determine whether another analyst obtains the same value from the same raw data and code.
2. **Consistent definitions:** Align the definitions and aggregation periods for good units, input volume, rework, and test lots.
3. **Separate impacts:** Put the statistical discrepancy, customer quality impact, and financial or disclosure materiality in separate columns.
4. **Source grading:** Treat the system of record and approved policies as facts, responsible employees' explanations as claims, and AI explanations of causes as inferences.

5. **Security:** Give the AI only de-identified aggregates and approved documents; exclude reporter, customer, and process-recipe information.
6. **Record disconfirmation:** State both the conditions under which 92% would be correct and those under which 89.8% would be correct, and specify the test that distinguishes them.

15.4.4. Core Questions for Debate

- How widely should an unverified warning be shared to protect both accuracy and confidentiality?
- Who determines when a 2.2-percentage-point yield gap is material for disclosure, customers, or quality?
- When process, quality, and production teams use conflicting definitions, who can mediate without owning the KPI?
- What access controls would preserve a dissenting view in the decision log while separating it from the reporter's performance evaluation and placement?

15.5. Case Discussion

You joined the yield analysis team four months ago. Management's proposed announcement reports a 92% yield for the flagship product family, but your recalculation of the raw data produces 89.8% when a particular low-yield product family is included. Your manager says the difference is a sampling-definition issue that can be resolved next quarter. The presentation will be finalized in 36 hours.

All additional figures below are hypothetical assumptions for discussion.

- The quarter's input was 100,000 wafers, using a simple aggregate without adjusting for product mix.
- Independent reproduction by one analyst requires eight hours; a preliminary estimate of the financial impact requires six hours.
- A hypothetical internal policy requires notification of the quality officer within 24 hours if the yield discrepancy is at least 0.5 percentage points or the expected loss is at least KRW 500 million.
- An audit channel bypasses the direct manager, but employees worry that the manager may learn that a report was filed.

The same raw data have produced different interpretations. The process team argues that the low-yield product is still in development and should be excluded from production yield. The production team wants to keep the announcement schedule and use a provisional figure until rework is complete. The quality team argues that any product destined for customer shipment belongs in the current population. Every claim requires verification, and no team is presumed to be engaging in misconduct.

Divide roles and authority. The new analyst preserves the raw data, code, and discrepancy-producing rows. The process, production, and quality teams submit their numerator, denominator, and governing policy within eight hours. A data-governance officer who does not own the KPI mediates; the quality officer has authority to hold shipments; and the disclosure committee approves any pause or resumption of the announcement. The audit officer controls access to the decision log and to the identity of the person who raised the concern. The log records each claim, the data and code versions used, results of attempts to disconfirm it, the final authority, its validity period, and the conditions for review.

The team must choose among **maintaining the announcement and correcting it afterward, conducting a 24-hour independent verification with a conditional hold, or postponing the announcement immediately**. It should also examine hypothetical stop conditions: an aligned-definition discrepancy of at least 0.5 percentage points, an expected loss of at least KRW 500 million, or a break in raw-data traceability. The final plan should specify the work for the first eight hours; the mediation meeting at 24 hours; the announcement decision at 36 hours; departmental authority; the decision log; access to the reporter's identity; a check for adverse performance consequences; and a recurrence test after 30 days.

15.6. Sample 90-Second Response

"I choose **Answer A: report immediately through the independent channel and pause the announcement first.** The OECD's 2023 Guidelines for Responsible Business Conduct treat disclosure and responsibility in science and technology as major areas. But the case figures—92% and 89.8% yield, a 2.2-percentage-point gap, an internal threshold of 0.5 percentage points, and an expected loss of KRW 500 million—are all hypothetical. The strongest counterargument is that a legitimate difference in definitions could be escalated too quickly into an allegation of wrongdoing. I would therefore avoid circulating suspicions across the company. I would freeze the raw-data hash and code version, then submit them only to the audit, quality, and disclosure officers. A mediator who does not own the KPI would align the numerators, denominators, and governing rules within 24 hours. If the discrepancy falls below 0.5 percentage points after definitions are aligned and raw-data traceability is complete, I would record both the basis for the correction and the dissenting view, then resume the announcement. If the discrepancy remains or traceability is broken, I would confirm the postponement. Reporting does not establish an allegation; it initiates verification. The first priority is preventing an irreversible misstatement, not preserving the speed of the announcement."

15.7. Follow-Up Questions

Cross-Examination and Rebuttal

- If the process, quality, and production teams each present a different yield definition supported by policy, how should the mediator determine the final population?
- If independent reproduction returns 90.4%, would you select one number or disclose separate figures under each definition?
- If the quality officer and disclosure committee reach different conclusions, how should authority over shipment and authority over the announcement be separated?
- Which fields must appear in the decision log, and which must be kept separate from the reporter's identity?

- How can the organization protect someone whose analysis proves wrong while distinguishing an honest error from deliberate manipulation?
- If the same error recurs within 30 days after the announcement proceeds, which authority or calculation rule should change?

15.8. Sources

- OECD, *OECD Guidelines for Multinational Enterprises on Responsible Business Conduct* (2023)
- OECD, Responsible Business Conduct—Overview of Risk-Based Due Diligence and Grievance Mechanisms (accessed 2026)
- Joseon Dynasty Chaekmun Archive, Gwanghaegun 1611, "The Most Urgent Task After the Imjin War"
- Academy of Korean Studies, 1611 Examination Record

Data note: The figures 92%, 89.8%, 100,000 wafers, 0.5 percentage points, KRW 500 million, eight hours, and 36 hours are all hypothetical assumptions created for this chapter's discussion; they are not actual corporate results. Do not conflate the content of the OECD Guidelines with the case figures.

16. Fab Data: Borders and Control

Today's Chaekmun

Should fab data be localized even if keeping it in-country raises suppliers' cloud costs by 25%?

Sejong's 1447 chaekmun on law and authority[1] held that the existence of rules alone does not constitute good governance. A rule

[1]The historical chaekmun (策問) connected to this chapter asked why laws repeatedly create harms that prompt new laws, and where authority should reside. The Chaekmun Archive condenses its meaning in the reconstructed Classical Chinese passage "法立而弊生□弊生而法又立。其所以救之者何歟□政權當歸於何所歟□." This is not a literal transcription of the original. It asks, "When a law is established, abuses arise; when abuses arise, another law is established. What can remedy this, and where should governing authority reside?" Today's question does not map the institutions of power at that time directly onto data centers. Instead, it borrows the question's structure: when a rule intended to strengthen control creates new costs and concentrations of power, where should authority reside, and what remedy should be available?

that keeps data in-country can increase control, but it can also create new harms by concentrating the market among a few providers or making recovery and collaboration more difficult.

Fab data are not a single category. Process recipes and equipment parameters, defect images, predictive-maintenance logs, employee personal data, purchasing and customer information, and de-identified production statistics pose different risks when they are disclosed, altered, or unavailable. Applying one location rule to all of them may be too weak for the most sensitive data and excessive for the least sensitive.

16.1. Why This Question Matters Now

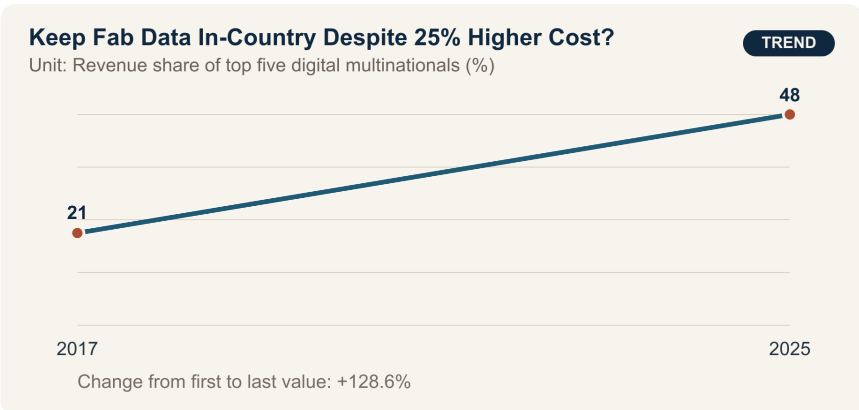
In 2025 reporting on concentration in digital markets, UNCTAD stated that the revenue share of the five largest digital multinational enterprises more than doubled from 21% in 2017 to 48% in 2025. It also reported that business e-commerce sales across 43 countries grew by about 60% between 2016 and 2022.

These figures provide context for concentration in global digital markets; they are not direct experiments showing that fab-data localization increases or decreases security. They instead signal that policy designers should examine both the power of a few global cloud providers and the risk of reconcentrating the domestic market among a few local providers.

Security can fail through encryption, key ownership, account privileges, operator access, patching, backup isolation, incident response, and the supply chain—not simply through the physical location of data. Location and jurisdiction are not irrelevant, however. They affect law enforcement during emergencies, seizure and access, requests from foreign governments, service disruption, and contract enforcement.

The practical task of a new data-governance employee is not to vote "domestic" or "overseas." It is to map data flows, classify sensitivity, and attach storage, processing, backup, cross-border transfer, key, operator, and deletion rules to each class—then test their cost and recovery time.

16.2. Data Lens



16.2.1. Evidence Table

Evidence	Figure or range in the primary source	Status	Significance for the debate
1	The revenue share of the five largest digital multinational enterprises rose from **21%** in 2017 to **48%** in 2025.	UNCTAD analysis of the global market	This is the context for concentration in cloud services, computing, and access to data.
2	Business e-commerce sales across 43 countries grew by **about 60%** between 2016 and 2022.	UNCTAD evidence cited in the project	As cross-border digital trade expands, the cost of transfer rules may affect a broader range of firms.
3	Data localization can increase control, but it can also raise the cost of cloud access for small and medium-sized enterprises.	Interpretive evidence in the project	Put the benefits of security and jurisdiction on the same scorecard as suppliers' costs and recovery.

Evidence	Figure or range in the primary source	Status	Significance for the debate
4	UNCTAD explains that access to capital, data, and computing can become barriers to entry in digital markets.	Policy analysis in the primary source	Treat access for small and medium-sized suppliers as a separate KPI when designing localization rules.
5	The case assumes that full localization raises suppliers’ costs by 25%, delays disaster recovery by three hours, and accelerates government emergency access by 24 hours.	**Hypothetical assumptions for discussion**	These are not official statistics; they are conditions for comparing the trade-offs of tiered rules.

16.2.2. How to Read These Numbers

The 21% and 48% figures are revenue shares for the five largest digital multinational enterprises as defined by UNCTAD. They must not be restated as shares of the entire cloud market or semiconductor fab software market. Revenue, assets, users, and data volume are different measures of concentration.

The increase between 2017 and 2025 shows that concentration grew; it does not identify a single cause. Economies of scale, mergers and acquisitions, network effects, AI investment, and regulation may all have contributed.

The 60% increase across 43 countries is not the growth rate for every business worldwide. The number of countries, nominal prices, currencies, and sample scope must be checked.

The security effect of localization requires separate measurement. Counting incidents alone can make an organization with better detection look worse. Examine detection time, misuse of privileges, key compromise, achievement of recovery objectives, third-party access, patch delays, independent audits, and actual harm together.

The 25%, three-hour, and 24-hour figures are case assumptions. Specify whether the cost increase uses storage cost or total IT cost as its denominator, whether recovery time means the recovery time objective or the actual average, and what lawful procedure "emergency access" entails.

16.3. Competing Answers

16.3.1. Answer A — Localize Critical Data

Process recipes, defect models, equipment control data, and security logs may directly affect manufacturing competitiveness and national security. Requiring domestic jurisdiction, domestic key management, approved operators, and isolated in-country backups can increase control over foreign service disruption and foreign legal process. It may also shorten the time needed for a joint government–industry response to an incident.

But if a system satisfies "domestic storage" while retaining weak accounts, encryption, and patching, it has only a safe address. Concentration in one domestic provider also increases common-mode failure and price-escalation risks.

16.3.2. Answer B — Permit Free Transfer Under Strong Security Controls

End-to-end encryption, customer-held keys, least privilege, immutable logs, multi-region backup, and provider portability matter more than data location. When global equipment suppliers can diagnose machinery on a shared platform, recovery is faster, and small and medium-sized firms can benefit from economies of scale. A 25% cost increase from full localization could weaken innovation and competition.

Yet encryption alone cannot eliminate nontechnical risks such as foreign jurisdiction, demands from overseas governments, sanctions, war, or network disconnection. Location and contracts still affect control and emergency access for the most sensitive data.

16.3.3. Answer C — Tiered Data Sovereignty

I would divide data into four tiers. Equipment control data, core recipes, and security keys should, in principle, be stored and processed domestically with isolated in-country backups. Sensitive quality and customer data may be transferred in encrypted form between approved countries using customer-held keys. De-identified statistics and public technical materials may use multi-region cloud services. Personal data and export-controlled data require separate legal review.

Tiers should be based on harm scenarios, not labels. Score confidentiality, integrity, availability, legal jurisdiction, recovery time, and provider switching; every exception should have an expiration date and an approver. Offer shared security services and standard transfer tools to small and medium-sized suppliers facing higher localization costs, but do not trade security exceptions for a cheaper contract.

16.4. AI Research Design

16.4.1. Prompt for the AI

> Design tiered rules for the storage, processing, and cross-border transfer of semiconductor fab data. Check sources in this order: □ the data inventory and actual flows, □ internal legal review of contracts, jurisdiction, privacy, and export controls, □ account, encryption, key, backup, and incident-response controls, □ cost,

latency, and recovery exercises, and □ UNCTAD material on concentration in digital markets. Distinguish process recipes, equipment control data, defect images, predictive-maintenance logs, employee personal data, purchasing and customer data, and de-identified statistics. Evaluate the confidentiality, integrity, and availability harm for each tier. Do not assume domestic storage automatically guarantees security; include key ownership, operator location, remote access, and backups. Separate verified facts, legal opinions, provider claims, AI inferences, and case assumptions. Do not enter original recipes, personal data, or security keys into the AI, and mark legal conclusions as requiring confirmation by responsible counsel.

16.4.2. Data Acquisition

Step	Source	What to verify
1	Data inventory and flow map	Originating equipment, fields, sensitivity, storage, processing and backup locations, recipients, retention period

Step	Source	What to verify
2	Access and security logs	Human and service accounts, key ownership, remote operators, administrator access, anomaly detection
3	Contract and legal review	Jurisdiction, subcontracting, notice of government requests, transfer and deletion, service termination, audit rights
4	Cost, performance, and recovery tests	Storage and transfer costs, latency, RTO and RPO, multi-region failure exercises, provider-switching time

Step	Source	What to verify
5	Supplier impact	Company size, incremental cost, technical staff, standard tools, security incidents, and support needs

16.4.3. Evaluating the Information

1. **Separate location from control:** Record the server address, legal jurisdiction, keys, operators, and backup location independently.
2. **Minimize first:** Remove unnecessary fields and shorten unnecessary retention periods before applying controls.
3. **Demonstrate recovery:** Examine actual disaster-recovery exercises, not only contractual RTOs.
4. **Provider concentration:** Determine whether moving to one domestic provider creates a new single point of failure.
5. **Cost denominator:** Specify whether the 25% increase applies to storage, total cloud cost, or product cost.
6. **Current law:** Have legal counsel reconfirm the original text, effective date, and exceptions of each jurisdiction's rules and contracts.

16.4.4. Core Questions for Debate

- Is data sovereignty defined by storage location, encryption keys, legal jurisdiction, or emergency-access authority?
- What costs arise if core process recipes and de-identified equipment-diagnostic statistics are governed by the same rule?
- Under what conditions should a large enterprise or government share the 25% cost borne by small and medium-sized suppliers?
- If localization strengthens a domestic provider monopoly, how can competition and security both be protected?

16.5. Case Discussion

You joined the data-governance team five months ago. An internal review estimates that full localization would raise small and medium-sized suppliers' total cloud costs by 25% and delay multi-region disaster recovery by three hours, while accelerating lawful government emergency access by 24 hours. All are hypothetical assumptions for discussion.

- The migration covers 2 PB: 5% core recipes and control data, 35% quality and defect images, 40% predictive maintenance, 10% personal and contract data, and 10% de-identified statistics.
- A dual-provider domestic architecture costs KRW 4 billion a year; the existing multi-region architecture costs KRW 3.2 billion.

- Twelve overseas equipment suppliers access predictive-maintenance logs; using a domestic relay adds an average of 90 minutes to diagnosis.
- Domestic key management and immutable logs could reduce the number of high-risk administrator accounts from 180 to 60.

The new employee must choose among **full localization, maintaining the current architecture with stronger security controls, or tiered localization with restricted transfers**. Fab operations, security, legal, small and medium-sized suppliers, overseas equipment suppliers, and the government-response team each present objections.

The final proposal should include a tiering table; storage, processing, and backup locations; key ownership; the overseas-access method; cost sharing; RTO and RPO; emergency-access procedures; exception expiration; and quarterly audit items.

16.6. Sample 90-Second Response

"I choose **Answer C: data sovereignty by tier, not full localization.** UNCTAD reports that the revenue share of the five largest digital multinational enterprises rose from 21% in 2017 to 48% in 2025. The concentration risk is real, but domestic storage alone does not guarantee security. The case figures—2 PB of data, a 25% cost increase, a three-hour recovery delay, and emergency access accelerated by 24 hours—are all hypothetical. I accept the counterargument that full localization can increase jurisdictional control and emergency authority. I would require domestic processing and isolated backups for the 5% consisting of core recipes and control data, as well as for security keys. Quality images would be split between domestic originals and pseudonymized diagnostic copies. Predictive-maintenance data would be minimized by field and made available overseas only under customer-held keys, approved sessions, and immutable logs. If a recovery exercise is more than one hour slower than the current architecture or administrator accounts exceed the target, I would redesign that tier. A single breach of key or logging requirements during overseas access would trigger an immediate switch to the domestic relay. Data sovereignty should be demonstrated through control over access and shutdown, and through recovery capability—not merely through the place of storage."

16.7. Follow-Up Questions

Cross-Examination and Rebuttal

- If core recipes make up only 5% of the data but can be reconstructed by combining other logs, how would you change the tiers?
- If two domestic providers rely on the same power grid and software, does that qualify as redundancy?
- How should the value of accelerating government emergency access by 24 hours be assessed if it ignores how often such access is actually needed?
- If a 90-minute delay in diagnosis by an overseas equipment supplier creates major fab losses, which exception should the security team allow?

- Would your conclusion change if the denominator for the supplier's 25% cost increase were storage cost rather than total IT cost?
- How should the documentation distinguish data that must be stored domestically by law from data localized voluntarily by the company?

16.8. Sources

- UNCTAD, "Highly Concentrated Digital Markets Put Consumers at Risk—Here's How to Change Course," revenue share of the five largest digital multinational enterprises (2025)
- UNCTAD, "Online Dispute Resolution Key to Boosting Consumer Trust," business e-commerce sales across 43 countries (2024)
- Joseon Dynasty Chaekmun Archive, Sejong 1447, "The Harms of Law and the Seat of Power"
- National Institute of Korean History, *Veritable Records of King Sejong*, entry for the fifth day of the eighth month in 1447

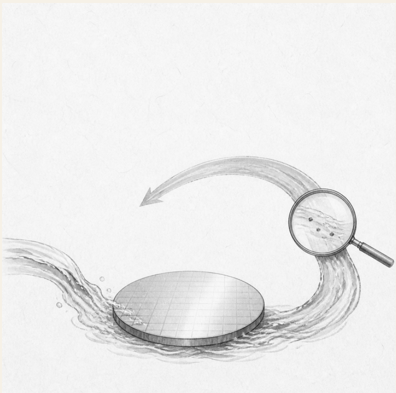
Data note: The figures 21%, 48%, 43 countries, and about 60% are market aggregates from UNCTAD. The 25% cost increase, three-hour recovery delay, 24-hour acceleration of emergency access, and all case figures concerning 2 PB, data composition, costs, equipment suppliers, account counts, and transition conditions are hypothetical assumptions for discussion. Actual legal obligations must be established by consulting the original rules in each jurisdiction and qualified legal experts.

Part IV.

Infrastructure and Sustainability

17. Ultrapure Water Reuse and Yield Risk

Today's Chaekmun



How much trace contamination and yield risk should an AI semiconductor fab accept as it increases ultrapure water reuse?

Jeongjo's 1798 chaekmun on the Honam region[1] does not equate abundant resources with improvements in residents' lives. Even

[1] The historical chaekmun (策問) connected to this chapter asked for measures that would sustain both Honam's resources and the lives of its people. The opening passage preceding the transmitted response reads "王若曰▫湖南一路卽我朝興王之地也。," or "The king says: the province of Honam is the land where our dynasty arose." Today's question is not a translation that transposes the regional policy of that era into a semiconductor plant. It borrows the underlying problem structure: when the benefits of production and the burdens of resources fall in different places, examine not only the total amount but also the distribution of the burden. See the individual entry in the Joseon Dynasty Chaekmun Archive.

if a fab reuses a great deal of water, its achievement may be overstated unless withdrawal volume, dry-season burdens, effluent quality, and yield risk are measured separately.

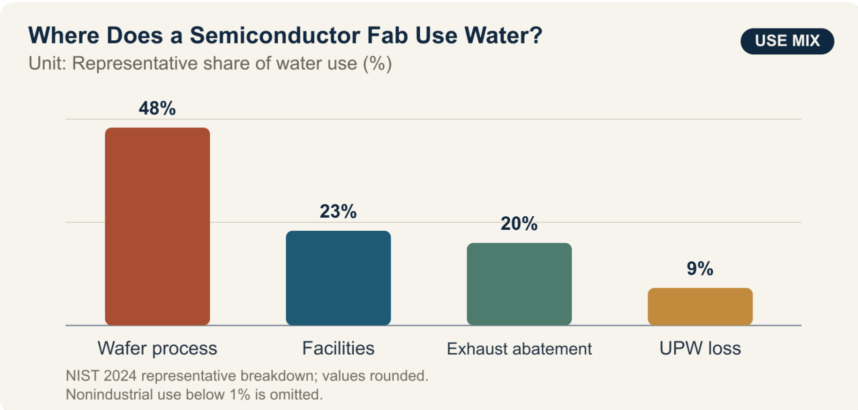
17.1. Why This Question Matters Now

Advanced logic and memory processes repeatedly wash particles, ions, and organic matter from wafer surfaces. Converting conventional water into ultrapure water (UPW), then recovering it after use, involves reverse osmosis, ion exchange, ultraviolet oxidation, degassing, and precision filtration. A higher reuse rate can reduce new withdrawals and discharge, but if contaminants become concentrated in the recovered water or monitoring develops blind spots, the result may be membrane fouling, piping contamination, and yield loss.

A 2024 NIST environmental assessment states that the average semiconductor fab recycles 40–70% of the water it receives on-site, while some large manufacturers recycle more than 80% of process UPW. The same document allocates 48% of representative water use to semiconductor processes, 23% to facility support, and 20% to exhaust-abatement systems. These figures indicate substantial scope for greater reuse; they do not mean that every effluent stream can be fed directly into final wafer cleaning.

Actual corporate pathways are also incremental. TSMC reported that it first used reclaimed water for secondary purposes, then expanded it from mature to advanced processes after validation, confirming application in 5 nm and 3 nm processes in 2024. In its 2026 report, Samsung Electronics stated that its domestic semiconductor sites reused 107.967 million tonnes of water in 2025. Reuse "volume" can count the same water each time it recirculates, so it must not be interpreted as equivalent to the reduction in new water withdrawals.

17.2. Data Lens



17.2.1. Evidence Table

Evidence	Figure or range in the official source	Significance for process and infrastructure decisions
1	NIST's 2024 environmental assessment summarizes representative water use as 48% for semiconductor processes, 23% for facility support, 20% for exhaust abatement, 9% lost in UPW treatment, and less than 1% for nonindustrial uses.	Water-quality requirements differ by use. A single reuse rate for high-purity cleaning water and cooling or scrubber makeup water conceals risk.
2	The same assessment gives a typical on-site recycling range of 40–70% of water received, and reports process-UPW recycling above 80% at some large manufacturers.	High percentages demonstrate feasibility, not the acceptance criterion for our process. Align the denominator and process scope first.
3	TSMC reported receiving 67,000 m^3 of reclaimed water per day in Tainan in 2024 and using 19.65 million m^3 during the year, reducing municipal-water use by 31% and achieving a 17% reclaimed-water substitution rate.	External reclaimed-water supply, internal recirculation, and reductions in municipal-water use are different metrics. One cannot replace another.

Evidence	Figure or range in the official source	Significance for process and infrastructure decisions
4	TSMC stated that, following validation, it applied reclaimed water to 5 nm and 3 nm processes in 2024.	Application to advanced processes shows that separation by water quality, pilots, and staged approvals are more realistic than "full deployment."
5	Samsung Electronics' 2026 report lists water reuse at domestic DS sites as 92.657 million tonnes in 2023, 101.106 million tonnes in 2024, and 107.967 million tonnes in 2025.	Higher reuse volume is an achievement, but without total withdrawals and the number of recirculation cycles, it cannot establish either a reuse rate or a reduction in regional withdrawals.
6	NIST's 2026 final environmental impact statement for Micron's New York project forecasts water demand rising from 7.85 million gallons per day in 2029 to 48 million gallons per day at full build-out in 2041.	These are forecasts for a future project plan, not current observed use. Regional infrastructure approvals should be tied to actual phased utilization and dry-season conditions.

17.2.2. How to Read These Numbers

Reuse rates cannot be compared when their denominators differ. "Recirculated volume as a share of total use," "recovered volume as a share of withdrawals," "recovery rate for process UPW alone," and "the share of municipal water displaced by externally reclaimed water" answer different questions. If the same water is counted each time it recirculates, reuse volume may exceed new withdrawals.

The chart's 48%, 23%, 20%, and 9% are a representative breakdown cited in the NIST environmental assessment. They are not proportions that can be applied unchanged to every site, process generation, cooling system, or abatement system. The 40–70% range and the figure above 80% are also industry ranges, not yield-safety limits. Micron's 7.85–48 million gallons per day are forecasts for a 2029–2041 build-out plan and must be distinguished from actual withdrawal.

A reuse-expansion proposal should connect the following measurements on a common timeline:

- Net withdrawals by source, recovered volume by process, number of recirculation cycles, and discharge volume
- Resistivity, total organic carbon, silica and boron, dissolved oxygen, and particle count at UPW points of use
- Membrane differential pressure and service life, ion-exchange regeneration cycles, and energy and chemical use
- Defect maps, yield, and process capability indices by product family and process stage
- Dry-season river flow, effects on household and agricultural water, and effluent quality

17.3. Competing Answers

17.3.1. Answer A — Accelerate Reuse Early

If recovered streams are separated by use and supported by sufficient advanced treatment, a fab can reduce new withdrawals, discharge, and the risk of water-supply interruption at the same time. Large fabs collect extensive water-quality sensor and process data, enabling early contamination detection and rapid expansion starting with low-risk uses. Where regional water stress is already high, repeatedly setting conservative targets also externalizes risk to the community.

But if management imposes a single reuse-rate target, teams may have an incentive to combine waters of different quality too aggressively or delay reporting alarms. If new withdrawals fall little while energy and chemical use and concentrated wastewater rise, the environmental gain may be smaller than expected.

17.3.2. Answer B — Defer Reintroduction into the Process

Trace contamination in advanced processes may appear as defects only after several process steps, making root-cause analysis difficult. Given the importance of customer trust and long learning cycles, reuse should remain in applications that do not contact wafers directly—such as cooling towers, scrubbers, and landscaping—while final cleaning water keeps its existing source until sufficient long-term data have accumulated.

This position also has costs. Waiting for “zero risk” prevents pilot learning from beginning and leaves the community to continue bearing the burden of new withdrawals. Existing municipal supplies also face drought, water-quality variation, and interruption, so the status quo is not risk-free.

17.3.3. Answer C — Staged Approval by Water-Quality Tier

Separate recovered water by contaminant profile, then validate it in sequence: cooling and abatement, pretreatment, initial cleaning, and final cleaning. Each stage must pass both water-quality criteria and defect-map and yield criteria. Provide dual piping and storage capacity so an anomaly automatically isolates the stream and returns the process to conventional makeup water.

A staged design is operationally complex and has a high initial capital cost. The number of approval stages must not become an end in itself. Put withdrawal reduction, total cost, carbon and chemical use, contamination events, and yield on the same review sheet, then decide whether to expand, hold, or retreat.

17.4. AI Research Design

17.4.1. Prompt for the AI

> Evaluate a proposal to increase ultrapure-water reuse at an AI semiconductor fab from 58% to 75%. Prioritize NIST environmental assessments and final environmental impact statements; materials from national and local

environmental authorities and water agencies; the latest semiconductor-company sustainability reports; and the site's original water-quality and yield data. For every figure, record the institution, document title, reference year, whether it is actual, a target, or a forecast, the numerator and denominator, water-quality tier, unit, section and page, and direct URL. Do not conflate water reuse volume, internal recycling rate, external reclaimed-water substitution rate, and reduction in new withdrawals. Analyze resistivity, TOC, silica, boron, and particles by recovered-water train; membrane differential pressure; energy and chemicals; defects and yield; and dry-season withdrawals and discharge on the same timeline. Do not infer yield causality from correlation; specify split lots, control trains, regression covariates, and statistical power. Conclude with the sequence of expansion by water-quality tier, automatic-isolation criteria, yield stop conditions, and metrics for public disclosure to the community.

17.4.2. Data Acquisition

Step	Source	Required record
1	Withdrawal, recirculation, and discharge meters	Source and train, hourly flow, net withdrawals, repeat recirculation, losses, discharge
2	UPW and recovered-water quality logs	Sampling point, resistivity, TOC, ions, silica, boron, particles, calibration, detection limit
3	Process and yield data	Lot, product, equipment, recipe, defect map, yield, rework, time lag
4	Equipment and cost data	Membrane differential pressure and life, energy, chemicals, sludge, maintenance, bypass-train capacity

Step	Source	Required record
5	Watershed and community data	Dry-season flow, household and agricultural water, withdrawal permits, effluent, resident complaints and consultation records

17.4.3. Evaluating the Information

1. **Distinguish water sources and treatment streams.** Do not combine municipal water, industrial water, external reclaimed water, internally recovered water, and UPW under a single label of "water."
2. **Fix the denominator.** Exclude from comparisons any rate that does not disclose its reuse formula and treatment of repeat recirculation.
3. **Account for time lags.** Align the process cycle between a water-quality anomaly and subsequent defects or yield changes; do not examine only a post hoc selected period.
4. **Distinguish not detected from absent.** Do not treat a value below the detection limit, a calibration failure, or missing data as normal.

5. **Examine local impact and process fairness together.** If new withdrawals do not fall, or if dry-season burdens, concentrated effluent, or energy use increase, do not declare success from reuse volume alone.

17.4.4. Core Questions for Debate

- Which train and product family would reveal a yield risk first?
- If water metrics remain within specification but yield falls, which confounding variable should be eliminated first?
- How much additional energy, chemical use, and waste should be allowed per 1 m^3 reduction in new withdrawals?
- Where should the boundary fall between indicators disclosed to the community and the company's trade secrets?

17.5. Case Discussion

You are a new engineer assigned to the infrastructure and process-integration team at a new AI semiconductor fab. The current reuse rate, measured on a total process-water recirculation basis, is 58%; the target one year from now is 75%. If total water demand is held constant at 45,000 m^3 per day, new withdrawals fall from 18,900 m^3 to 11,250 m^3 per day—a reduction of 7,650 m^3. This is a simplified assumption for discussion that ignores leakage, evaporation, repeat recirculation, and production changes.

In a 12-week pilot, the share of recovered water used as UPW makeup rose from 20% to 40%. Resistivity remained within the internal specification, but four TOC alarms occurred, and the count of particles measuring at least 20 nm was 15% higher than in the control train. Yield in the pilot lots was 0.18 percentage points lower, but with a sample of only 18 lots, neither causality nor statistical significance was established. The membrane-replacement interval shortened by 10%, while new withdrawals fell by 12%. All figures and internal specifications are hypothetical assumptions for discussion.

The production team wants to slow the expansion because the yield signal is uncertain. The infrastructure team wants to secure a withdrawal buffer before the dry season. The environmental team wants to consider energy, chemicals, and effluent concentration alongside withdrawals. No side denies the legitimacy of the others' objectives.

You must recommend one of three options: **expand immediately to 75%, stop reintroducing recovered water into the process, or expand in stages by water-quality tier**. A new employee's role is not final approval, but checking data consistency and drafting the split-lot design, automatic isolation and bypass conditions, and decision log.

17.6. Sample 90-Second Response

"I choose **Answer C: staged expansion by water-quality tier.** NIST describes typical on-site water recycling at fabs as 40–70%, with some facilities exceeding 80%, and TSMC reported applying reclaimed water to 5 nm and 3 nm processes in 2024 following validation. These facts show that high reuse is possible, but they do not set our yield-safety threshold. The case figures—an increase from 58% to 75% reuse, 45,000 m^3 per day, and a 0.18-percentage-point yield decline—are all hypothetical. The counterargument that immediate expansion can reduce withdrawals is valid, but the TOC alarms and higher particle count cannot be ignored. I would expand first in cooling, scrubbers, and noncritical cleaning. Final cleaning would advance only after 30 matched lots and eight consecutive weeks of passing water quality. The train would isolate automatically if TOC alarms occur twice in one week or particles rise more than 10% above the control train. If a yield decline of at least 0.10 percentage points relative to the control group is confirmed, the process would revert to the previous stage. I would disclose monthly net withdrawals, dry-season withdrawals, effluent quality, and yield together, so that water-saving benefits and process risks can be judged on the same scorecard."

17.7. Follow-Up Questions

- How should the numerator and denominator of the 75% reuse rate be defined so that sites can be compared?
- If resistivity is normal but TOC and particles rise, which sampling point and equipment should be isolated first?
- What additional statistical information is needed to decide whether to stop expansion when yield is 0.18 percentage points lower across 18 lots?
- If new withdrawals fall but membrane replacement and energy and chemical use rise, in what unit should the total effect be compared?

- If regional withdrawals approach their dry-season limit, what advance rule should govern production volume, the reuse stage, and residential supply?
- If contamination recurs in production after eight passing weeks, what automatic action should precede any assignment of blame?

17.8. Sources

- NIST, *Final Programmatic Environmental Assessment for Modernization and Expansion of Existing Semiconductor Fabrication Facilities* (2024)
- NIST, *Micron Semiconductor Manufacturing Project, Clay, New York—Final Environmental Impact Statement* (2026)
- TSMC, *2024 Sustainability Report* (2025)
- TSMC, *2024 Annual Report*, Water Management (2025)
- Samsung Electronics, *Sustainability Report 2026* (2026)
- Samsung Semiconductor, “MOU for Phase 1 Water-Reuse Project at Semiconductor Sites in the Gyeonggi Region”
- Joseon Dynasty Chaekmun Archive, Jeongjo 1798, “Seven Problems in Honam and Their Remedies”
- Honam Institute for Korean Studies, “Gwangju Hyanggyo and the Cultivation of Gwangju’s Public Morality” (2021)
- National Institute of Korean History, *Annals of King Jeongjo*, entry for the 18th day of the sixth month in Jeongjo’s 22nd year (1798)

Data note: The chart's 48%, 23%, 20%, and 9%, along with the 40–70% range and figure above 80%, come from NIST's 2024 environmental assessment. The TSMC figures are 2024 results for Tainan, the Samsung Electronics figure is a 2025 result for domestic DS sites, and the Micron figures are forecasts for 2029–2041. All case values concerning water demand, reuse rates, water quality, yield, membrane life, approval, and stop conditions are hypothetical and are not the actual operating values of any particular company.

18. The Paradox of Compute Efficiency and Total Electricity Use

Today's Chaekmun

If a new AI chip reduces energy per computation by 40% but service usage triples, can it still be described as environmentally sustainable?

A Jungjong-era chaekmun on restraint[1] did not call for eliminating something useful. It asked how to control the point at which a beneficial use becomes excessive in aggregate.

AI computing presents the same problem. A chip that produces the same answer with less electricity is a genuine technical advance. But if cheaper computation drives a sharp increase in advertising recommendations, video generation, search, and agent calls, total data-center electricity consumption and peak demand may still rise. Efficiency and total consumption are not competing metrics; they answer different questions.

The IEA's *Energy and AI* analyzes AI demand as a system extending beyond servers to data centers, electricity grids, and generation sources. An efficiency assessment should therefore not end at the chip data sheet. It should extend to service request volume and total data-center electricity consumption.

18.1. Why This Question Matters Now

In the IEA's 2025 *Energy and AI* base case, global data-center electricity consumption more than doubles from about 415 TWh in 2024 to about 945 TWh in 2030. That is just under 3% of global electricity

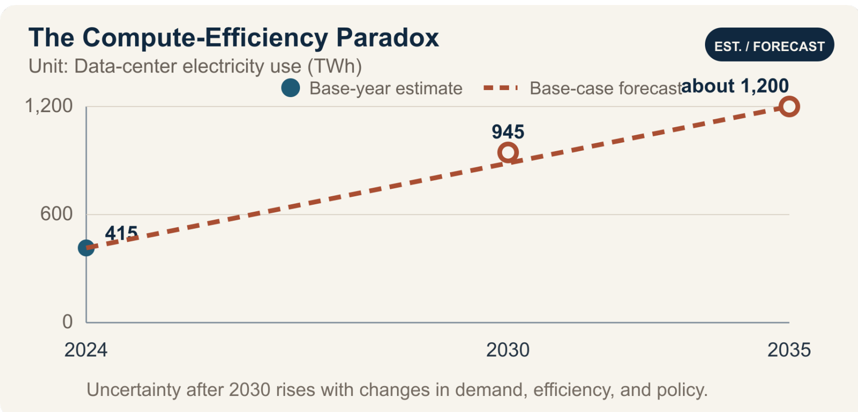
[1] The historical chaekmun (策問) connected to this chapter asked why wine, despite its beneficial use in ritual and fellowship, becomes an abuse that harms both state and household, and how it should be restrained. The Chaekmun Archive condenses its meaning in the reconstructed Classical Chinese passage "酒有禮用□而其弊敗國喪家。何以節之□." This is not a literal transcription of the original. It asks, "Wine has its proper ritual uses, yet its harms can ruin the state and destroy the household. How should it be restrained?" This chapter does not equate wine with AI. It borrows the question's structure: acknowledge a useful technology while designing limits for its total use.

consumption in 2030. The IEA projects that electricity consumption by accelerated servers will grow by about 30% per year and account for almost half of the net increase.

This forecast does not mean that every AI chip consumes the same amount of electricity or that conditions are identical in every country. Data centers cluster according to grids, climate, cooling methods, semiconductor supply, and service demand. Even if the global share remains below 3%, new loads can substantially affect interconnection, electricity prices, water use, and reserve capacity in particular regions.

A new sustainability analyst should not attach a green label based only on a chip's TOPS/W. The analysis must connect measured server power, utilization, memory and networking, total facility power including cooling, task quality per request, total request volume, and hourly electricity use and carbon intensity.

18.2. Data Lens



18.2.1. Evidence Table

Evidence	Figure or range in the primary source	Status	Significance for the debate
1	Global data-center electricity consumption is estimated at about **415 TWh** in 2024.	IEA base-year estimate	This is total data-center consumption, not AI computing alone.
2	The IEA base case forecasts about **945 TWh** in 2030.	Conditional forecast	The result changes if policy, efficiency, demand, or supply changes.
3	The 945 TWh forecast for 2030 is **just under 3%** of global electricity consumption.	Global share in the IEA base case	Even if the global average appears small, regional grid impacts require separate analysis.

Evidence	Figure or range in the primary source	Status	Significance for the debate
4	Electricity consumption by accelerated servers is projected to grow by **about 30% per year** through 2030 and account for almost half of the net increase in data-center electricity use.	IEA technology-level forecast	Track both AI server efficiency and deployment volume.
5	The IEA base case puts data-center electricity consumption at **about 1,200 TWh** in 2035.	Long-term conditional forecast	The IEA also warns that uncertainty increases beyond 2030.

Evidence	Figure or range in the primary source	Status	Significance for the debate
6	In the case, a 40% reduction in energy per computation means each request uses 60% of the previous amount. If request volume triples, total electricity use becomes 0.6×3=1.8 times the previous level—an **80% increase**.	**Arithmetic using hypothetical figures for discussion**	Efficiency gains alone do not guarantee lower total electricity consumption.

18.2.2. How to Read These Numbers

TWh measures energy consumed over a period; GW measures instantaneous or contracted capacity. Do not restate 945 TWh as 945 GW. Converting annual energy to average power requires dividing by the hours in a year, but actual peak demand will exceed the average.

The 415 TWh figure is a 2024 base-year estimate; 945 TWh and about 1,200 TWh are base-case forecasts for 2030 and 2035, respectively. The three points must not be read as equally certain observed results. Total data-center electricity use includes servers, cooling, power conversion, and other loads, and it is difficult to separate AI electricity fully from traditional workloads. A separate IEA supply analysis reports 460 TWh of generation in 2024; that figure exceeds consumption because it accounts for transmission and distribution losses, so the two values should not be combined.

The annual 30% rate is compounded growth, not the same absolute increase each year. "Almost half of the net increase" is also different from "half of total consumption."

The calculation $0.6 \times 3 = 1.8$ is a simplified scenario for this chapter. It assumes no change in task quality, model size, batching, caching, request length, server utilization, or cooling. In reality, a more efficient chip may be used for a larger model or higher quality, changing the amount of computation per request itself.

Carbon is not the same as electricity consumption. The emissions from 1 MWh depend on the generation mix at its time and location. Examine hourly carbon-free energy matching, peak electricity demand, and additional generation capacity together.

18.3. Competing Answers

18.3.1. Answer A — Efficiency-Led Innovation

A 40% reduction in energy per computation lowers cost, electricity use, and cooling demand at the same service level. AI can also create benefits in medicine, science, and industrial optimization, so rising usage should not automatically be called waste. Rapid deployment of efficient chips can replace older servers and support more valuable work within a constrained grid.

But it is necessary to verify whether the “same service level” is actually maintained. If every efficiency gain is spent on larger models and unlimited automated calls, total consumption and peak demand will rise. Product-level metrics cannot substitute for the environmental performance of the facility.

18.3.2. Answer B — Cap Total Consumption

Environmental impacts arise from final electricity consumption, water use, and emissions. If total electricity rises 80% despite lower energy per request, calling the outcome green is difficult. Establish a data-center electricity budget, peak-demand cap, and restrictions on nonessential inference first, then treat efficiency as a means of allocating use within those limits.

But a fixed cap that treats the social value of every service as equal may also block worthwhile new applications. The cap could become a barrier to entry that favors incumbent providers.

18.3.3. Answer C — Two-Ledger Approval

I would require both energy per computation and total service energy in chip approval. The product team would be responsible for electricity per standard task at equal quality; the service team would be responsible for monthly inference volume, total electricity use, and peak demand. Set a budget that allows only part of the efficiency savings to fund demand growth and preserves the rest as an absolute reduction.

Do not stop at the annual volume of renewable electricity purchased. Put hourly carbon-free energy, additionality of new generation, grid congestion, demand response, and emergency curtailment into the contract. If total electricity exceeds the approved budget or carbon intensity during peak hours exceeds the threshold, delay low-value requests and lower the cap on automated calls.

18.4. AI Research Design

18.4.1. Prompt for the AI

> Analyze how a 40% reduction in energy per computation for a new AI chip and a projected tripling of service usage would affect total data-center electricity, carbon, and peak demand. Check sources in this order: □ server measurements under an equal-quality benchmark, □ actual request logs with model, token, and batch data, □ facility meters, PUE, and cooling, □ hourly electricity and carbon intensity and power contracts, and □ the original IEA *Energy and AI* report. Compare chip, memory, network, and facility electricity for the baseline and new servers using the same functional unit. Calculate total electricity and peak demand under usage sensitivities of 1×, 2×, 3×, and 5×. Distinguish measurements, modeling, company forecasts, IEA forecasts, and hypothetical

discussion assumptions. Do not equate the purchase of renewable-energy certificates with actual hourly carbon-free supply. Conclude with an efficiency KPI, total electricity budget, peak reduction, demand response, and conditions for withdrawing approval.

18.4.2. Data Acquisition

Step	Source	What to verify
1	Server benchmarks	Model, accuracy, latency, batch size, input and output length, chip, memory, and network power
2	Service request logs	Request count, tokens or image size, retries and cache, automated calls, value by customer and function
3	Facility metering	Server and facility electricity, PUE, cooling water, peak demand, time and location

Step	Source	What to verify
4	Power procurement	Grid electricity, PPA, certificates, additionality, hourly matching, demand-response obligations
5	Original IEA and grid sources	Scenario, scope, base year, regional concentration, forecast uncertainty

18.4.3. Evaluating the Information

1. **Functional equivalence:** Confirm that the systems process the same quality, latency, and amount of work.
2. **Consistent boundary:** Record chip, server, and facility electricity at separate layers rather than mixing them.
3. **Check the total:** Multiply unit efficiency by actual usage to recalculate total electricity use.
4. **Time and location:** Separate annual averages from peaks, and regional carbon intensity from grid constraints.
5. **Additionality:** Determine whether the power purchase resulted in new carbon-free generation.
6. **Cause of rebound:** Distinguish demand growth driven by price, new features, and repeated automated calls.

18.4.4. Core Questions for Debate

- Should the denominator of an environmental claim be a computation, a useful task, a customer, or total facility electricity?
- How much of the efficiency saving should fund greater demand, and how much should remain as an absolute reduction?
- Who should have authority to classify, delay, or stop a low-value request: the product team or the customer?
- If the annual renewable-energy purchase is unchanged but peak-hour electricity comes from fossil generation, has the target been met?

18.5. Case Discussion

You are a new sustainability analyst for an AI server product. In the same benchmark, the new chip uses 40% less energy per computation, but customer inference volume is projected to triple because of lower prices and new features. All figures below are hypothetical assumptions for discussion.

- The existing service uses 100 GWh of facility electricity per year. If request volume alone triples, the new equipment uses 180 GWh under a simple calculation.
- Current peak demand is 20 MW, and the grid operator requires a KRW 20 billion interconnection-upgrade payment if demand exceeds 25 MW.
- The annual renewable-electricity contract covers 120 GWh, but actual request peaks occur between 6 p.m. and 10 p.m.

- Delay-tolerant batch work accounts for 30% of total electricity use, and half of that load can be shifted to daytime hours.

The new employee must choose among **launching on efficiency performance alone, launching conditionally with a 150 GWh annual cap and demand management, or deferring until demand is confirmed**. The product team owns performance; the service team owns request value; the data-center team owns peak demand; procurement owns power contracts; and finance owns the upgrade cost.

The final contract should include energy per computation at equal quality, monthly total electricity, the 25 MW peak, hourly carbon-free share, batch shifting, request caps, customer notification, and reapproval conditions.

18.6. Sample 90-Second Response

"I choose **Answer C: a conditional launch with a 150 GWh total-electricity cap.** In the IEA base case, data-center electricity consumption rises from about 415 TWh in 2024 to 945 TWh in 2030, while electricity use by accelerated servers grows by about 30% per year. This is real-world evidence that efficiency alone does not guarantee lower total consumption. The case figures—a 40% reduction in energy per computation, triple the requests, an increase from 100 GWh to 180 GWh in total electricity, and a 25 MW peak—are all hypothetical. I accept the counterargument that greater usage may provide more high-value computing. I would shift delay-tolerant work to daytime hours and restrict low-value automated calls when peak demand reaches 25 MW. Six months after launch, I would expand if electricity per useful task has fallen by at least 30% and total electricity remains within 150 GWh. If the cap is exceeded by 5% for two consecutive months or the peak is breached, I would stop new automated calls and require reapproval. I would also publish the hourly carbon-free share to keep total electricity and carbon distinct. An environmental claim should apply not to a fast chip alone, but to a service that controls both efficiency and total demand."

18.7. Follow-Up Questions

Cross-Examination and Rebuttal

- If the tripled request volume supports high-value work such as medical diagnosis, should it be allowed to exceed the 150 GWh cap?
- Would it be arbitrary for the product company to define the value of a "useful task"?
- If PUE improves but memory power outside the chip rises, how should energy efficiency per computation be recalculated?
- If the company purchased all 120 GWh of annual renewable electricity but the evening peak relies on fossil generation, how should its carbon claim change?
- If customers reject usage caps, which instrument should be used first: price, delay, or contract terms?
- If total electricity stays within 150 GWh but local water use and noise rise, should the launch still be considered successful?

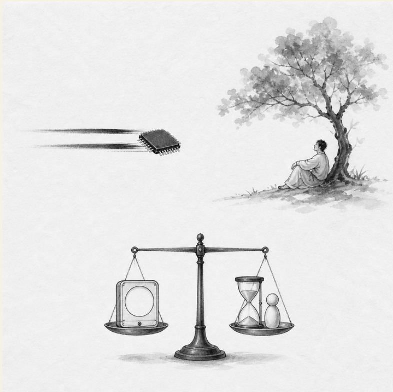
18.8. Sources

- IEA, *Energy and AI—Energy Demand from AI* (2025)
- IEA, *Energy and AI—Executive Summary* (2025)
- IEA, *Global Data Centre Electricity Consumption by Sensitivity Case, 2020–2035* (2025)
- IEA, *Energy and AI—Energy Supply for AI* (2025)
- Joseon Dynasty Chaekmun Archive, Jungjong-era "Discuss the Harms of Wine"
- Encyclopedia of Korean Culture, Gim Gu and the Jungjong-era Examination Response

Data note: The 415 TWh figure is the IEA's 2024 base-year estimate; 945 TWh and about 1,200 TWh are base-case forecasts for 2030 and 2035. The figure just under 3% and the annual 30% growth rate are also elements of the IEA forecast. The IEA supply analysis reports 460 TWh of generation in 2024 to serve data-center demand. The 40% reduction, tripled demand, and the case figures of 100, 180, and 150 GWh; 20 and 25 MW; KRW 20 billion; 120 GWh; 30%; and all transition conditions are hypothetical assumptions for discussion.

19. Product Success Beyond Performance

Today's Chaekmun



Even if a new chip doubles performance, can it be considered a success if electricity use, replacement cost, and work intensity all rise?

Jungjong's 1515 chaekmun on ideal government[1] did not stop at listing the elements of a good society. It required candidates to state

[1]The historical chaekmun (策問) connected to this chapter asked what should come first when putting an ideal into practice in present circumstances. The Chaekmun Archive condenses its meaning in the reconstructed Classical Chinese passage "欲致堯舜之治於今日□何者爲先□子若孔子□將何以治國歟□." This is not a literal transcription of the original. It asks, "If the government of Yao and Shun were to be realized today, what should come first? If you were Confucius, how would you govern the state?" Today's question does not translate ideal government into product KPIs. It borrows the question's structure: do not stop at praising a desirable condition; state real priorities and criteria for implementation.

what should come first and by what results the ideal would be verified.

A product can likewise be declared successful with one sentence: "Its performance is higher." But benchmark performance and improvements in life diverge if users do not spend less time waiting, if electricity, cooling, and replacement costs rise sharply, or if higher throughput merely results in more work being assigned.

OECD indicators measure labor productivity and working hours separately. Product evaluation should also look beyond benchmark performance and independently assess user time and health, electricity and cooling, and replacement costs. Only then can we tell whether good components have produced a good system in practice.

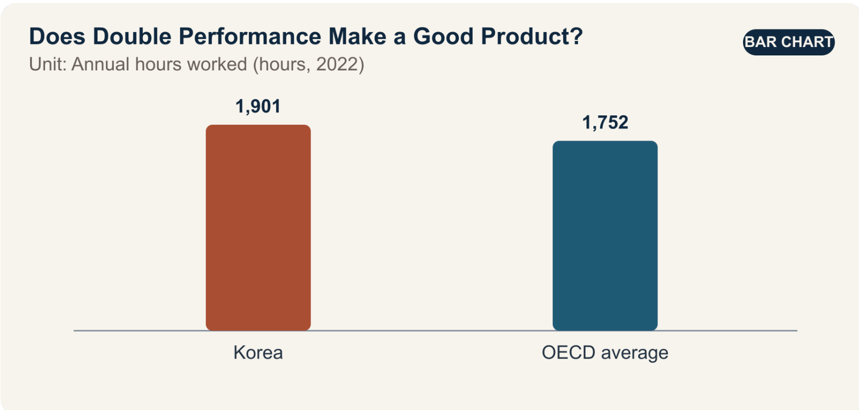
19.1. Why This Question Matters Now

In the comparison cited by the *OECD Economic Surveys: Korea 2024*, Korea's annual hours worked in 2022 were 1,901—149 hours, or about 8.5%, more than the OECD average of 1,752 hours. This does not mean that every worker in Korea worked exactly 1,901 hours; it is a national-average comparison.

OECD productivity data also show that Korea's hourly labor productivity rose from $13.9 in 1995 to $52.0 in 2024—almost quadrupling. The unit is constant 2020 US dollars at purchasing power parity. Productivity growth creates a foundation for higher living standards, but it does not mean that working hours, health, distribution, and the environment automatically improved.

A new engineer in a product-review meeting is not expected to discount performance. The engineer should trace how higher performance translates into outcomes: Does benchmark performance reduce actual waiting and completion time? What are system electricity use and total cost of ownership? How much earlier will existing equipment be discarded? Did night and emergency work increase? Do older users and users with disabilities share in the benefit?

19.2. Data Lens



19.2.1. Evidence Table

Evidence	Figure or range in the primary source	Status	Significance for the debate
1	Annual hours worked in Korea were **1,901 hours** in 2022.	OECD national-average statistic	This is social context for asking whether technological benefits actually reduce users' time.
2	The OECD average in the same year was **1,752 hours**. Korea's figure was 149 hours, or about **8.5%**, higher.	Arithmetic comparison of OECD figures	Institutions and industry structures differ across countries, so the gap must not be linked directly to chip performance.
3	Korea's hourly labor productivity rose from **$13.9** in 1995 to **$52.0** in 2024.	OECD, constant 2020 PPP dollars	A near quadrupling of productivity does not mean every quality-of-life indicator changed at the same rate.

Evidence	Figure or range in the primary source	Status	Significance for the debate
4	The case assumes double the benchmark performance, 35% more system electricity, 20% higher replacement cost, and an 8% reduction in actual work time.	**Hypothetical assumptions for discussion**	The case practices decisions in which performance, cost, and user outcomes move in different directions.
5	If performance is represented as 2 and system electricity as 1.35, theoretical performance per unit of electricity is about **1.48 times** the baseline.	**Arithmetic using hypothetical figures for discussion**	Data-sheet efficiency improves, but it is a different metric from an 8% reduction in actual work time.

19.2.2. How to Read These Numbers

Annual hours worked are an average. The distribution differs by employment status, gender, occupation, and income, and long hours cannot be explained by semiconductors or AI alone. The national average is not a direct KPI for product success; it is a reason to measure user time separately.

The hourly labor-productivity figures of \$13.9 and \$52.0 are not dollars converted at nominal exchange rates; they are in constant 2020 PPP terms. The ratio is about 3.74, so "almost four times" is more accurate than "exactly four times." Productivity measures output per labor hour; it is not a unit of pay, life satisfaction, or environmental quality.

A twofold benchmark result depends on the task, precision, power limit, and software. Separate maximum performance, sustained performance, and average user task time. The calculation $2/1.35 \approx 1.48$ also applies only to a scenario in which maximum performance and system electricity are measured under the same conditions.

An 8% reduction in work time does not automatically create 8% more leisure. Determine whether the saved time became additional work, waiting, reporting, training, or rest. Measure team workloads, nighttime calls, and error-correction time alongside it.

19.3. Competing Answers

19.3.1. Answer A — Performance Leadership

Computing performance is a general-purpose foundation that expands the possibilities of science, manufacturing, medicine, and creative work. If performance doubles and performance per unit of electricity rises 1.48 times, organizations can solve larger problems in the same time and reduce the number of servers required. Even if electricity use and replacement costs rise initially, new features, revenue, and shorter customer waiting times may create long-term benefits.

But mistaking a peak benchmark for actual demand leads to premature replacement and idle capacity. Unless the organization verifies that saved time was returned to users, performance remains an internal target.

19.3.2. Answer B — Human Outcomes First

If actual work time falls by only 8% while electricity rises 35% and replacement costs rise 20%, it is difficult to call a twofold performance gain a success. Users' lives should improve when work intensity, night work, health, affordability, product life, and disposal are included. Keeping an existing product whose performance is sufficient can also be innovation.

But delaying a launch until every long-term human outcome is known may forfeit needed performance gains and market opportunities. A focus on time savings alone can also miss qualitative benefits such as higher medical accuracy or research that was not previously possible.

19.3.3. Answer C — Multiple Thresholds

Instead of allowing performance, electricity, cost, user, and environmental outcomes to offset one another in a single weighted average, set minimum thresholds first. Safety and reliability, improvement in real core work, power and thermal limits, total cost of ownership, accessibility, and product life must all pass before a weighted product-family score is calculated. Even extremely high performance does not justify a launch that fails a safety or grid threshold.

After launch, examine who receives the saved time. If work time falls but workloads and nighttime calls rise, change the organizational process; if actual utilization remains low, defer replacement. Accept higher costs for customers whose research or medical work delivers exceptional benefits, but do not force the same replacement cycle on general office work.

19.4. AI Research Design

19.4.1. Prompt for the AI

Design a dashboard that evaluates the success of a new semiconductor product through actual user, cost, and environmental outcomes as well as benchmark performance. Check sources in this order: □ chip and system benchmarks under equivalent conditions, □ customer work logs and before-and-after time studies, □ electricity, cooling, failure, replacement, and disposal data, □ price, total cost of ownership, accessibility, and health surveys, and □ original OECD sources on working hours and productivity. Mark double performance, a 35% increase in electricity, a 20% increase in cost, and an 8% reduction in work time as hypothetical case assumptions, and conduct sensitivity analysis by function and customer group. Present distributions and the worst-affected group as well as averages, and distinguish whether saved time became additional work or rest. Put verified facts, company claims, user observations, AI inferences, and forecasts in separate columns. De-identify safety and personal logs, and test causality through a comparison group or phased rollout.

19.4.2. Data Acquisition

Step	Source	What to verify
1	Chip and system testing	Task, accuracy, sustained performance, chip, memory, and cooling power, temperature, errors
2	User time-and-motion study	Type of work, before-and-after completion time, waiting and correction, throughput, nighttime calls, learning time
3	Economic data	Product, installation, training, electricity, cooling, maintenance, downtime, replacement, residual value

Step	Source	What to verify
4	Access, health, and quality	Adoption by price tier, disability access, fatigue and stress, errors, customer outcomes
5	Life cycle and environment	Manufacturing, transport, use, disposal, actual service life, early replacement, take-back and reuse

19.4.3. Evaluating the Information

1. **Equivalent conditions:** Align benchmark accuracy, software, and power limits.
2. **Real-world use:** Separate maximum performance from customer utilization and completion time.
3. **Distribution:** Break down average benefits and costs by user group, shift, and company size.
4. **Destination of saved time:** Determine whether saved time became rest, additional production, meetings, or reporting.
5. **Full life cycle:** Check whether lower use-phase electricity is offset by manufacturing or premature disposal.

6. **Causality:** Distinguish the chip's effect from other institutional and demand changes before and after adoption.

19.4.4. Core Questions for Debate

- Which safety, electricity, cost, or time thresholds cannot be offset by higher performance?
- How can the organization determine whether an 8% reduction in actual work time improved user well-being or merely enabled more work?
- If researchers capture the performance benefit while society bears the electricity cost, what should be reflected in the price?
- Is it fair to use different success KPIs for different customer groups?

19.5. Case Discussion

You are a system engineer seven months into the job, attending a new-product review meeting. The new chip doubles performance on a core benchmark, but system electricity rises 35%, replacement costs rise 20%, and pilot users' actual work time falls 8%. All figures below are hypothetical assumptions for discussion.

- The pilot observed 60 design-validation employees for eight weeks, with a comparison group of 60 using existing equipment.
- Median utilization of the new chip was 42%, while utilization among the top 10% of users was 78%.

- Nighttime error calls from automated jobs rose from 10 to 16 per week, while daytime waiting fell by an average of 50 minutes.
- Existing equipment has two years of remaining accounting life, and replacing 1,000 units early would add KRW 12 billion in cost.

The new engineer must recommend one of three options: **full launch, limited launch for high-workload customer groups, or deferral for one product generation**. The performance team, user representative, finance, sustainability, safety and quality, and sales each propose KPIs.

The final proposal should include non-offsettable minimum thresholds, weights, pilot bias, early replacement, actual time savings, nighttime calls, customer-specific launch scope, six-month reevaluation, and stop conditions.

19.6. Sample 90-Second Response

"I choose **Answer C: a limited launch only for high-workload customer groups.** OECD data show that annual hours worked in Korea were 1,901 in 2022, about 8.5% above the OECD average of 1,752. This is social context for measuring technological performance and user time separately. The case figures—double chip performance, 35% higher electricity, 20% higher replacement cost, an 8% reduction in work time, and 42% utilization—are all hypothetical. I accept the counterargument that delaying peak performance risks losing a competitive opportunity. But a faster benchmark does not make a successful product if actual work and total costs do not improve. I would launch only for customers expected to achieve at least 60% utilization, and I would not replace existing equipment early. The review would combine safety and errors, actual work time, total cost of ownership, electricity and service life, and the benchmark. After six months, I would expand if work time has fallen by at least 10% and nighttime calls and electricity increases remain within their thresholds. If any threshold fails, the product would be modified and reviewed again. A good chip is not merely a faster chip; it is one that demonstrably improves users' time and society's costs."

19.7. Follow-Up Questions

Cross-Examination and Rebuttal

- If the quality of medical or scientific work improves substantially, would you broaden the launch even if work time falls by only 8%?
- If the 60 pilot users were all high-performance users, how would you adjust the observed 8% time saving?
- If nighttime calls rise by 60% but daytime waiting falls by 50 minutes, which time should receive greater weight?
- Would a benchmark weight of only 10% understate a semiconductor company's core competitiveness?
- If existing equipment has two years of life remaining but competitors adopt the new product, would you allow early replacement?

- If the company captures the productivity gain while users and communities bear the electricity and health costs, what should be included in price and compensation?

19.8. Sources

- OECD, *OECD Economic Surveys: Korea 2024*—Working-Hours Comparison (2024)
- OECD, *Compendium of Productivity Indicators 2026*—Long-Term Change in Korea's Hourly Labor Productivity (2026)
- Joseon Dynasty Chaekmun Archive, Jungjong 1515, "How to Make Ideal Government a Reality"
- Academy of Korean Studies, 1515 Special Palace Examination Roster

Data note: The 1,901-hour and 1,752-hour figures are 2022 OECD comparisons; the $13.9 and $52.0 productivity figures are for 1995 and 2024 in constant 2020 PPP dollars. The doubled performance, 35% more electricity, 20% higher cost, 8% time saving, and all case figures concerning the pilot, utilization, calls, replacement cost, and sample-answer KPIs are hypothetical assumptions for discussion.

Epilogue — When Data Becomes a Sentence

An engineer's day often resembles a crime scene.

A defect has occurred, but the culprit is nowhere to be seen. We establish when and where it appeared, search the process history, and examine changes in materials and equipment. We collect measurements, draw time-series charts, separate the mean from the variation, and look for outliers. We even suspect that two factors may have acted together.

Then we gather in a conference room and construct possible sequences of events.

- Was the material the principal cause?
- Were the process conditions an accomplice?
- Did a change in equipment pull the trigger?
- Or is there a variable we have not yet measured?

Years ago, in a Brunch Story essay titled "Find the Culprit", I wrote that engineers and detectives have something in common.[2] A detective gathers evidence to reconstruct a crime; an engineer gathers data to reconstruct the path by which a defect emerged. The detective arrests the culprit. The engineer removes the cause and designs a countermeasure so that the same problem does not return.

At the end of that process, there was always writing.

[2]Kakao describes Brunch Story as its writing and content-publishing platform. Kakao, "Kakao Story Services", January 9, 2024.

The job was not finished when the chart was drawn. Even after calculating regression coefficients and checking statistical significance, the analysis remained trapped on one person's computer if colleagues could not understand what it meant. Data needs sentences before it can move into organizational judgment.

- What happened?
- What do we know?
- What do we still not know?
- What is the most plausible explanation?
- Which experiment could show that explanation to be wrong?

Answering those questions is an engineer's form of writing.

Writing exposes gaps in thought. A causal relationship that seemed obvious in the mind becomes uncertain when it is put into a sentence. Missing logic appears between two charts. Something we treated as a fact may reveal itself to be no more than an estimate drawn from experience.

Writing is therefore not decoration added after problem-solving is complete. It is **the final verification step in the problem-solving process**.

Why Engineers Should Write

For many years, I have written in the Brunch Story magazine "Stories of a Display Engineer" about technology and people, organizations and the future. I wrote not only about what I learned while developing LCD, OLED, and Micro LED technologies, but also about organizational culture, diversity, generational change, and AI.

Writing taught me that engineers do not work with matter alone. People who create technology must also reckon with the humans who use it and the society in which it operates. The performance of semiconductors and displays can be described in numbers. Numbers alone cannot decide whose lives those technologies will make easier, who will bear their costs, or what kind of future they will create.

The first reason engineers should write is that writing tests their own thinking.

The second is that it leaves experience behind as organizational memory. Knowledge that disappears when one person goes home or changes departments remains a private memory, not a technical asset. Failed experiments and mistaken hypotheses must also be recorded, so that the next person does not have to lose their way on the same road.

The third is that writing connects people from different disciplines. A process engineer, a device researcher, a data analyst, and an executive can look at the same graph and read different meanings. Writing builds a bridge between the languages they use.

There is one more reason.

Writing is a pleasure.

After staring at numbers and defects all day, the expression on the work changes when I begin to write. A data point labeled an "outlier" in a report becomes an event that happened on a particular day in the fab. Behind one sentence noting increased variation, materials, equipment, people, and schedules appear, moving in relation to one another.

For a moment, the engineer ceases to be only a person who calculates and becomes someone who discovers a story.

This pleasure is different from the satisfaction of finding the correct answer. It comes from knowing that the time I have lived and the lessons I learned on the manufacturing floor will not disappear, but will remain in a sentence. It is also the pleasure of rediscovering familiar technology from an unfamiliar angle.

What Remains After a Reorganization

When a reorganization is announced, only a few boxes and lines change on the organization chart. For the people inside those boxes, accumulated roles and relationships are being severed. When I had to leave an unfinished project behind, I sometimes thought of Admiral Yi Sun-sin serving as an ordinary soldier after disgrace and then returning to command at Myeongnyang, or of unnamed people who crossed strange borders along the Silk Road. I did not compare myself to a hero. I was trying to understand the feeling of having to begin the next day's work even when orders and relationships have become uncertain.

I first recorded these emotions in the Brunch Story essays "Reorganization", "Serving in the Ranks ... and Dreaming of Myeongnyang", and "A Tactical Retreat". Personal experience is not objective evidence about an industry. It can, however, become a record that keeps failed hypotheses and hard-won lessons from disappearing when a project stops. I leave a trace of that feeling in this book not to tell applicants my personal story, but to remember that every technical decision is also work performed by people.

Practicing How to See Another World with AI

A weekly AI study group with colleagues gave me the same pleasure. With AI, we traveled outside our own specialties. We interpreted historical questions through engineering and looked at technical problems through sociology. We did not accept AI's first answer. We asked again, produced objections, and compared our interpretations.

In that group, a younger colleague who works for a semiconductor and display-equipment company once gave an unexpected answer—0.864 milliseconds—to a king's question about nature's cycles and the finitude of human life.

It was not the correct answer. It made us laugh, made a familiar question strange again, and showed that one question could travel freely across the humanities, physics, astronomy, and signal processing.

AI took part in finding material for this book, expanding its questions, and producing counterarguments. Humans still had to decide what to ask, which sources to trust, and which values mattered most when they collided.

From Brush-Written Answers to a Web Page

While making this book, I also created the Joseon Royal Policy Question Archive so that contemporary readers could explore the *chaekmun* directly. It brings together questions posed by kings, answers written by candidates, and material that helps readers reconsider those questions in today's language.

Questions asked by kings centuries ago no longer end inside Classical Chinese records. They return on a web page built with data and code. Answers once written with a brush appear in a browser, and the royal policy questions of the civil service examination continue into the semiconductor industry.

The process itself was a joyful form of writing for me.

I read questions from the past, found data from the present, developed objections with AI, and wove charts and sentences together with R and Quarto. Writing and data analysis, history and semiconductors, human judgment and AI computation all met inside one book.

In the essay "Living as a Hardware Engineer", I found the engineer's spirit in the Neo-Confucian expression *gyeokmul chiji* (格物致知): investigating things deeply in order to extend knowledge. Engineering writing adds one more step to that practice.

Look closely at things,
verify them with data,
explain them in sentences,
and share them with colleagues so that judgment can improve.

An experience left unwritten ends as one person's memory. An experience placed on the record becomes knowledge that colleagues can examine. Knowledge read and debated together becomes part of an organization's—and a society's—culture.

Writing is therefore not an optional skill for engineers. It is how they offer the world they have seen to another person. It translates technology into human language. Above all, it is a joyful way for someone who has worked for a long time to rediscover the meaning of that time.

It will be all right if readers close this book and still cannot produce a perfect answer in an interview. I hope instead that they can state the data they verified and the standards behind their judgment, discover a new perspective in another person's argument, and explain the conditions under which they would change their minds.

The *chaekmun* did more than test one person's knowledge. It asked how that person understood the country's problems and what form of life with others they proposed.

The same question remains for engineers today.

> What will you make with technology?
> And in what words will you describe the world that technology will create?

Now it is time for each of us to write an answer.

Final Interview Checklist

After working through all nineteen royal policy questions, read this page **once, immediately before the interview**. Do not try to memorize every sentence. Check only that your conclusion, evidence, objection, and reversal condition form one connected argument.

30-Second Preparation

- Have I reframed the prompt as a **decision that must be made**, rather than a slogan for or against something?
- Have I chosen the single most important value to protect and identified the group that will bear the cost of that choice?
- Have I named one number or event—a **reversal condition**—that would change my conclusion?

Data Check

- Can I state a number together with its **source, base year, and unit**?
- Have I distinguished forecasts from observations, nominal from real values, averages from distributions, and correlation from causation?

- Am I directly comparing numbers that cover different populations or accounting scopes?
- If I cannot remember a number, am I ready to name the dataset I would verify instead of inventing a value?

Structure for a 90-Second Answer

1. **Conclusion:** "My judgment is ..."
2. **Evidence:** Present two pieces of evidence that differ in kind.
3. **Objection:** Acknowledge the strongest reason supporting the other side.
4. **Condition:** State which indicator, at what threshold, would make you revise the decision.

Group-Discussion Conduct

- Did I briefly summarize the previous speaker's central point before adding a missing indicator?
- Did I examine assumptions, data, and criteria rather than attacking a person?
- Did I help clarify the structure of the discussion instead of merely maximizing how often I spoke?
- Did I invite a quiet participant into the discussion or connect them with an opportunity to speak?

Final Royal Policy Question

What evidence would show that my conclusion is wrong, and would I actually change my judgment if that evidence appeared?

About the Author

Nakcho Choi

Semiconductor and display manufacturing engineer with 26 years of experience

Nakcho Choi has spent his career analyzing complex problems in manufacturing and R&D and turning the results into reports that colleagues can use.

After earning a professional data-analysis certification, he expanded his use of R for manufacturing-data preprocessing and visualization. His work focuses on how to analyze large volumes of production-line data and Design of Experiments (DOE) results, then translate them into clear charts and decision materials.

Since the emergence of GPT, he has studied multi-agent analysis and writing with tools including Claude, GPT, and Gemini. He sees AI not as a substitute writer, but as a collaborator that broadens questions, produces counterarguments, and supports human judgment.

Observing the growing importance of group discussions that connect technology with the humanities in semiconductor recruitment, he applied the Joseon civil-service examination's *chaekmun* format to contemporary industry problems. This book uses data to debate AI, manufacturing processes, chip design, equipment, supply chains, labor, and the environment.

He writes in the hope that researchers and applicants in the semiconductor and display fields will grow beyond explaining technology alone: that they will see the world more broadly and work with colleagues to solve problems that have no single correct answer.

For many years, he has written about technology and people, organizations and the future in the Brunch Story magazine "Stories of a Display Engineer".

> **Note**
>
> This independent work presents the author's personal research and views. It does not represent the official position or hiring guidance of his employer, and it uses no confidential company data or internal recruitment information.